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MITRE TECHNICAL REPORT

DARPA Automating Scientific Knowledge Extraction and Modeling (ASKEM) Final Program Evaluation Report

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Abstract

This report contains an overview of the DARPA Automating Scientific Knowledge Extraction and Modeling (ASKEM) final program evaluation, designed and executed by the MITRE Test and Evaluation (T&E) team. The evaluation exercised functionality from all four program Technical Areas (TAs) and for the two main program use cases - epidemiology and climate modeling. The report describes: (1) the methodology for the evaluation and processing of results; (2) qualitative and quantitative results, including timing, quality and completeness scores, decision-maker panel scores, and accuracy metrics; (3) challenges, successes, and lessons learned; and (4) end-of-program transition efforts, partnerships, and collaborations.

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All modelers (as listed in Table 1) who dedicated more than a week of their valuable time to participate in the final evaluation described in this report; their efforts enabled the MITRE Test and Evaluation (T&E) team to successfully execute the evaluation and collect detailed quantitative and qualitative outcomes

- All decision-maker panelists (as listed in Table 1) who dedicated valuable time to participate in the final evaluation described in this report, and Rakhee Palekar, who helped the T&E team connect with and recruit several panelists for the epidemiology use case; their efforts enabled the MITRE T&E team to obtain critical decision-maker metrics needed for the program
- The performer teams funded by the program, who developed the technology that was evaluated and with whom the MITRE T&E team worked closely over the course of the program, through numerous hackathons and evaluations
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Table 1. All modelers and decision-maker panelists who helped make the final DARPA ASKEM evaluation possible.

Use Case	Modelers	Decision-maker Panelists
Epidemiology	Lilly Boyer Ben Harris Kitt Burch Mikaela Meyer David D'Auria Jeff Rewley Ellen Badgley Joshua Stadlan	Mohsen Malekinejad Carla Bezold Alex Wu Justin Crow Sam Malloy
Climate	Justin Lieffers Kristy Weber Carlos Ramos Jason Veneman Drew Lipman Brian Pettegrew Matthew Enlow Michael Postol	Susan Boa Jenn Richkus Ted Watters

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1 Executive Summary

1.1 Purpose and Scope

DARPA's Automating Scientific Knowledge Extraction and Modeling (ASKEM) program aimed to develop Artificial Intelligence (AI)-enabled tools and a next-generation knowledge-modeling-simulation ecosystem for the agile creation, sustainment, and enhancement of complex models and simulators that support expert knowledge- and data-informed decision-making in diverse missions and scientific domains.

This report contains an overview of the DARPA ASKEM final program evaluation, designed and executed by the MITRE Test and Evaluation (T&E) team.

1.2 Final Evaluation Design and Structure

The objectives for ASKEM's final program evaluation were multifold:

- Assess progress and validate ASKEM technology and tools by obtaining quantitative/semi-quantitative metrics in comparison to a baseline
- Leverage evaluation results and takeaways to compile best practices and lessons learned, which can shape the design of future experiments with AI technology (including on other programs)

The MITRE team exercised functionality from all four program Technical Areas and for the two main program use cases - epidemiology and climate modeling. For each use case, there was one experiment with an extended decision-maker scenario with multiple parts, and one experiment with a collection of independent questions, resulting in four experiments being run concurrently: **1) Epidemiological Decision-maker Scenario, 2) Epidemiological Independent Questions, 3) Climate Decision-maker Scenario, and 4) Climate Independent Questions**.

Decision-maker scenarios were designed to test ASKEM technologies under realistic and open-ended scenarios, where outcomes and solutions presented to decision-maker panels intended to emulate the decision maker/modeler interface and real-world decision-making processes. Independent questions involved constrained question sets designed to test specific aspects of ASKEM technology without worrying about cascading effects and errors.

For each experiment, there was 1) a **Baseline modeling team** working on the scenario or questions using state-of-the-art non-ASKEM technology and whose output constitutes the baseline results, and 2) a **Workbench modeling team** using ASKEM technology (either as a collection of tools or an integrated workbench, depending on the program use case and experiment), and whose output constitutes ASKEM results. All teams consisted of two modelers working collaboratively.

Separately from the four experiments, the MITRE team also designed a new influenza **forecasting challenge** specifically to obtain accuracy metrics for the epidemiology use case and leverage a natural baseline that already exists through a commonly-known epidemiological forecasting challenge platform.

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In total, the final evaluation supported the following metric categories:

- Accuracy (based on the forecasting challenge)
- Timeliness (based on the independent question sets)
- Decision-making Confidence (based on the decision-maker scenarios)
- Generalizability (no baseline required)
- Scalability (no baseline required, partially supported by independent question sets)

1.3 Key Findings and Opportunity Areas

We highlight several key results from each of the four experiments, and forecasting challenge.

Epidemiological Independent Questions:

- The Workbench team was more efficient with model extraction of more complicated models, correctly ingesting and reproducing unit tests almost three times as quickly as the Baseline team, and additionally identifying a population conservation error which the Baseline team did not catch.
- Some critical and fundamental tasks for editing models (and which are important for iterating on and developing models) remained slow with ASKEM tools, highlighting an important opportunity for further improvements.
- For nearly all simulation-related questions, tasks were done more quickly and efficiently using the automated workflows in the ASKEM workbench, at a comparable or higher quality compared to Baseline tools, achieving perfect scores for almost all questions. However, the only question that the Workbench team didn't achieve a perfect score on, highlights an improvement opportunity - this was a multi-objective optimization problem with multiple constraints, and although the Workbench team found an answer faster, it was not the correct one. The ASKEM workbench optimization operators generally struggled with optimization questions due to basin-hopping and unpredictable behavior.
- For model comparison and summarization, ASKEM technology provided a meaningful speed increase over humans manually reading papers and creating summarization tables, with no significant differences in the quality and completeness of the outputs.
- Questions under Technical Areas 3 and 4 were in general completed more quickly using ASKEM tools, regardless of question type or difficulty. The time to finish the questions in the ASKEM workbench did not have any clear correlation with question difficulty or complexity level.

For the **Epidemiology Decision-maker Scenario**, the Baseline and Workbench modeling teams ended up taking very different approaches, making it less meaningful to compare on a question-by-question basis.

- The Baseline team's approach was to create a complex model, and faced challenges in finding the right parameters and munging data into the right formats. The team chose not to use off-the-shelf large language models (LLMs) for literature search due to lack of familiarity with LLMs at the time, but later mentioned that they would have appreciated

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AI assistance on this topic. Despite the model's complexity, simulation results were largely reasonable. However, some decision-maker panelists pointed out concerns about model overcomplexity and the possibility of parameter overfitting, and the difficulties in understanding uncertainty propagation with a large number of model compartments.

- In contrast, the Workbench team chose to ingest and extend one of the published models described in the scenario, but ran into many challenges with ASKEM tools during ingestion and stratification, and also inherited some awkward assumptions around vaccination that were built into the original published model. Due to these challenges, many of the team's simulation outcomes were not plausible, were not explainable, and did not match decision-maker panelists' intuitions, leaving them skeptical of the outcomes and the reasons behind them. However, the Workbench team noted and appreciated easier metadata extractions (e.g. of parameter descriptions and units) from publications. They also found that having multiple ways to view/inspect/verify models, was very helpful for debugging.

Climate Independent Questions:

- Overall, the Baseline team spent more time to answer fewer questions
- For model configuration problems, there was one problem where the Workbench team could not demonstrate turbulent behavior, potentially due to the specific solvers used with ASKEM tools (Decapodes), or user error. Additionally, ASKEM tools have limits on the types of topology supported, and required some simulations during the evaluation to be run on torus or spherical geometries.
- Another area of difficulty was meshes used for multiphysics simulations. In one instance, the Workbench team spent several hours unsuccessfully attempting to create the required geometry (a 3D pipe mesh) to move forward with the questions. Based on this experience, the T&E team recommends that ASKEM tools include additional documentation and guidance to help new users of their tools to create geometries, a critical step in modeling and simulation with physics-based partial differential equation (PDE) models.
- An example where ASKEM technology provided an advantage was in applying real-world data to a physics-based model – this was a simple exercise with ASKEM tooling, but extremely difficult for the Baseline team using software like MATLAB.
- The integration of ASKEM tooling with the Julia open-source computational ecosystem provided huge benefits to the Workbench team for completing questions using pre-existing documentation, or in allowing modelers to ask how to use various Julia packages, or ask for help in writing Julia code. Although ASKEM technology outputs were not completely free of errors, the Workbench team noted that the tools got them 60-70% towards a final solution. The Baseline team did not have access to this type of functionality or assistance using non-ASKEM tools.
- For questions related to finding and downloading climate data, ASKEM tools helped the Workbench team find the name of relevant datasets more quickly; the team also noted

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that one of the returned dataset names did not actually exist, hinting at LLM hallucination.

For the **Climate Decision-maker Scenario**, both Baseline and Workbench teams ended up taking similar approaches to the problem, although using very different toolsets.

- For processing large climate datasets and identifying trends, the Baseline team painstakingly processed data with a number of scripts, requiring interpolation and an equation transformation. The Workbench team used ASKEM tools to investigate climate data files and then create very precise prompts for LLM agents, to subset data and organize it in certain formats. Although the format of their plots were different, both teams ended up observing similar trends (e.g. temperature decreasing with altitude, temperature increasing with year, moisture/humidity decreasing with altitude).
- For some of the multiphysics modeling questions, both teams achieved roughly equivalent outcomes, but simulating the Navier Stokes model under turbulent flow conditions was a challenge for the Workbench team. Using ASKEM tools, model convergence was chaotic and failed to show the expected outputs, despite the team exploring multiple mesh sizes, solvers, time steps, boundary conditions, and vortex conditions. The modelers suspected the reason was the handling of boundary conditions, although it is unclear to what extent user error contributed to these issues. Inability to complete this key question led to limitations downstream in later parts of the scenario, and also detracted from decision-maker confidence scores.
- In implementing radar equations for this scenario, it should be noted that the Workbench team encountered a valuable lesson on not over-relying on ASKEM's AI tools to generate code, without checking for scientific reasonableness. The tools initially generated Python code from Matlab functions that upon first glance seemed complete and equivalent, but upon further inspection did not generate sensible results, although once identified, the problem was corrected.
- For the final section of the scenario which involves generating radar performance results, the Workbench team generated distributions while the Baseline team did not incorporate uncertainty; however, the Workbench team's results were unrealistic in its limited sensitivity to relevant factors. This may potentially be due to challenges in implementing turbulent flow earlier in the scenario, in additional to possible unit conversion mistakes which may have contributed to the issue.
- During the decision-maker panels, for all parts scored, the mean confidence score was higher for the Baseline team. Most likely the biggest detractors of confidence for the Workbench team trace back to the challenges with the Turbulent Flow model (e.g. inability to validate the model with a unit test, and not being able to implement periodic boundary conditions).
- Decision-maker panelists also wanted to see more validation of approaches from both teams, which would have increased their confidence in outcomes, but understood the limitations created by time constraints.

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Forecasting Challenge:

- For the forecasting challenge, ASKEM performers were asked to make single model and ensemble model forecasts for two states - Pennsylvania and Florida. The model performances were compared against results from the CDC's FluSight Forecast Hub (which collectively represented the non-ASKEM Baseline modeling effort).
- While previous forecasting challenges had ASKEM performer forecasts in the general neighborhood of Baseline forecast scores, in the final forecast challenge, there were clearly several timepoints and circumstances where one or more of the ASKEM performer forecasts outperformed all Baseline forecasts.
- For the state of Florida, both ASKEM performer forecasts (single and ensemble models) outperformed all Baseline forecasts, for part of the forecast period.
- For the state of Pennsylvania, the ASKEM ensemble forecast outperformed all Baseline forecasts, for part of the forecast period.

Lessons Learned from Test and Evaluation (T&E) Experiment Design

In reviewing the final evaluation's experimental design and based on feedback from participants and program performers, we highlight best practices that can be leveraged for other programs with similar evaluation structures, as well as opportunities for improvement.

Best Practices	Opportunities for Improvement
• In-person teamed approach • On-site team of performers for support and debugging • Have data preprocessed, accessible, and ready to use • Comparable 'tool learning curves' for users of both DARPA and baseline technologies	• More hands-on exercises for Baseline teams during training/preparation phase • Future evaluations of new AI technology should consider ***requiring*** the use of state-of-the-art AI models by Baseline teams (or recruiting users who are familiar enough with such models to use them in their daily work) • Consider flipping scenario design development so the evaluation has challenging scenarios that are known to be completable with DARPA performer tools, and see whether state-of-the-art baseline tools can achieve the same functionality in the same time

1.4 Potential for Future Impact

Finally, in Table 2 we summarize the ASKEM program's current and potential future for impact, based on the capabilities demonstrated during the final program evaluation, and stakeholder engagements during the last phase of the program.

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Table 2. Potential for technology impact by program use case

Program Use Case	Potential for Impact
Epidemiology Use Case	• The ASKEM epidemiology workbench Terarium (developed by the Uncharted team and integrating technology from a number of other performer teams) was built with epidemiologists in mind, and many of its capabilities exercised during the final evaluation, have been demonstrated to a number of stakeholders in the public health space, including academic and government institutions like the University of California, San Diego, the Centers for Disease Control and Prevention, and the California Department of Public Health • By modeling the impact of interventions on health care costs, the workbench has demonstrated the ability to extend disease modeling to include economic components • There is potential for ASKEM technology to be applied to other domains with similar modeling and simulation requirements to epidemiological compartmental models, including: population growth, ecology and predator-prey systems, biological signaling, migration flows, game theory models, economic models, and supply chains. These are topic areas which all have the potential to be represented by ordinary differential equations and modeled in the ASKEM workbench. The wide array of potential use cases leads to a large set of possible stakeholders and transition pathways.
Climate Use Case	• Decapodes, a physics modeling framework developed by ASKEM performers, has the potential to demonstrate an advantage over state-of-the-art baseline software tools for developing and implementing new physics models and equations through its multiphysics model coupling capabilities and ability to easily apply complex models to arbitrary geometries. This technology is also being expanded on to apply to space weather (including, magnetohydrodynamics and the chemistry and physics of lightning) • JuliaHub, one of the ASKEM performer teams, has become a corporate partner of the National Science Foundation (NSF) Colorado-Wyoming (CO-WY) Climate Resilience Engine, and will support R&D projects with its technical expertise in scientific modeling and simulation, high-performance computing, and scientific machine learning • AI-powered co-scientist tool Beaker (developed by the Jataware team), is capable of generating bespoke analytical and computational workflows and visualizations, and has demonstrated applicability and generalizability to the wildfire, critical minerals, biomedical, and climate/weather domains.

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2 Introduction

DARPA's Automating Scientific Knowledge Extraction and Modeling (ASKEM) program aimed to develop Artificial Intelligence (AI)-enabled tools and a next-generation knowledge-modeling-simulation ecosystem for the agile creation, sustainment, and enhancement of the complex models and simulators that support expert knowledge- and data-informed decision-making in diverse missions and scientific domains.

This report contains an overview of the DARPA ASKEM final program evaluation, designed and executed by the MITRE Test and Evaluation (T&E) team. The team exercised functionality from all four program Technical Areas (TAs) and for the two main program use cases - epidemiology and climate modeling. In addition to designing decision-maker scenarios as done in previous evaluations, the T&E team also developed independent question sets. With a few exceptions, these questions were designed to be independent of each other so that various aspects of ASKEM technology could be tested without worrying about cascading effects. This resulted in four experiments being run concurrently with a total of eight modeling teams:

- Epidemiological Decision-maker Scenario (Baseline and ASKEM Workbench teams)
- Epidemiological Independent Questions (Baseline and ASKEM Workbench teams)
- Climate Decision-maker Scenario (Baseline and ASKEM Workbench teams)
- Climate Independent Questions (Baseline and ASKEM Workbench teams)

Section 3 of this report (Evaluation Design, Execution, and Post-Processing) describes the methodology for the evaluation and processing of results. Section 4 (Evaluation Outcomes) covers qualitative and quantitative results from the evaluation, including timing, quality, and completeness scores, decision-maker panel scores, and accuracy metrics from the forecasting challenge. It also includes challenges and successes from the perspective of the modeling teams, and lessons learned regarding the design and execution of the evaluation itself. Section 4 (Features and Capabilities Not Evaluated) mentions recently developed capabilities not evaluated during the formal program evaluation for various reasons. Section 5 (Potential for Future Impact) focuses on the future by describing end-of-program transition efforts, partnerships, and collaborations. And finally, Appendix A: Metrics Tables contains metrics tables for both use cases, and Appendix B: Performer-Specific Metrics contains team-specific metrics.

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3 Evaluation Design, Execution, and Post-Processing

3.1 ASKEM T&E Design, Evaluation Structure, and Logistics

The ASKEM program evaluation metrics have been designed around a comparison of 'Baseline' modelers using existing modeling approaches (including any state-of-the-art tools but not ASKEM technology), against modelers using the ASKEM tool suite (or workbench). The ASKEM Broad Agency Announcement (pp. 16-18) outlines MITRE's responsibilities as the independent Test and Evaluation (T&E) team along with a timeline and associated metrics for

> "… program-level assessments comparing performance on end-to-end modeling tasks, with and without the ASKEM system. [The T&E team] will be responsible for providing the state-of-the-art modeling baseline used to measure overall ASKEM impact on the modeling pipeline."

The ASKEM Broad Agency Announcement, Table 2 (pp. 17) also indicates the program metrics for each Technical Area (TA), for various metric dimensions or categories. These metrics have had some adjustments and updates throughout the course of the program, and the Appendix A: Metrics Tables contains the final program metric definitions and targets.

Shaped by the Broad Agency Announcement and evolution of the program, the test objectives for ASKEM's final program evaluation milestone were twofold:

- Assess progress and validate ASKEM technology and tools by obtaining quantitative/semi-quantitative metrics in comparison to a baseline
- Leverage evaluation results and takeaways to compile best practices and lessons learned, which can shape the design of future experiments with AI technology (including on other programs)

The final program milestone consisted of the following events and activities:

- A week-long in-person hackathon session preceding the actual evaluation, for program performers to develop and strengthen capabilities that would be needed for the evaluation. This took place from December 2nd – 6th, 2024, in Fort Collins, Colorado. All hackathon scenarios and supplementary materials were posted at the program GitHub repository: https://github.com/DARPA-ASKEM/program-milestones/tree/main/final-program-milestone/hackathon
- Pre-evaluation orientation and training week, January 27th - 31st, 2025. The T&E team delivered an orientation session to all modeling teams participating in the evaluation. Workbench training sessions were delivered by ASKEM performers on each of the tools or technology stacks they developed, and Baseline training sessions covering equivalent types of functionality (but using baseline tools) were delivered by the T&E team.
- A week-long in-person evaluation that took place from February 3rd - 7th, 2025, in Arlington, Virginia. The structure of the evaluation was as follows.
    - Four experiments were run concurrently across the two use cases. For each use case, there was one experiment with an extended **Decision-maker (DM)**

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Scenario with multiple parts, and one experiment with a collection of **Independent Questions (IQ)**. The four experiments were:
    - Epidemiology Decision-maker Scenario
    - Epidemiology Independent Questions
    - Climate Decision-maker Scenario
    - Climate Independent Questions
  - For each experiment, there was 1) a **Baseline modeling team** working on the scenario or independent questions using state-of-the-art non-ASKEM technology and whose output constitutes the baseline results, and 2) a **Workbench modeling team** using ASKEM technology (either as a collection of tools or an integrated workbench, depending on the program use case and experiment) and whose output constitutes ASKEM results. All teams consisted of two modelers working collaboratively. See Figure 1 for the organization of teams across experiment and use case.
  - The evaluation scenarios were released to all modeling teams (and performers) at the beginning of the one-week in-person evaluation event. All evaluation scenarios and supplementary materials were posted at the program GitHub repository: https://github.com/DARPA-ASKEM/program-milestones/tree/main/final-program-milestone/evaluation.
  - To capture data on timing, all modeling teams were asked to log their activities on each scenario question during the evaluation. There was one member of the T&E team present with each modeling team, supervising the evaluation process, tracking timing and progress, providing clarification on scenarios, and taking contextual notes to: (1) support post-event analysis of results, (2) understand the modeling artifacts being created, (3) capture the challenges experienced by modelers, and (4) capture feedback from the modelers throughout the event.
- A decision-maker panel for each of the decision-maker scenarios, held at the culmination of the evaluation event, on Friday, February 7th, 2025.
- Post-evaluation debrief sessions between T&E team, modeling teams, and performers, as well as post-evaluation finalizing of timing logs and scenario artifacts
- A post-evaluation forecasting challenge for performers, posted at the program GitHub repository: https://github.com/DARPA-ASKEM/program-milestones/tree/main/final-program-milestone/march-25-forecasting-challenge

Use Case	Baseline (Using Non-ASKEM tools)		ASKEM Tools	
	Decision-maker Scenario	Independent Questions	Decision-maker Scenario	Independent Questions
Epi	*Team 1A-Baseline, Decision-maker Scenario*	*Team 2A-Baseline, Independent Questions*	*Team 1B-Workbench, Decision-maker Scenario*	*Team 2B-Workbench, Independent Questions*
Climate	*Team 3A-Baseline, Decision-maker Scenario*	*Team 4A-Baseline, Independent Questions*	*Team 3B-Workbench, Decision-maker Scenario*	*Team 4B-Workbench, Independent Questions*

Figure 1. Evaluation structure indicating eight modeling teams working concurrently across two use cases.

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3.2 Scenario Design

The T&E team designed four experiments, each meant to exercise a set of key functionalities aligning with the four program Technical Areas (TAs), and supporting a different metrics dimension/category, as described below.

3.2.1 Epidemiology Independent Questions (Epi IQ)

Table 3 describes the tasks and functionality exercised in the Epi IQ experiment, organized by Technical Area (TA).

Table 3. Organization of the Epidemiology Independent Questions.

Section	Tasks and Functionality
Section 1 (TA1)	• Model extraction for COVID-19 and non-COVID-19 epidemiological modeling papers, involving extracting a representation of models from publications and providing it in executable form. • Questions also involved executing extracted models, for purposes of debugging and unit testing. • ***Note***: TA1 was not part of ASKEM Phase 2, and the alignment of Section 1 with the TA1 label is a remnant of task division in the original design of the program. For this evaluation, there were some aspects of TA1 performer technology used to answer extraction questions, but the majority of the extraction features actually reflect the TA4 technology stack (with contributions from TA2 performers as well).
Section 2 (TA2)	• Structural comparison of two or more models and providing a description of the similarities and differences • Model editing • Model stratification
Section 3 (TA3)	Simulation tasks with a given model under different conditions, including: • Basic forward simulation/forecasting • Calibrating model parameters with data • Implementing and optimizing interventions • Creating ensembles
Section 4 (TA4)	• Summarization and comparison of models based on publications and supporting material, according to various aspects features like distinguishing characteristics, assumptions, strengths, limitations, etc. • Data wrangling, processing, and generating decision-maker-friendly visualizations and annotations • Executing commonly used epidemiology analyses (such as horizon-scanning, sensitivity analysis and more)

Each set of questions exercising a particular functionality consisted of 2-3 questions of varying degrees of difficulty (from “easy” to “hard”), where degree of difficulty could be reflected in the size/scale of the problem, or complexity of the models or task involved. The purpose of this setup was to allow for a comparison to be made between the Workbench and Baseline outcomes

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at different levels of problem difficulty. Varying degrees of difficulty also helps support the '***Scalability'*** program metric category, for specific functionality.

Except in the case of the Stratification set of questions, each question was formulated to be independent of every other question, so that all questions could be attempted and evaluated regardless of whether preceding questions were successfully completed; this also avoided the propagation of errors between questions (a common issue in past evaluations with longer scenarios that had many interdependent parts). Additionally, each question was structured to have well-defined inputs, tasks, and outputs, so that it was clear when a problem was completed and what format the answer should be provided in; this supported precise timing measurements for the ***'Timeliness'*** program metric category, for the epidemiological modeling use case.

3.2.2 Epidemiology Decision-maker Scenario (Epi DM)

The Epi DM scenario takes place in September 2025, at the start of the typical respiratory season and the back-to-school period and asks modelers to play a role where they are working for a public health department serving an area with 1 million residents. The scenario centers around the following *decision-maker priorities*:

- Understand role of coinfection when seasonal influenza, RSV, and COVID-19 are cocirculating. What is the impact of viral interaction type (synergistic, antagonistic) on timing and size of infection waves?
- Navigating uncertainty around avian influenza. What could happen if a new strain emerges with pandemic potential, under best- and worst-case scenarios?
- Understand the impact of public health interventions on disease trajectories. Interventions considered include:
    - NPIs like masking, social distancing, and testing policies that impact transmission
    - Immunizations for specific age groups or the general population

The Epi DM Scenario was divided into three main parts, each designed to exercise different capabilities, and aligned to a different ***'Decision-maker Confidence'*** metric, as described in Table 4.

Table 4. Organization of the Epidemiology Decision-maker Scenario

Epi DM Scenario Part	Modeler Tasks	Decision-maker Confidence Metric Supported
Part 1	• Modelers consider various viral coinfection models, and arrive at a final selection based on model capabilities, gap analysis, and their fit-for-purpose (i.e. which model is best suited to address decision-maker priorities). • During the model selection process, modelers are asked to understand the various capabilities of candidate models, and ensure they can reproduce some basic results with the models.	Decision-maker confidence in modelers finding and selecting an appropriate starting point for the scenario
Part 2	• Modelers are asked to build upon their selected model and modify it in such a way that it's capable of being used to address the series of decision-maker questions in Part 3.	Decision-maker confidence in modelers developing an appropriate model and associated parameter space to

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Epi DM Scenario Part	Modeler Tasks	Decision-maker Confidence Metric Supported
	• As updates are made, modelers can incorporate model checks to validate their approach. • Model edits include ○ adding hospitalization and deaths compartments ○ adding an ability to model vaccination ○ age stratification ○ incorporating antagonistic and synergistic viral interactions for coinfections	sufficiently explore the scenario
Part 3	• Modelers forecast what October 2025 to February 2026 will look like when RSV and COVID-19 are cocirculating • Modelers are asked how the forecast changes when they assume antagonistic vs synergistic viral interactions for coinfections • Modelers are also asked to do a Value-of-Information analysis with three intervention options • (*Optional Task*) Forecasting with seasonal influenza and COVID-19 cocirculating • (*Optional Task*) Horizon scanning with a new emergent strain of avian influenza and COVID-19 cocirculating • (*Optional Task*) Optimize intervention strategies to improve outcomes	Decision-maker confidence in understanding model results and tradeoffs between potential interventions

3.2.3 Climate Independent Questions (Climate IQ)

Table 5 describes the types of tasks and functionality exercised with the Climate IQ questions, by Technical Area (TA). ***Note:*** TA1 performers left the program before they were able to develop meaningful capabilities for the climate use case, and therefore this experiment does not include TA1 in any capacity.

Table 5. Organization of the Climate Independent Questions.

Section	Tasks and Functionality
Section 1 (TA2)	• Model configuration and execution with physics models • Editing of physics models • Multiphysics model composition/coupling
Section 2 (TA3)	• Forecasting • Sensitivity analysis • Calibration
Section 3 (TA4)	• Integrated Assessment Model (IAM) understanding • IAM forecasting • IAM interventions • IAM optimizations • Climate data search and transformations

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The Climate IQ questions were designed and structured in the same manner as the Epi IQ experiment, with the only exception being no TA1 functionality exercised in this evaluation. As with the Epi IQ experiment, the Climate IQ questions support the ***'Timeliness'*** program metric category for the climate modeling use case.

3.2.4 Climate Decision-maker Scenario (Climate DM)

The Climate DM Scenario focuses on air defense radar performance as it relates to climate conditions. The scenario considers various locations relevant to the North Warning System (Anchorage, AK; Seattle, WA; Honolulu, HI), historical and future climate conditions (aligned to years 1970, 2025, 2050, and 2075), and a range of altitudes (mean sea level, 750 m, 1500 m, 3050 m, 6100 m).

Similarly to the Epi DM Scenario, this scenario is divided into three main parts, each designed to exercise different capabilities. Parts 1-2 and part 3, are aligned to two different ***'Decision-maker Confidence'*** metrics, as described in Table 6.

Table 6. Organization of the Climate Decision-maker Scenario

Climate DM Scenario Part	Modeler Tasks	Decision-maker Confidence Metric Supported
Part 1	Setup with multiphysics modeling • Find relevant climate data on temperature and humidity, relevant to the time periods and locations under consideration • Implement various component physics models that are being used as approximations of atmospheric behavior (Cahn-Hilliard and Navier Stokes)	Decision-maker confidence in modelers having/developing an appropriate model (or set of models) and associated parameter space to sufficiently explore the scenario/problem
Part 2	Update models to add realism • Build upon component physics models and couple them to increase realism of the modeling capabilities • Replace laminar flow with turbulent flow (more realistic assumption for atmospheric behavior) • Update model parameters and geometry to be more physically realistic	
Part 3	• Determine impact of cloud formation on radar performance (specifically max range of detection) considering various time periods, climate conditions, altitudes, and locations • Summarize findings for DoD decision makers: ○ Are there instances where performance improves, and by how much? ○ Are there instances where performance degrades, and by how much? ○ Which locations are most impacted by future climate conditions in 2050 and 2075? ○ Recommended course of action for each location discussed in the scenario	Decision-maker confidence in understanding model results

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3.2.5 Forecasting Challenge

The four experiments that took place during the February evaluation week were not designed to obtain ***accuracy*** metrics, because those do not require a separate Baseline modeling team to be compared against a Workbench team. Accuracy metrics for the epidemiology use case have a natural baseline available through various epidemiological forecasting challenge platforms (e.g. for COVID-19, influenza, RSV, etc.). Therefore, we designed a new forecasting challenge for influenza for a historical time period, where the Workbench team was represented by ASKEM performers, and the non-ASKEM Baseline effort was collectively represented by forecasts previously submitted to the CDC's FluSight Forecast Hub.

The forecasting challenge for this evaluation was made available on March 5th, 2025, and asked performers (specifically the TA3 Causal Inference Engine for Multi-scale Simulations [CIEMSS] team led by the Pacific Northwest National Laboratory [PNNL]) to forecast seasonal influenza hospitalizations. Performers were asked to begin their forecast on December 9th, 2023, and forecast one-week incident hospitalizations over the next four weeks. The target forecast dates were December 16th, 2023; December 23rd, 2023; December 30th, 2023; and January 6th, 2024. The target forecast locations were the states of Pennsylvania and Florida. For training their models, performers were given hospitalization data for the entire 2022-2023 influenza season (October 2022 – April 2023), as well as the beginning of the 2023-2024 season (October 7th – December 9th, 2023).

Performers were instructed to produce forecasts from both a single model and an ensemble of at least three models. Forecasts were evaluated using the weighted interval score (WIS) metric as done in previous challenges. Performer results were compared with all models which had previously submitted forecasts over the same timeframe, obtained from the FluSight Forecast Hub. The outputs from all the comparison models and forecasters (representing the Baseline) are available here.

Performers submitted their results at the beginning of April 2025, and the T&E team processed their results and compared against the FluSight forecasters, using the *zoltar* and *covidHubUtils* R packages developed by the Reich Lab (at the University of Massachusetts Amherst), for pulling down the data and evaluating the forecasts. WIS, the main metric used, "can be interpreted as a generalization of the absolute error to probabilistic forecasts and allows for a decomposition into a measure of sharpness [spread] and penalties for over- and underprediction." A lower score indicates better performance. See the Delphi Evaluation Dashboard (from the Carnegie Mellon University's Delphi Group) and cited publications in the dashboard, for more details.

3.3 Recruitment of Workbench and Baseline Modelers, and Onboarding Process

For each of the four experiments run during the week-long evaluation, the T&E team attempted to recruit four MITRE staff with the appropriate background and modeling experience, who were not part of the ASKEM program. Because it was challenging to find enough MITRE employees with the skill sets required, in two instances we contracted with modelers outside of MITRE to fill out our roster of modelers. We recruited a total of sixteen (16) modelers for this evaluation.

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All modelers were provided with onboarding materials that covered the following topics:

1. Overview of the DARPA ASKEM program
2. Evaluation Structure
    - MITRE's role and approach to the evaluation
    - The modeler's role in the evaluation
    - Pre-evaluation, evaluation, and post-evaluation tasks
3. Evaluation Preparation (topics the modelers should be familiar with and preinstallation of relevant software, for each use case)
4. Training Details (schedule and topics covered)
5. Logistics and Administrative Items
    - Information about scenarios, time budgeting, and expectations for output products
    - Storage (where to save outputs and also find resources)
    - Record keeping (how and where to log time, with examples)
    - Information on reach-back SME support teams for each use case
    - Communications channels for asking questions of the T&E team
    - Evaluation agenda
6. Resources and Background Materials
    - Orientation materials folder (which included starter kit materials, prior program briefs on the use cases, previous evaluation scenarios, etc.)
    - Information on a virtual machine which was pre-configured with several common scientific computing, data science, and visualization packages. For the epidemiology use case, some models from the Starter Kit were pre-installed. This was available as a resource if the modelers wanted to utilize it.

The orientation and training week took place from January 27th-31st, 2025, and included

1. A general orientation session about the overall evaluation, for all modelers across both use cases
2. An orientation session for each use case
3. Various training sessions depending on the specific toolsets each team would be using during the evaluation

In particular, the Climate Baseline teams were asked to use COMSOL, MATLAB, various Julia packages, and potentially Python or R computational libraries to support the other tools. Installation instructions and tutorials for these were included in the onboarding materials and training sessions. The T&E team also brought two loaner laptops capable of supporting all these software applications and packages; each Climate Baseline team made use of one of these laptops, particularly for the questions and scenario sections requiring COMSOL, and in some cases installing some of the additional software packages and libraries.

Figure 2 and Figure 3 describe the orientation and training topics and schedules for each use case.

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	Baseline (Using Non-ASKEM tools)		ASKEM Tools	
	Decision-maker Scenario	**Independent Questions**	**Decision-maker Scenario**	**Independent Questions**
Team	*Team 1A-Baseline, Decision-maker Scenario*	*Team 2A-Baseline, Independent Questions*	*Team 1B-Workbench, Decision-maker Scenarios*	*Team 2B-Workbench, Independent Questions*
Use Case Mtg 1/27 2-3pm ET	• Walk through example modeling and simulation tasks for epi models • Walk through scenario design templates for infectious diseases (Runge et. al)			
Training	Training on baseline modeling and simulation tools • ODE solvers (odeint, deSolve) • forecasting • interventions • calibration • sensitivity analysis • optimization • Visualizing simulation results **Scheduled for 1/29, 1-2pm ET**		Training on ASKEM tools (Terarium Workbench) • Day 1: Gathering resources, model extraction, configuration and simulation • Day 2: Model edits, stratification, model checks • Day 3: Simulation workflows (forecasting, interventions, calibration, sensitivity analysis, optimization, ensembles) • Day 4: Scenario templates, etc. • Each session will have pre-reading and homework (try some exercises on your own) **Scheduled for 1/27-1/31, 12-1pm ET each day**	

Figure 2. Orientation and training topics and schedules for the epidemiology modeling teams.

Use Case	Baseline (Using Non-ASKEM tools)		ASKEM Tools	
	Decision-maker Scenario	**Independent Questions**	**Decision-maker Scenario**	**Independent Questions**
Team	*Team 3A-Baseline, Decision-maker Scenario*	*Team 4A-Baseline, Independent Questions*	*Team 3B-Workbench, Decision-maker Scenario*	*Team 4B-Workbench, Independent Questions*
Use Case Mtg 1/27 4-5 pm ET	• Overview of climate modeling use case (overview of climate and earth sciences, modeling landscape, example models, how models are used, DoD priorities) • Walk through climate modeling workflows (Mural)			
Training	Training on baseline modeling and simulation tools • ESGF • Mimi framework (IAMs) • COMSOL (multi-physics) • Matlab and other tools (model edits, forecasting, calibration, sensitivity analysis) **Scheduled for 1/30, 3-5:30pm ET**		Training on ASKEM tools • Beaker (ESGF data, IAMs) **1/28, 1-2:30pm ET** • Decapodes (multi-physics) **1/31, 11am-1pm ET** • Julia simulation tools (forecasting, calibration, sensitivity analysis) **1/31 4:30-5:30pm ET**	

Figure 3. Orientation and training topics and schedules for the climate modeling teams

3.4 Decision-maker Panels

For each of the two use cases, we recruited individuals from state and local government, academia, or MITRE staff with the relevant policy or operational backgrounds and experience working with modelers, to play the role of decision makers and review select output from the modeling teams working on the decision-maker scenarios. These decision-maker panels were intended to emulate the decision maker/modeler interface and real-world decision-making

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processes related to policies or courses of action (CoAs) supported by modeling and simulation output. For the epidemiology use case, panelists had backgrounds in epidemiology and public health, and a couple were heavily involved in various local and state government roles during the height of the COVID-19 pandemic. For the climate use case, panelists had backgrounds in oceanography, climate and environmental science, applied math and data science, and have supported various government sponsors in weather or climate-related roles.

The decision-maker panels were held in-person on the final day of the evaluation event - February 7th, 2025. For each use case, the Baseline team presented to the respective panel in the morning session, and the Workbench team presented in the afternoon. Decision makers were asked to evaluate modelers' approaches and outputs, according to the decision-maker confidence metrics described in the Scenario Design section, and according to the metrics tables in Appendix A: Metrics Tables.

3.5 Debrief Sessions and Processing Evaluation Results

3.5.1 Debrief Sessions

During the final day of the evaluation week, once decision-maker panel scores were collected, there were debrief sessions for both the Epi DM and Climate DM experiments, with the modeler teams, panelists, and the T&E team, discussing how the panels went, modelers' experiences, and general feedback from the decision makers. Much of this feedback, takeaways, and lessons learned, is summarized later in the Evaluation Outcomes section of the report.

The T&E team also held separate debrief sessions with each of the modeling teams. For the IQ teams, this took place during the final day of the evaluation itself. For the DM teams, these debrief sessions were scheduled after the evaluation week. During these debrief sessions, the T&E team clarified any questions they had as they processed the scenario outputs, timing logs, or other topics. The T&E team also asked modelers for their feedback on their biggest challenges and highlights from the eval, aspects that were unexpected, things they would do differently if they were to repeat the experience, and feedback on evaluation structure and logistics. Much of the feedback, takeaways, and lessons learned, is summarized later in the Evaluation Outcomes section of the report.

Following the completion of the evaluation event, the T&E team held additional tool feedback debriefing sessions with the Workbench modeling teams together with the performers whose tools were used during the evaluation; these sessions gave modelers an opportunity to directly provide feedback to the performers, and gave performers an opportunity to ask questions of the modelers. The TA4 team from Uncharted Software asked the Epi Workbench teams (for IQ and DM scenarios) to complete a survey, and reviewed the results with them in detail, taking careful notes on their experiences and feedback.

3.5.2 Timing, Quality, and Completeness Logs

Following the completion of the evaluation, the T&E team performed a detailed post-processing analysis of timing logs and scenario results. This included:

- Correcting mistakes in the original timing logs, and filling in missing times based on reconstruction of events, leveraging notes and recordings

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- Converting from wall-clock time to person-time (i.e. accounting for number of people contributing to various tasks) to enable fairer comparisons
- Separating critical path solving time (time spent directly solving the problem) from non-critical path time (time spent on exploratory or sidebar work that did not end up being required for the final answer), and time spent learning a particular tool (i.e. not solving the problem but this time was required for obtaining the final answer)
- Assigning a quality and completeness score for every question, each according to a 4-point scale.

We used the following rubric for scoring quality and completeness (Table 7). It should be noted that the rubric is ***not linear.*** To achieve a 4 rating, everything had to be correct or fully complete. Any errors or skipped aspects of the problems dropped the score down to a 3, even if the solution was very close to completely correct. Therefore a 3 should not, for example, be interpreted as a score of 75% out of 100% total.

Table 7. Quality and completeness scoring rubric

Quality			Completeness	
Score	**Definition**		**Score**	**Definition**
1	Execution was almost completely wrong		1	Barely started
2	Execution was mix of correct/incorrect, mostly wrong (more wrong than right)		2	Needs a lot of work (made some progress but less than 50% complete)
3	Execution was more reasonable than incorrect (more right than wrong)		3	Mostly or almost finished (50% to 99% complete)
4	Execution was completely reasonable		4	Complete (100%)

For quality, we avoided penalizing later tasks for propagating errors from earlier tasks. This wasn't a big factor for the IQ experiments due to the way the problems were structured. However, error propagation was something that came up for the decision-maker scenarios, with mistakes during earlier parts of the scenarios, having implications for later sections.

If teams ran out of time to do sufficient debugging but created outputs for all parts of the task, we graded results as complete but of poor quality.

For certain tasks, completeness may reflect the utility of the tools used to complete the work in a timely manner, and quality may reflect the utility of the tools for inspecting results, identifying sources of error, and helping to de-bug the issues. In this sense, completeness of results and quality of results can reflect different dimensions of the utility of the Workbench/Baseline tool sets. For this reason it is not advised to create or rely solely upon a combined score of the two; however, for the sake of concise visualization and ease of comparison, we sum up both scores for each problem to communicate broad outcomes in the Evaluation Outcomes section.

Future assessments with the tools and user testing can help untangle the primary sources of user error and inability to complete tasks in timed settings.

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3.5.3 Metrics Tables

The T&E team constructed a full Technical Area (TA)-level metrics table for each use case. This includes, where applicable for the TA and use case: (1) metric definitions, (2) metric targets (established only for the epidemiology use case), and (3) Baseline/Workbench outcomes from the evaluation, for each of the TAs, and for the following metric categories:

- Accuracy (based on the post-evaluation forecasting challenge done by performers)
- Timeliness (based on the timing logs from the independent question experiments)
- Decision-making Confidence (based on scores from the decision-maker panels)
- Generalizability (no baseline required)
- Scalability (no baseline required, partially supported by independent question experiments)

The full metrics tables can be found in Appendix A: Metrics Tables

4 Evaluation Outcomes

4.1 Timing, Scores and Discussion

4.1.1 Epidemiology Independent Questions (Epi IQ)

4.1.1.1 Highlights in Approaches and Results

The Baseline and Workbench teams took differing strategies for completing the Epi IQ problems. The Baseline team used a pair programming approach, working on each of the questions together. The Workbench team employed a divide-and-conquer approach, working on most problems separately while using their partner as a resource or sounding board. As described in section 3.5.2, the T&E team accounted for this difference by tracking time spent on each question by each person, and not just wall clock time; this enabled apples-to-apples time comparisons between teams who approached the questions with different strategies.

As a result, it is impossible to discern whether the Baseline teams or Workbench teams were more efficient just by looking at how many questions each team completed. However, we do note that the Workbench team completed 36 out of the 37 questions while the Baseline team completed 30 out of the 37 questions. Table 8 below indicates the questions both teams completed (green cells), the questions only the Workbench team completed (red cells), the questions neither team completed (white cells), and the questions that were not included in the set (gray cells).

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Table 8. Epi Independent Questions completion distribution

Question Set	Easy	Medium	Hard
1.1 (Model Extraction, Covid)			
1.2 (Model Extraction, Non-Covid)			Workbench Only
2.1 (Structural Model Comparison)			N/A
2.2 (Model Editing)			
2.3 (Model Stratification)			Neither Team
3.1 (Parameter Calibration)			Workbench Only
3.2 (Forecasting)			
3.3 (Interventions)			Workbench Only
3.4 (Optimizations)			
3.5 (Ensembles)			N/A
4.1 (Model Summarization and Comparison)		Workbench Only	Workbench Only
4.2 (Visualization and Data Comparison)			
4.3 (Template Scenarios)			Workbench Only

In Figure 4 we plot the cumulative critical path times (in person-minutes) for both teams against each other; the Workbench modeling efforts are on the y-axis and the Baseline modeling efforts are on the x-axis. The dotted diagonal line shows where the two teams' cumulative times would be equal. Where the plotted lines fall into the region where $x > y$, the Baseline modelers had cumulatively taken more time to solve the questions than the Workbench modelers using ASKEM tools. We can see that by the end of the evaluation there was a large gap in the cumulative time taken to answer the questions between the two teams.

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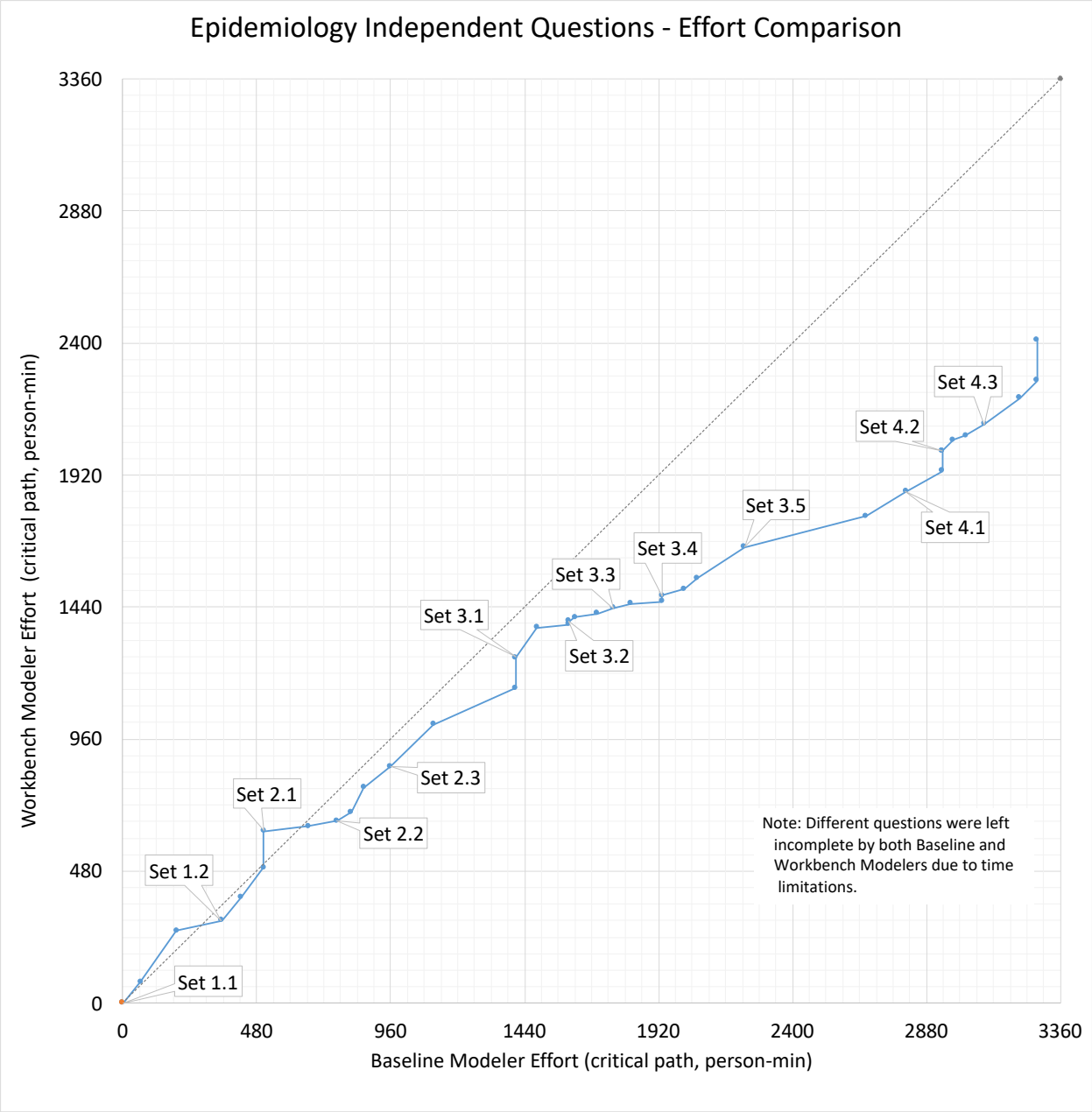


Figure 4. Epidemiology Independent Questions (Epi IQ) cumulative critical path times for the Baseline and Workbench modeling teams. The time spent on each question is represented by a line segment, and the beginning of a question set is annotated with a label. The dotted diagonal line shows where the cumulative times for each team would be equal.

4.1.1.1.1 *Section 1 (TA1)*

Section 1 contained two sets of questions exercising TA1 functionality: Set 1.1, model extraction for COVID-19 models, and Set 1.2, model extraction for non-COVID-19 models. The full breakdown of timing and combined quality and completeness scores can be found in Figure 5. For the easy and medium-level questions for both COVID-19 and non-COVID-19 models, the Baseline modelers were quicker to finish the problems with somewhat similar quality and

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completeness scores compared to the Workbench team. However, the Workbench team was more efficient when extracting more complicated models in the hard questions, completing one such question that Baseliners did not attempt, and completing another question much more quickly than the Baseline team.

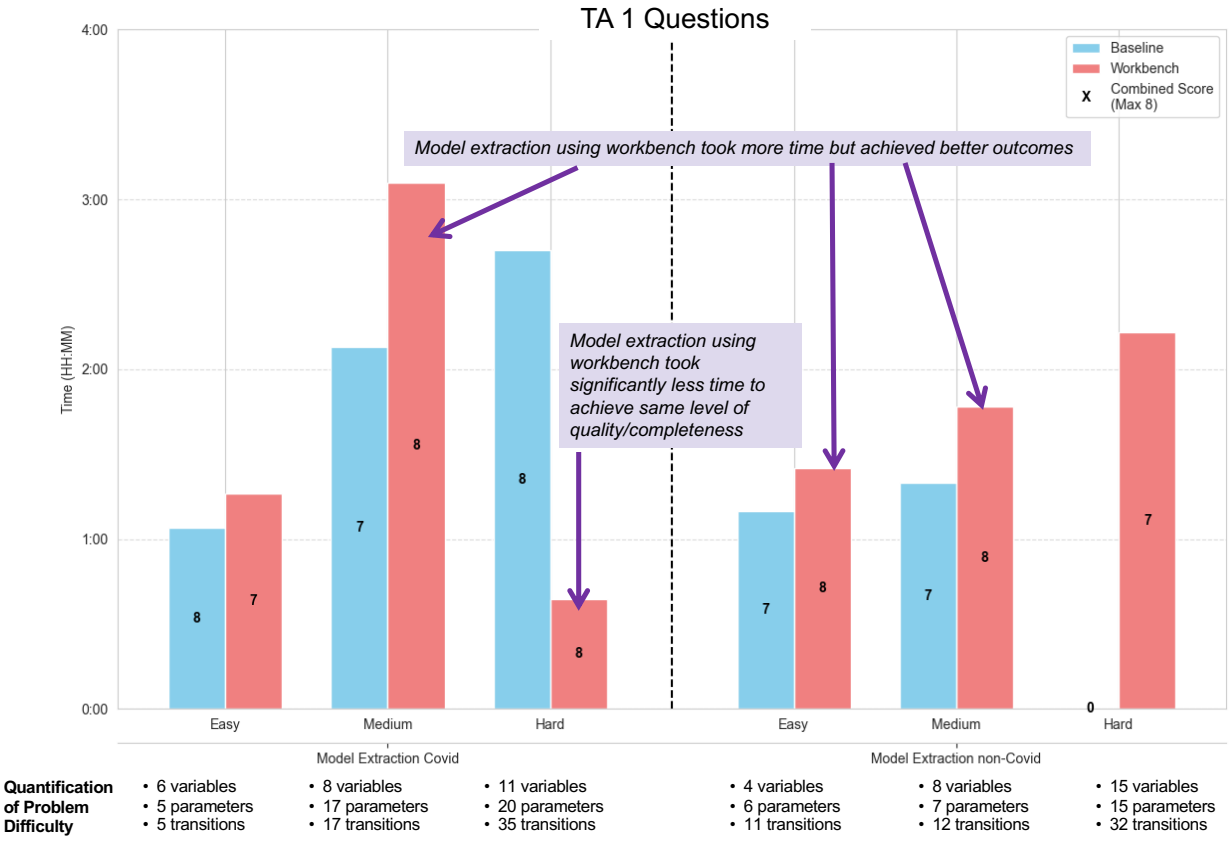


Figure 5. Timing and scores for Epidemiology Independent Questions (Epi IQ), Section 1 (TA1) problems.The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty, in terms of model size and complexity.

One example of the benefits of the Workbench can be seen in the hard COVID-19 model extraction question, the output of which is shown in Figure 6. For this question, both teams received perfect completeness and quality scores. While the Baseline modelers correctly ingested the model as written in the paper, they did not realize that there were issues with population conservation in the original paper. The published model contained an incoming transition into state "I" that did not seem to be distributed into any outgoing transition pathways. This flaw in the model did not affect the Baseline team's ability to correctly reproduce the unit test in the question. The Workbench team also correctly reproduced the unit tests, and while doing so, used ASKEM tools to recognize the population conservation issues in the model and document the problems. Even while identifying and documenting these issues, the Workbench team completed the question using ASKEM tools almost three times more quickly than the Baseline team.

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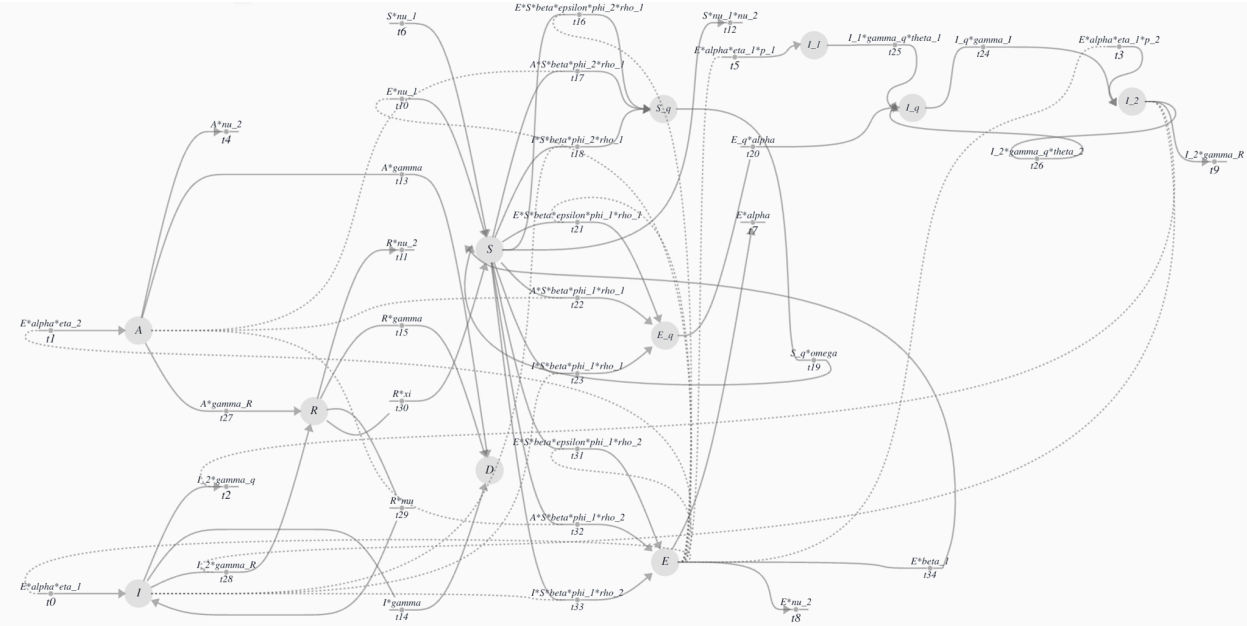


Figure 6. Extracted model diagram in workbench (Q1.1.3, hard COVID-19 model extraction)

4.1.1.1.2 *Section 2 (TA2)*

Section 2 contained three sets of questions exercising TA2 functionality: Set 2.1, model comparison (which only had an easy and a hard question); Set 2.2, model editing; and Set 2.3, stratification. The full breakdown of timing and combined quality and completeness scores can be found in Figure 7. For model comparison and stratification, the Workbench modelers were generally faster than the Baseline modelers with similar quality and completeness outcomes. For model editing, on average the Workbench and Baseline teams took similar amounts of time and produced results of comparable quality. The Baseline modelers did not attempt the hard stratification problem; the Workbench team attempted but could not complete the question, receiving a 0/8 rating for it. This question was the only one that the Workbench team was unable to solve; they deemed it 'impossible' because of its complexity – it would have required 196 strata, which Terarium was unable to handle. However, they did attempt the problem where the Baseline team did not, suggesting that solving the problem with the ASKEM workbench may have felt easier than trying to solve the problem without it.

The Workbench team was able to complete the model comparison problems (Set 2.1) 5-9x more quickly than the Baseline modelers. However, the Terarium model comparison diagrams, as shown with an example in Figure 8 below, were challenging for naïve users to read. This represents an improvement opportunity for performer technology, since new users of Terarium may not be familiar with how to understand the model comparison output diagram, and the diagram format is unlikely to be a useful representation to share with supervisors or decision makers.

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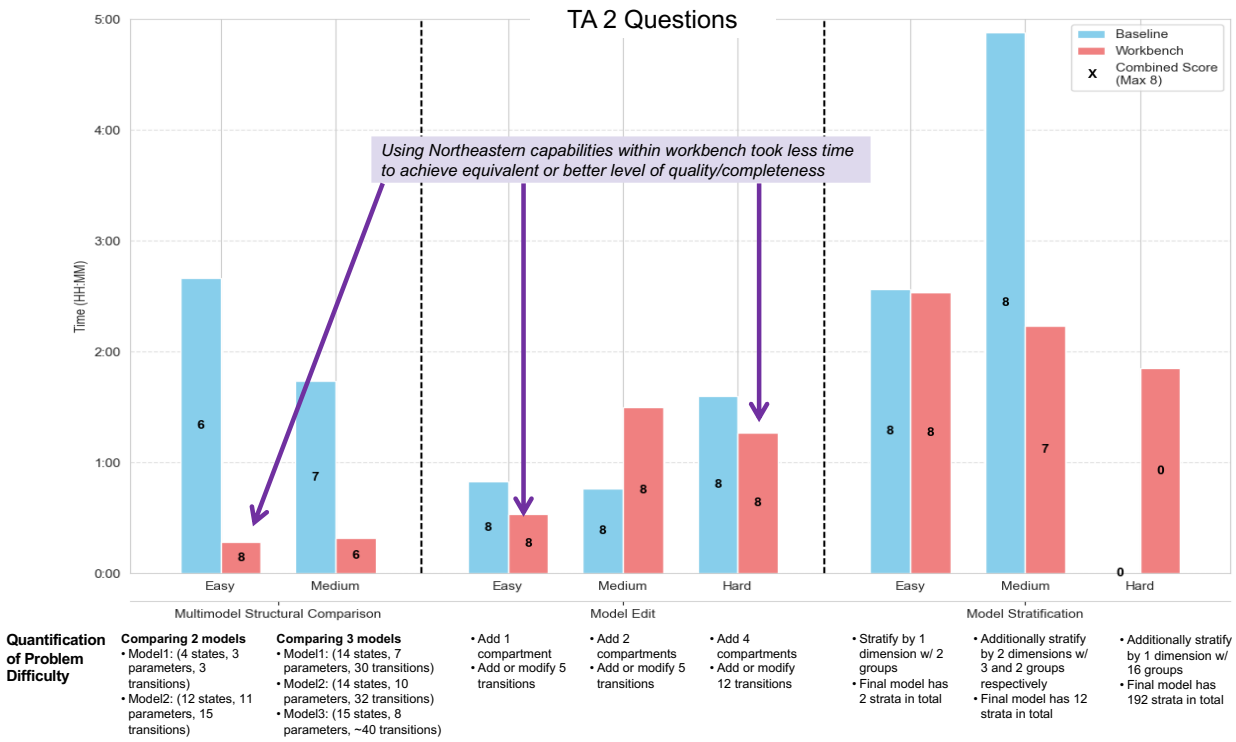


Figure 7. Timing and scores for Epidemiology Independent Questions (Epi IQ), Section 2 (TA2) problems.The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty, in terms of task and model complexity and size.

For the model editing questions (Set 2.2), both teams received perfect scores for every question. As shown in Figure 7, there was some speedup that came from using the Workbench tools, but it was not as significant as the speedups for the model comparison questions and the medium-level stratification question. Some critical, fundamental, and very commonly done tasks for editing a model (like changing variable names) remain slow in the Terarium workbench; this is an important opportunity for further improvement. Because iteration is critical to developing an appropriate model, having the ability to edit a model quickly and efficiently could substantially accelerate the model building process.

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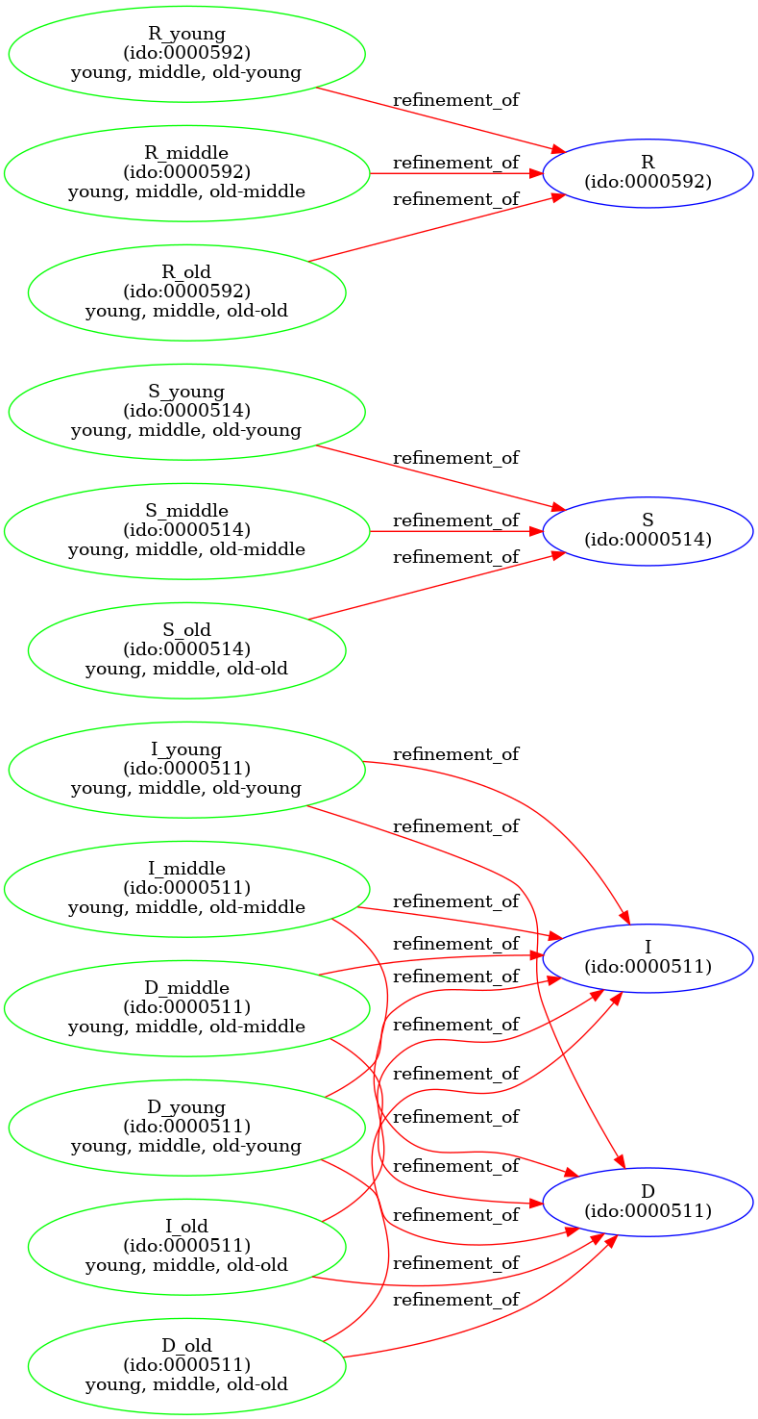


Figure 8. Workbench team's diagram for Q2.1.1 (easy model comparison problem)

4.1.1.1.3 *Section 3 (TA3)*

Section 3 contained five sets of questions (3.1 to 3.5) exercising TA3 functionality – parameter calibration, forecasting, interventions, optimization, and ensembles. The full breakdown of timing and combined quality and completeness scores can be found in Figure 9. For all but one of the problems (easy calibration question), the Workbench team completed the questions in less

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time than the Baseline team. Interestingly, the Workbench team took longer for the easy calibration question but finished the medium-level calibration question almost four times more quickly than the Baseline team. The time advantage for the Workbench team was significant (at least three times faster) for several questions (medium calibration, forecast, and intervention questions, and easy ensemble). The Workbench team finished all questions, while the Baseline team only finished 12 of the 14 questions; they did not have time to work through the hard calibration and intervention questions.

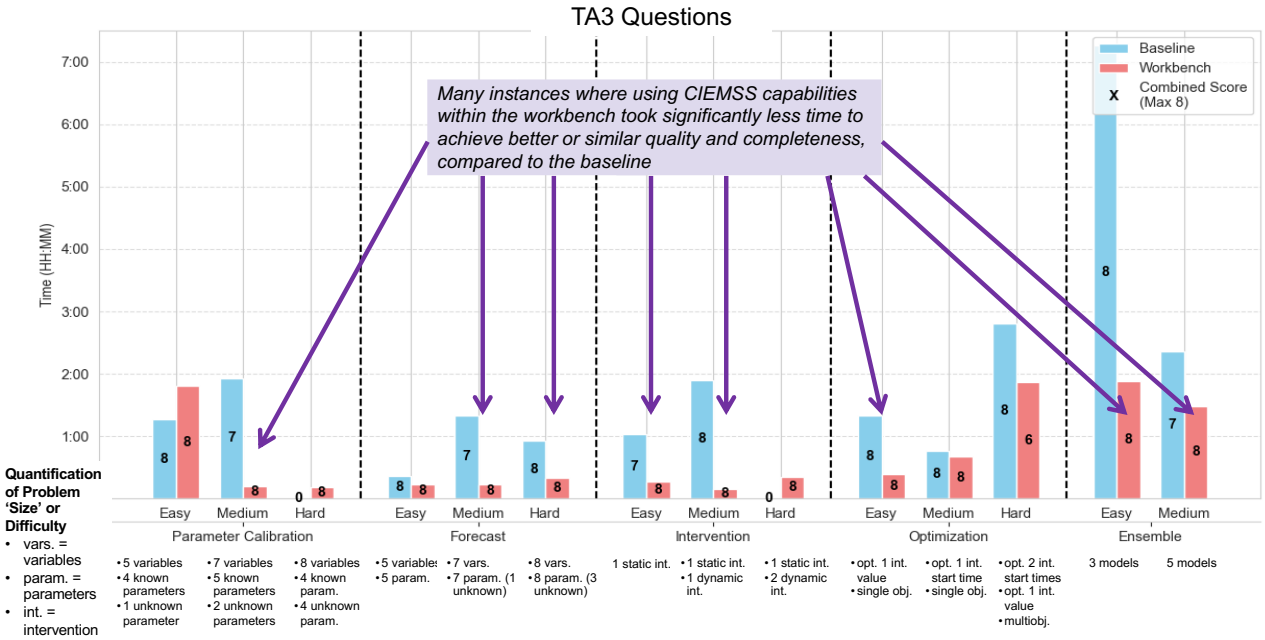


Figure 9. Timing and scores for Epidemiology Independent Questions (Epi IQ), Section 3 (TA3) problems.The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty.

For nearly all the Section 3 questions, the Workbench team achieved quality and completeness scores equal to or higher than the Baseline team (Figure 9), in fact achieving perfect scores for 13 out of the 14 questions. These times and scores indicate that simulation tasks can be done more quickly and efficiently using the automated workflows in the Workbench, at a comparable or higher quality compared to Baseline tools in R or Python.

The only question that the Workbench team didn't achieve a perfect score on was the hard optimization question, which we use to highlight an improvement opportunity. For this question (Q3.4.3), the Workbench team scored a 2/4 on quality, with work product shown in Figure 10. This was a multi-objective optimization problem with multiple constraints, involving two parameters: p_T, the proportion of the population treated with antivirals, and λ, the force of infection. The question asks to find 1) the latest possible day to start the antiviral treatment intervention (represented by a change in p_T), 2) the lowest possible value of p_T, and 3) the latest possible day to start the quarantine intervention (represented by a change in λ). The modelers were tasked with identifying whether there were multiple solution sets that could achieve the

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desired result, and if so, to choose the solution set that contained the lowest possible p_T value. The modelers could then choose days for treatment and quarantine interventions (if there was more than one set that worked) that satisfied the optimal outcome and were asked to justify their choice for those days. The Baseline team successfully implemented the initial unoptimized intervention policy, optimized the intervention policy by creating a grid of possible values of p_T, days for starting treatment intervention (p_T), and days for starting quarantine intervention (λ); from the multiple solution sets, the team found an optimal value of $p_T \approx 0.311$ that is reasonably close to the true solution. The Baseline team then only had one set of options for the two days of interventions in the optimization problem to consider. Their output plot is shown in Figure 10A.

The Workbench team successfully implemented the initial unoptimized intervention policy. However, they struggled to run the Terarium workbench operator for optimizing the intervention policy. The operators available in the workbench did not allow the team to mix the objectives ("low"/minimize value and "high"/maximize day). The Workbench team was eventually able to overcome these issues with the help of performers, but then were unable to see if there were multiple solution sets that met the constraints. Therefore, their plot in Figure 10B shows the output plot of the single answer they obtained. The answer they obtained was not reproducible and actually did not satisfy the constraints. So, even though the Workbench team was able to obtain *an* answer about an hour faster than the Baseline team despite their struggles, they did not arrive at a correct solution and were unable to identify multiple solution sets. In addition, the optimize operators in the workbench struggled with all of the optimization questions due to basin-hopping and unpredictable behavior.

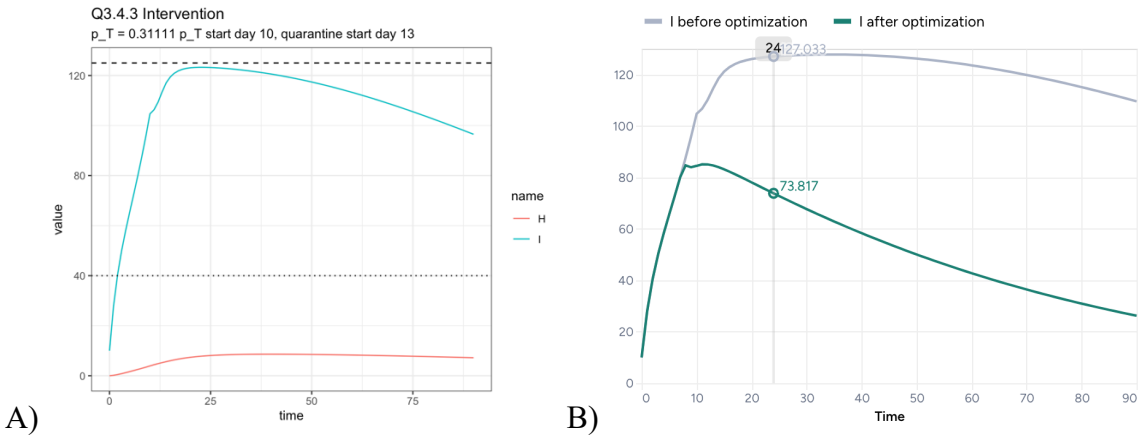


Figure 10. Output for hard optimization problem, Q3.4.3.A) Baseline solution with H and I variable outputs after optimized intervention is applied. B) Workbench solution showing I variable trajectory before and after optimization (but their optimized parameter values were not correct).

4.1.1.1.4 *Section 4 (TA4)*

Section 4 contained 3 sets of questions (4.1 to 4.3) exercising TA4 functionality. The full breakdown of timing and combined quality and completeness scores can be found in Figure 11. For almost all questions in this section, both teams received perfect scores on the questions they

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completed. The only exception was for the easy scenario template question, where the Workbench team received a 3/4 on quality due to an incorrect parameter.

The Baseline team finished 6 out of 9 questions, while the Workbench team finished all 9 questions in equivalent or less time. The time advantage for the Workbench team was significant (twice as fast) for the easy model comparison/summarization question, due to the underlying capabilities of the LLMs utilized by ASKEM tools. The Baseline team did not find the model summarization and comparison questions to be interesting and felt reading and comparing papers was tedious and took too long given the limited time for the evaluation, so they did not focus on the medium-level or hard questions for this set.

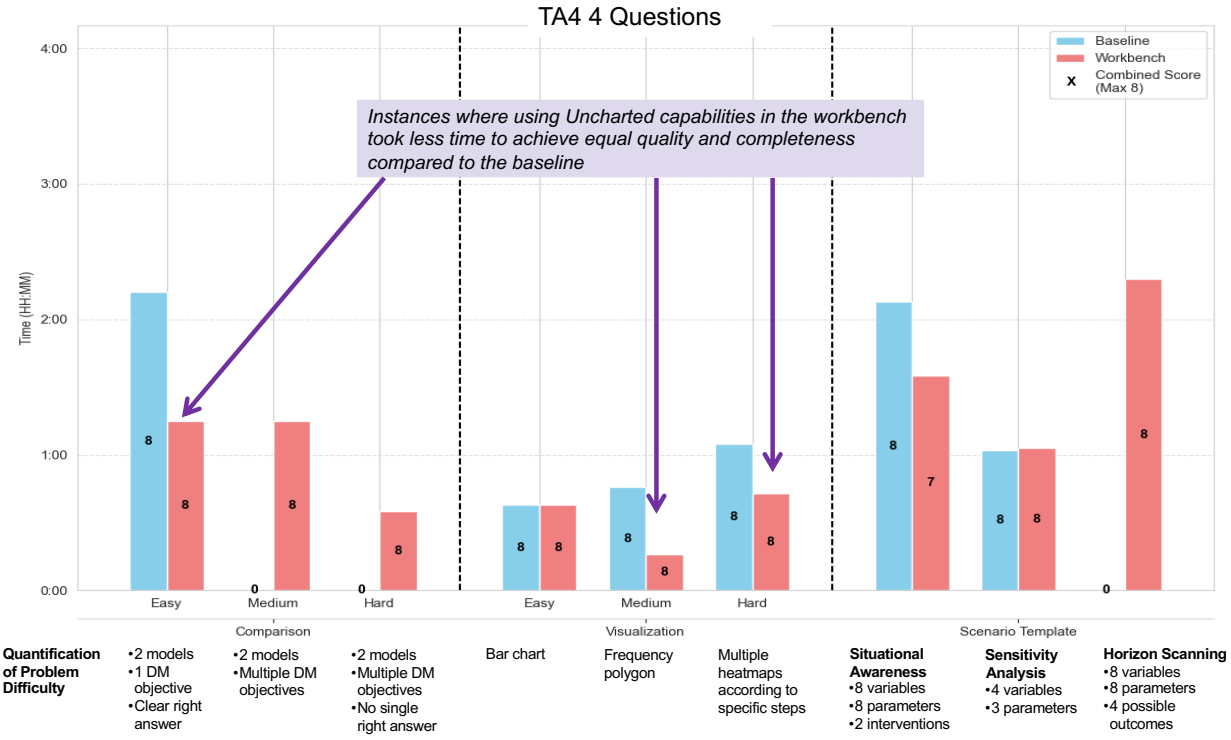


Figure 11. Timing and scores for Epidemiology Independent Questions (Epi IQ), Section 4 (TA4) problems.The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty.

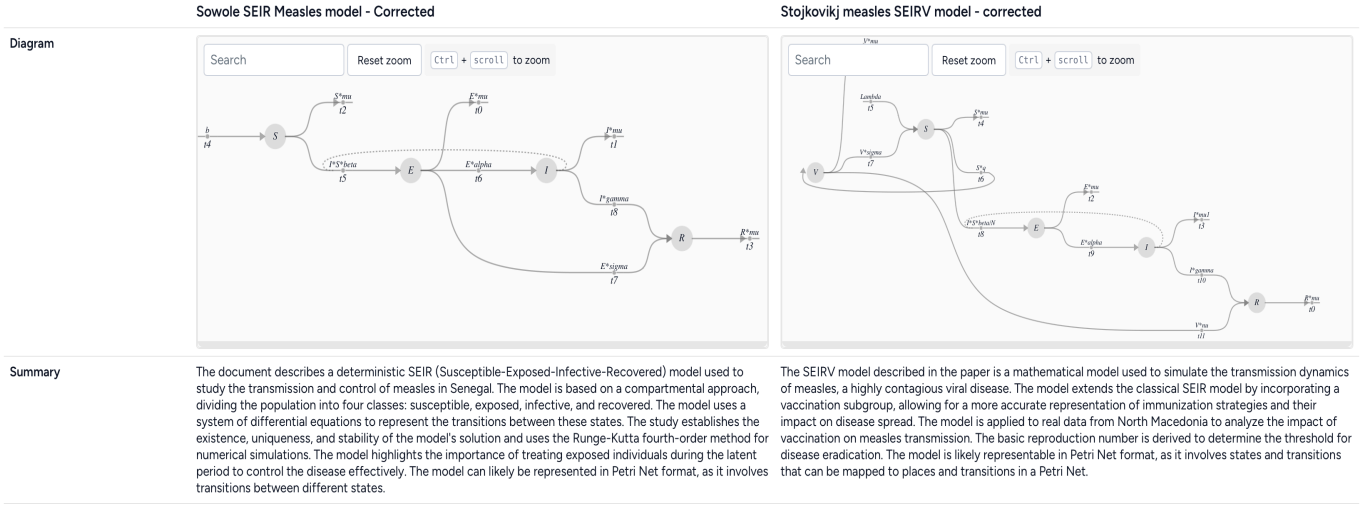
The easy model summarization and comparison problem (Q4.1.1), asked modelers to compare two measles models, for similarities, differences, gaps, and limitations (see Figure 12 for partial example outputs). Modelers were asked to identify which model was a better fit with respect to a decision-maker objective (understanding how vaccination affects the model outcomes and how discontinuation of an existing vaccination program might affect the population). The Workbench team finished this question about twice as quickly as the Baseline team, and both teams achieved perfect scores, identifying the same gaps, limitations, and which model was a better selection. The Baseline team had slightly more nuanced answers in places (e.g., vaccination mechanism should be remedied to allow for different transmission rates between vaccinated and

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unvaccinated individuals), but the overall quality of the solutions were very similar. The Baseline team manually read both papers and discussed the answers together when building the table, while the summarization in Terarium was done automatically by the Compare Models operator. The performance of Terarium functionality was thus not compared to commercial-off-the-shelf Generative AI tools, such as OpenAI's ChatGPT or Google's Gemini models, so we are unable to make conclusions on whether other baseline tools might have yielded better time results. This is due to the Baseline team being unfamiliar with and unaccustomed to using such Generative AI tools in their daily workflows (at the time of the evaluation); if done again, future evaluation setups would likely incorporate state-of-the-art commercial or open-source AI tools into the baseline more formally. However, as demonstrated by the time advantage for the Workbench team, ASKEM technology definitely does provide a speed increase over humans manually reading papers and creating summarization tables, with no significant differences in the quality and completeness of the outputs.

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A)



B).

Model	Gap analysis relative to objective	Proposed remedies for gap	Rank fit for purpose with reasoning
A	Doesn't include vaccine compartment, so impacts of this measure aren't observable	Include vaccination compartment; stratification by age would also be useful based on what the papers say about complications related to age	2: difficult to measure objective without a vaccination compartment
B	Includes vaccine compartment; however, this model doesn't consider whether someone could be vaccinated and still able to transmit the virus even if they're not infectious	Include vaccine stratification such that people who are vaccinated but potentially still carrying the virus transmit the virus at a different rate than unvaccinated infected individuals	1: important to include vaccination compartment

Figure 12. Workbench (A) and Baseline (B) solutions to easy model summarization/comparison question, Q4.1.1.

4.1.1.2 Epidemiology Independent Questions Results Summary

Questions exercising TA3 and TA4 functionality were in general completed more quickly using ASKEM tools, regardless of question type or difficulty (see Figure 9 and see Figure 11). As question difficulty increased for each topic area, the time to finish the questions in the ASKEM workbench did not increase significantly or have any clear correlation with increasing complexity.

Table 9 shows the overall "critical path" completion times for both teams by section. To compute the critical path times, we excluded any time that was spent which did not directly contribute to addressing the problem, and any time spent learning the tools but not actually answering the

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question. For the Workbench team, we excluded 1.5 hours from Q1.1.3 since the performers asked them to revisit that problem at a later date after the Workbench team had begun the problem. Similarly, the Workbench team spent 20 minutes on Q2.2.1 before deciding to start over with a different base model, so that time was also excluded. The Baseline team members spent 35 minutes each (70 person-minutes total) on Q3.5.1 searching for R packages.

Although Baseline total times are less than Workbench total times for Sections 1 and 4 (see Table 9), fewer questions were completed by the Baseline team in both sections (Section 1: 5 out of 6 for Baseline compared to 6 out of 6 for Workbench; Section 4: 6 out of 9 for Baseline compared to 9 out of 9 for Workbench). Figure 5 and Figure 11 indicate that the smaller amount of time for the Baseline teams in Sections 1 and 4 is partially due to unfinished questions.

Table 9. Epidemiology Independent Questions total times by section. All times are 'critical path' times except for the Additional Time category.

Part	Baseline Time In Person-Hours (Critical Path, in hh:mm format)	Workbench Time In Person-Hours (Critical Path, in hh:mm format)
Section 1 (TA 1)	8:24	10:26
Section 2 (TA 2)	15:03	10:31
Section 3 (TA 3)	23:18	10:02
Section 4 (TA 4)	7:51	9:48
Additional Time (Planning, Slides, Noncritical Path, Etc.)	0:14	1:11
Total time spent	**54:50**	**41:58**

The Baseline modelers achieved higher average quality scores for Sections 1, 2, and 4 (see Table 10 for score breakdown). Note that the average scores were only taken across finished questions. In Section 3, the Workbench team scored slightly higher in average quality and completeness scores, and completed all 14 questions (compared to 12 out of 14 questions for the Baseline team). For all sections, the Workbench team had equivalent or better completeness scores, and finished as many or more questions.

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Table 10. Epidemiology Independent Questions overall quality and completion scores

Part	Baseline Quality Score Avg	Baseline Completeness Score Avg	Baseline Questions Finished	Workbench Quality Score Avg	Workbench Completeness Score Avg	Workbench Questions Finished
Section 1	4	3.4	5/6	3.67	4	6/6
Section 2	3.86	3.71	7/8	3.71	3.86	7/8
Section 3	3.75	3.92	12/14	3.86	4	14/14
Section 4	4	4	6/9	3.89	4	9/9
Average Scores Across All Completed Questions	**3.87**	**3.80**	30/37 Questions Finished	**3.81**	**3.97**	36/37 Questions Finished

4.1.2 Epidemiology Decision-maker Scenario (Epi DM)

While the Epidemiology Independent Questions were intended to get very direct comparisons of time and quality for particular aspects of the epidemiological modeling process, the Epidemiology Decision-maker Scenario was intended to support the decision-maker metrics on a more realistic and slightly more open-ended multi-part scenario. Because the workbench and traditional toolsets might support or encourage different problem-solving approaches, we were less prescriptive in instructing the modelers on how to approach the problems. As it turned out, our two modeling teams took very different approaches, making it less meaningful to compare on a question-by-question basis. Here we present some high-level timing information along with descriptions of the two different approaches, and then we present the outcomes of the decision-maker panel.

4.1.2.1 The Scenario

See the Epidemiology Decision-maker Scenario (Epi DM) section under Evaluation Design, Execution, and Post-Processing, for structure and summary of the scenario.

4.1.2.2 Baseline Approach

For Part 1 of the scenario, the Baseline team noted that the workflow of replicating models exactly as they see in papers was not part of their normal modeling workflow and opted not to attempt any unit tests. They read the papers, wrote out the models for each, and listed the problems they observed in those models. When they reached Question 4 (Gap Analysis), they noticed the option to “Extend Single Virus SIR-Type model” and decided that this was the route they would take.

For Part 2, The team created a full SEIRHD model for one virus and then extended it to include a second virus in a “matrix” format. A simplified version is shown in Figure 13. The benefit of this model is that it eliminates nonphysical simplifications (for example, becoming temporarily immune to one disease while exposed to another, or losing one’s vaccination status after

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recovery, aspects that were present in some other models from the literature). However, without the nonphysical simplifications, the model had 36 compartments and 16 strata, for a total of 576 states, and multiple transitions between most of them. This is extraordinarily complex, and one decision maker commented that the model was almost closer to an agent-based model in level of complexity, compared to more standard types of compartmental models. Running this model required time on a high-performance computing (HPC) cluster.

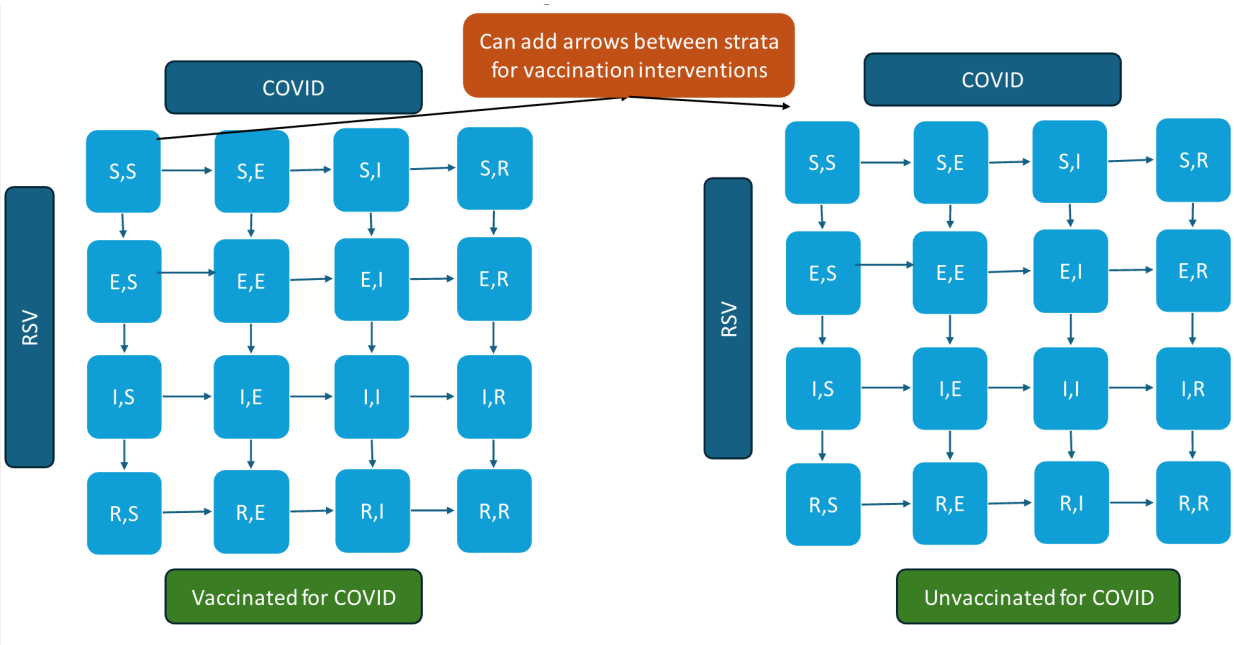


Figure 13. Simplified model from Baseline team only showing the SEIR compartments. The full model had H and D compartments as well. Each compartment has a state for both RSV and COVID-19.

One additional challenge with a model this complicated was the number of parameters required. The Baseline team did an admirable job finding relevant parameters for their model, but they took over twice as long as the Workbench modelers on this step.

Despite model complexity, the simulation results for completed questions in Part 3 were largely reasonable, as shown in Figure 14. The peaks for the number of hospitalizations were too high, but the position of the peaks for COVID and RSV were sensible. The Baseline team was unable to complete all of the intervention simulations (in particular, the RSV vaccination for children intervention), due to the amount of time it took to run the simulations on the HPC cluster.

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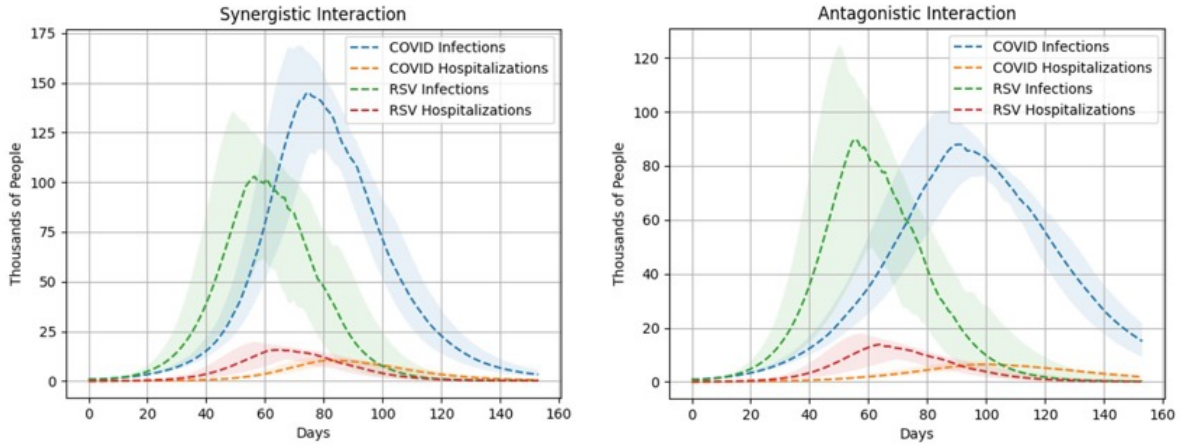


Figure 14. Baseline team's simulation results for synergistic (left) and antagonistic (right) models.

4.1.2.3 Workbench Approach

The Workbench team chose to extend Model B, one of the input models to consider as the starting point for the scenario. In doing so they inherited some awkward assumptions, particularly about modeling vaccination. Perhaps in part because of this modeling choice, many of their projections were not plausible and left decision makers at the panel skeptical.

In Part 1, the team used the Terarium workbench to ingest the three models as instructed in the scenario. This ended up taking significantly longer than expected due to unforeseen challenges with the model ingestion functionality, particularly for their selected model, Model B. They were able to successfully ingest and reproduce the unit tests for Models A and C, but never quite managed to reproduce Model B's unit test. As mentioned in the previous section, the Baseline team chose to skip most of this part of the exercise, leaving them more time to work on their final model.

Because the Workbench team spent so long ingesting the models, they did not spend much time with Terarium's model comparison features, since they felt they got to know all three models very well through the ingestion process, and could already describe their strengths, weaknesses, and assumptions.

In Part 2, the team modified Model B to better address the needs of the scenario, and to undo some of the assumptions the original model included that they felt were not appropriate. The original model only supported hospitalization for COVID-19, and not for influenza or (most surprisingly) co-infection. The team added hospitalization and the distinction between severe and mild illness for all three infection types. They added the ability to become co-infected at multiple points in the disease progression process, and added the ability for people to transition from being vaccinated against one infection to being infected with either of the two viruses. Their updated model assumed that co-infection did not happen simultaneously. They added a multiplier on the force of infection parameter to permit them to model synergistic and antagonistic models of co-infection. The Workbench team's final model structure is shown in Figure 15.

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Unfortunately, the team did not modify the way that Model B handled vaccination. This model has a separate *Vaccinated* compartment that individuals can move to from *Susceptible*. However, once a *Susceptible* individual moves to *Exposed*, the model "forgets" the person's vaccinated status. This type of model of vaccination is most useful in a single-virus model where *Recovered* is an absorbing state, and with a vaccine whose primary value is reducing the probability of infection. In that case, once the person is infected, their vaccine status becomes largely irrelevant. But with a multi-virus model, forgetting all of a person's vaccination information when they get any of the diseases is particularly unfortunate. It means, for example, that a person vaccinated against both viruses and then gets infected with one of them effectively loses the value of their vaccine against the other virus, making them more prone to co-infection than they ought to be. It also doesn't allow modelers to make hospitalization or death from the virus less probable in vaccinated individuals. This way of representing vaccination also made it impossible to count how many people were vaccinated at any given time, making it hard to implement the scenario's vaccination policy interventions in Part 3.

The Workbench team also ran into significant technology challenges with stratifying the model, and ultimately had to use an unstratified model for their projections. (Among other challenges, once the team did manage to create a stratification in the workbench, they were unable to run simulations with it, which we believe was due to the complexity being too great for the computing resources provided by Terarium. It should be noted that they did not have access to HPC computing which the Baseline team used to complete their simulations.) Their final model structure is illustrated in Figure 15.

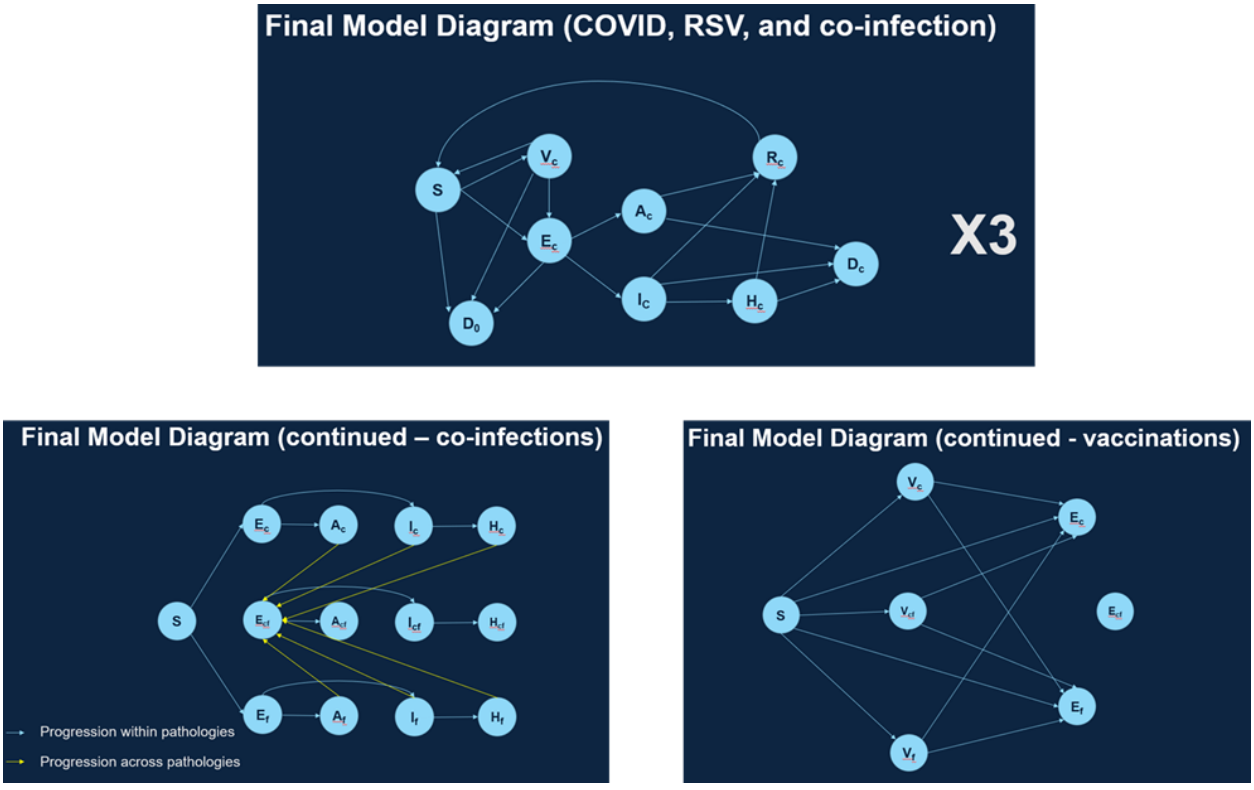


Figure 15. Workbench team's updated model from Part 2 of the scenario.

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In Part 3, the Workbench modelers ran into difficulty implementing the requested vaccine implementation scenarios (Question 14) due to the choice to stick with Model B's handling of vaccination. Like the Baseline team, the Workbench team completed 2 out of 3 of the requested intervention scenarios. While they were able to show more co-infection and later peaks with synergistic versus antagonistic infection, their infection trajectories did not peak as expected during simulations, leading to significant skepticism from decision makers.

4.1.2.4 Decision-maker Panel Outcomes

After four days of work on the Epi DM Scenario, on the final day of the evaluation event, the decision-maker panel (consisting of 5 panelists) heard presentations from the Baseline team in the morning session and the Workbench team in the afternoon session. Panelists were asked to judge the two teams independently rather than comparatively. They were asked to score each of the three parts on a scale from 1 (very low) to 7 (very high) indicating their confidence in the answers and outputs for that part.

For each part, the panelists were given a high-level confidence question and sub questions to support their assessment, as described below.

Part 1: Choosing the Right Starting Model

How confident are you that the modeling team selected a model and associated information appropriate for the decision-making questions under consideration?

Metric Explanation: Decision makers will evaluate whether results include an appropriate model, with relevant parameters, starting values, etc.

The decision-maker confidence score should be supported by the answers to the following questions:

- Did modelers clearly communicate key differences between the chosen model and other candidate models? What kinds of information were provided to help you understand the differences? What kinds of information would you have wanted?
- Are results traceable to sources and did modelers assess source quality?
- Is the model chosen for the scenario appropriate/fit-for-purpose for the given problem?
- Were assumptions and data associated with models clearly communicated?

Part 2: Developing/Updating the Model

How confident are you that the modeling team appropriately modified the model and its parameter space to sufficiently explore the scenario/problem?

Metric Explanation: The scenario involves updating or modifying a model. Decision makers will evaluate whether this was done in a sensible way and whether the final model can support all the questions asked in the scenario.

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The decision-maker confidence score should be supported by the answers to the following questions:

- Did modelers clearly explain the changes being made and key differences between the original and updated models? Did the modifications/extensions the modelers made make sense and were they reasonable to you?
- Are you confident that the starting model was updated in ways that make sense? Is the final model structurally sound?
- As the model was updated, was the parameter space being explored reasonable and broad enough/complete enough to support the questions required by the scenario?

Part 3: Addressing Decision-maker Priorities

With what degree of confidence do you feel that you understand the modelers' results and the potential tradeoffs between the interventions they implemented in their model(s)?

Metric Explanation: Based on the information presented to you by the modelers, determine your confidence in being able to assess effectiveness of all interventions considered in the scenario and understand how uncertainty factors into model results and their implications.

The decision-maker confidence score should be supported by the answers to the following questions:

- Did simulation results make sense? If results did not match your intuition or expectations, did modelers explain the reason(s)? Did their explanations make sense?
- Do you understand the effects of interventions on trajectories? Was the effectiveness of interventions communicated?
- Is it clear how to interpret uncertainty in the results? Do you understand the key drivers of uncertainty in the results?

Figure 16 shows the score distributions and mean scores for each of the 3 scenario parts. For part 1, the mean confidence score was higher for the Workbench team but for parts 2-3, the mean confidence was higher for the Baseline team. As with the climate use case, the order of the presentations (Baseline first) could have influenced scores – ideally, we would want to have two different panels see the presentations in different orders, although this was not logistically feasible. The scores were quite variable between panelists, with surprising outliers for Part 3.

Table 11 shows a summary of panelists comments for each part of the two presentations.

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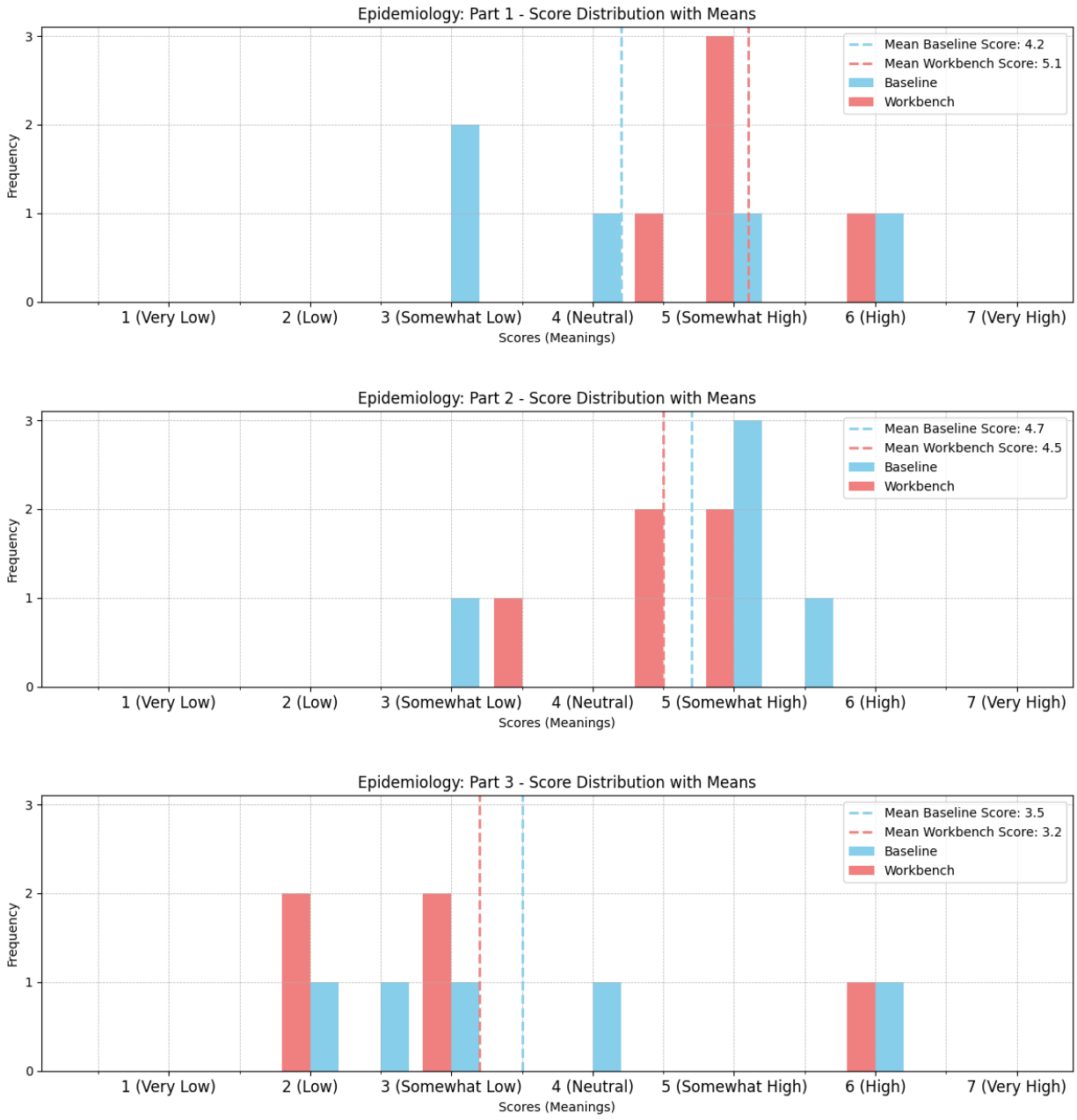


Figure 16. Epidemiology Decision-maker Panel scores.

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Table 11. Overview of Epidemiology Decision-maker Panel comments

Part	Baseline Avg Score	Summary of Panelist Comments	Work-bench Avg Score	Summary of Panelist Comments
Part 1: Choosing the Right Starting Model	4.2	• Did a good job highlighting some differences • Their choice is flexible (good scaffold to build on) though data limitation can affect utility • Assumptions were (partially) explained • No assessment of source quality • Additional info on parameterization desired	5.1	• Clear communication of differences and assumptions • Model's quality at high level (peer review or not, type of journal) was assessed • Model is relatively fit for purpose given time • Should have considered building fresh
Part 2: Developing/ Updating the Model	4.7	• Model changes clearly explained and made sense • Broad enough, wide enough parameter space • Found vaxed people more likely to be hospitalized? • Concern about overcomplexity, given the uncertainty of many parameters; overfitting? • Approach requires fine-grained parameterization – research burden and opportunity for error • Missing structure for social interaction	4.5	• Model changes and assumptions clearly communicated • Vax handling a weakness; vax status not remembered feels really consequential • The model got more complex/broad but not comprehensive enough to answer the main questions • Age stratification is necessary to answer the questions
Part 3: Addressing Decision-maker Priorities	3.5	• Data is uncertain or unclear but acknowledged • Model outputs didn't make sense (magnitude) • Little info given about how calibration was done • Uncertainty was well explained, both statistical uncertainty and uncertainty with the model || Areas of uncertainty not presented || With so many compartments, how does uncertainty propagate? • Intervention effects minimal and hard to understand • Intervention plots didn't address the key question being asked	3.2	• Confident in direction but not magnitude • Results don't match my intuition and it's hard to understand why • No calibration against real data, lack of age stratification limit confidence in outputs • Some discussion of uncertainty but I'm still unclear on what is driving the findings and how to communicate that; visualizations displayed uncertainty at cost of interpretability • Use of parameters from papers as opposed to surveillance data limited certainty of results

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4.1.2.5 Epidemiology Decision-maker Scenario Summary Times and Scores

The Epidemiology DM experiment was not used for program timing metrics. However, we make some high level notes on time spent by the two teams. The Baseline team chose to work during the evenings and ended up spending roughly 18 person-hours more overall on the scenario than the Workbench team did. However, we do note that the Workbench team's person-hours documented in Table 12 below do not include time spent by the performers helping them wrangle the technology. Most notably, the Uncharted team spent time the first night completing the ingestion of Model B since they had determined that Terarium's back-end LLM was running into an unexpected challenge with the model, that they were able to get around using a direct interface with the LLM (outside of the workbench).

As noted above, the two teams took very different approaches to the problem and focused their attention on different aspects of the scenario. Thus, direct comparison of time taken on any portions of the scenario are not particularly meaningful. The summary in Table 12 is meant only to give a high-level view of where time was spent by each team. Similarly, the T&E team also did not score the results produced by each team, as a comparison of scores would also not be meaningful, given the very different approaches taken by each team.

Table 12. Overview of Epidemiology Decision-maker Scenario timing and approaches

Part	Baseline Time Person-Hours	Baseline Summary	Workbench Time Person-Hours*	Workbench Summary
Part 1: Choosing the Right Starting Model	20.3	• Did not attempt execution or unit tests • Very thorough on parameters	20	• 11+ hours spent on extraction; reproduced 2 of 3 unit tests • Did not do structural model comparison
Part 2: Developing/ Updating the Model	32.9	• Built model from scratch • Possibly too complex	25.6	• Extended Model B, inheriting some awkward assumptions • Did not complete age stratification (could not execute)
Part 3: Addressing Decision-maker Priorities	7.2	• Height of peaks were off, but overall shape of trajectories still reasonable • Intervention implementations were reasonable	4.8	• Trajectories did not peak as expected during simulations • Vaccination rate used was questionable (possibly due to structure of the model)
Additional Time (Planning, Slides, Etc.)	14.7		6.3	
Total Time	75.1		56.8	** Total times listed here do not include untracked performer help*

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4.1.3 Climate Independent Questions (Climate IQ)

4.1.3.1 Highlights in Approach and Results

There were three sections in the Climate IQ experiment, each corresponding to a different Technical Area (TAs) (TA2, TA3, and TA4).

The Workbench team was required to use tools developed in ASKEM through the Beaker interface, and any additional tools they needed. Beaker is an AI-assistant code notebook that allows the user to both chat with a LLM and also write and update code as they would in a Jupyter notebook. In addition, there are specialized contexts that the user can set which contain extra instructions that the LLM can access. This can include documentation or general instructions.

For Sections 1 and 2 (corresponding to TA2 and TA3 functionality), the Beaker context used was the Decapodes context, which contained a preinstalled version of Decapodes in a Julia notebook along with the Decapodes documentation in the LLM's context window. For Section 3 (corresponding to TA4 functionality) content related to Integrated Assessment Models (IAMs), the Mimi Beaker context was used; this context included the MimiFUND and MimiFAIR models preinstalled and their documentation and code made available in the LLM context window. For the Data Search and Data Transformation questions, the Beaker context used was the general climate data context, which included the Earth System Grid Federation's API documentation.

The Baseline team used state-of-the-art programs like COMSOL and Matlab, along with Julia and Python computational libraries, to answer the Climate IQ questions.

In general, both teams chose a divide-and-conquer strategy, with each team member working independently from their partner. The split of questions differed per team. For the Baseline team, one team member focused solely on the multiphysics and IAM questions using COMSOL and Julia, while their partner solved the rest. For the Workbench team, one team member solved the Decapodes-related multiphysics questions, while the other solved the IAM and climate data questions.

Overall, the Baseline team completed 15 out of 33 questions while the Workbench team completed 19 questions. Table 13 below indicates the questions both teams completed (green cells), the questions only the Workbench team completed (red cells), the questions only the Baseline team completed (blue cells), the questions neither team completed (white cells), and the questions that were not included in the set (gray cells). However, completion does not tell a full story and in the next sections, we present an in-depth look at the teams' performances.

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Table 13: Climate Independent Questions completion distribution

Question Set	Easy	Medium	Hard
1.1 (Model Config & Execution)		Workbench Only	Neither Team
1.2 (Model Editing)		Workbench Only	Neither Team
1.3 (Multiphysics Model Composition)	Baseline Only	Baseline Only	Baseline Only
2.1 (Forecasting)		Workbench Only	Neither Team
2.2 (Sensitivity Analysis)		Neither Team	Neither Team
2.3 (Calibration)	Workbench Only	Workbench Only	Neither Team
3.1 (IAM Understanding)			Baseline Only
3.2 (IAM Forecasting)	Workbench Only	Neither Team	N/A
3.3 (IAM Interventions)	Workbench Only	Workbench Only	N/A
3.4 (Optimizing IAM Interventions)	Workbench Only	Neither Team	N/A
3.5 (Data Search)			Baseline Only
3.6 (Data Transformation)			Neither Team

Figure 17 plots the cumulative critical path times (in person-minutes) for both teams against each other; the Workbench modeling efforts are on the y-axis and the Baseline modeling efforts are on the x-axis. The dotted diagonal line shows where the two teams' cumulative times would be equal. Where the plotted lines fall into the region where $x > y$, the Baseline modelers had cumulatively taken more time to solve the questions than the Workbench modelers. It is clear from this graph that overall the Baseline team spent more time to answer fewer questions.

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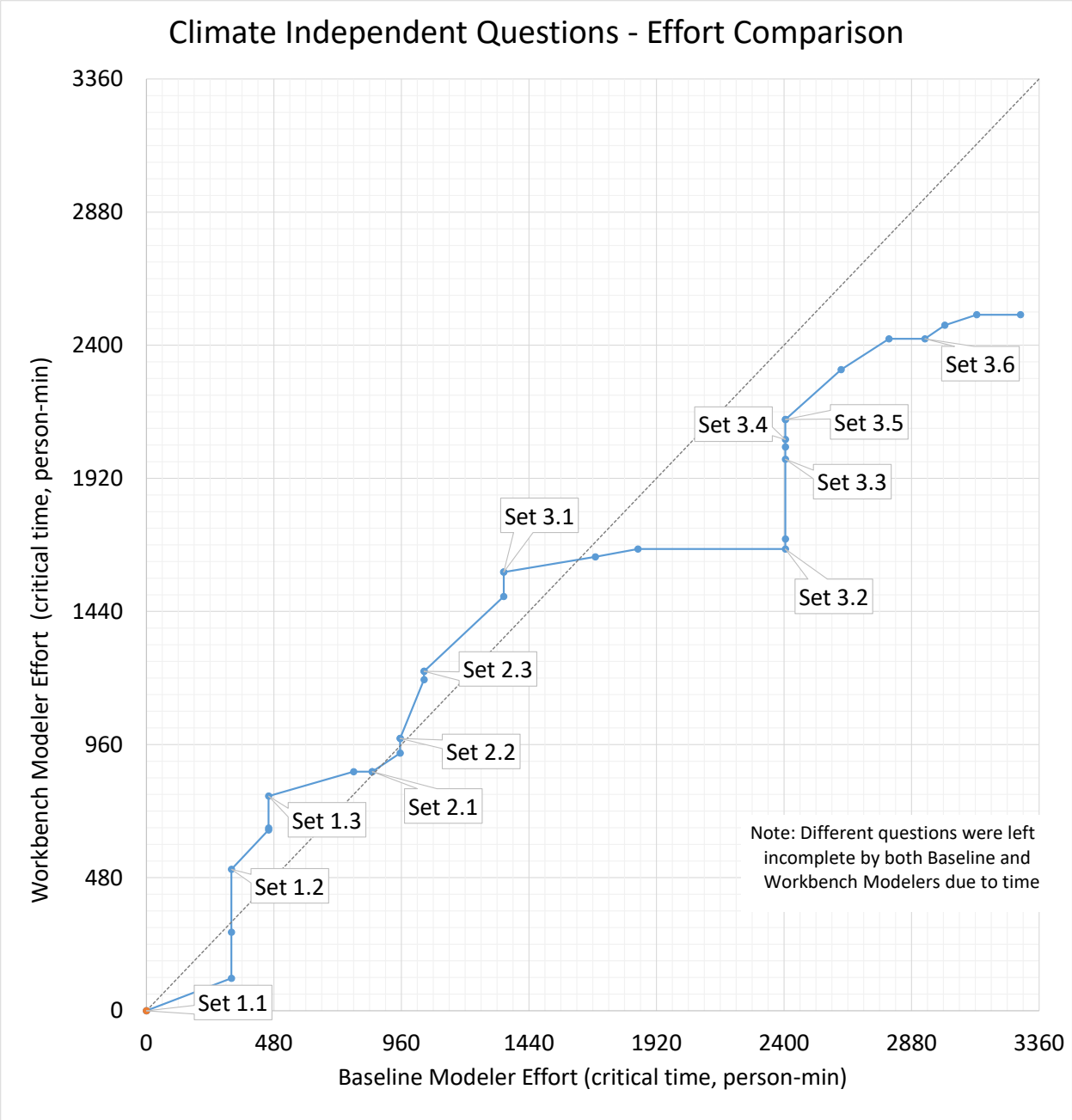


Figure 17. Climate Independent Questions (Climate IQ) cumulative critical path times for the Baseline and Workbench modeling teams. The time spent on each question is represented by a line segment, and the beginning of a question set is annotated with a label. The dotted diagonal line shows where the cumulative times for each team would be equal.

4.1.3.1.1 *Section 1 (TA2)*

Section 1 contained three sets of questions exercising TA2 functionality: Set 1.1, model configuration and execution with physics models, Set 1.2, model editing, and Set 1.3, multiphysics model coupling/composition. The full breakdown of timing and combined quality and completeness scores can be found in Figure 18. Overall, the Baseline team was able to

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complete one more question than the Workbench team, mostly due to challenges with Set 1.3, described in more detail later in this section.

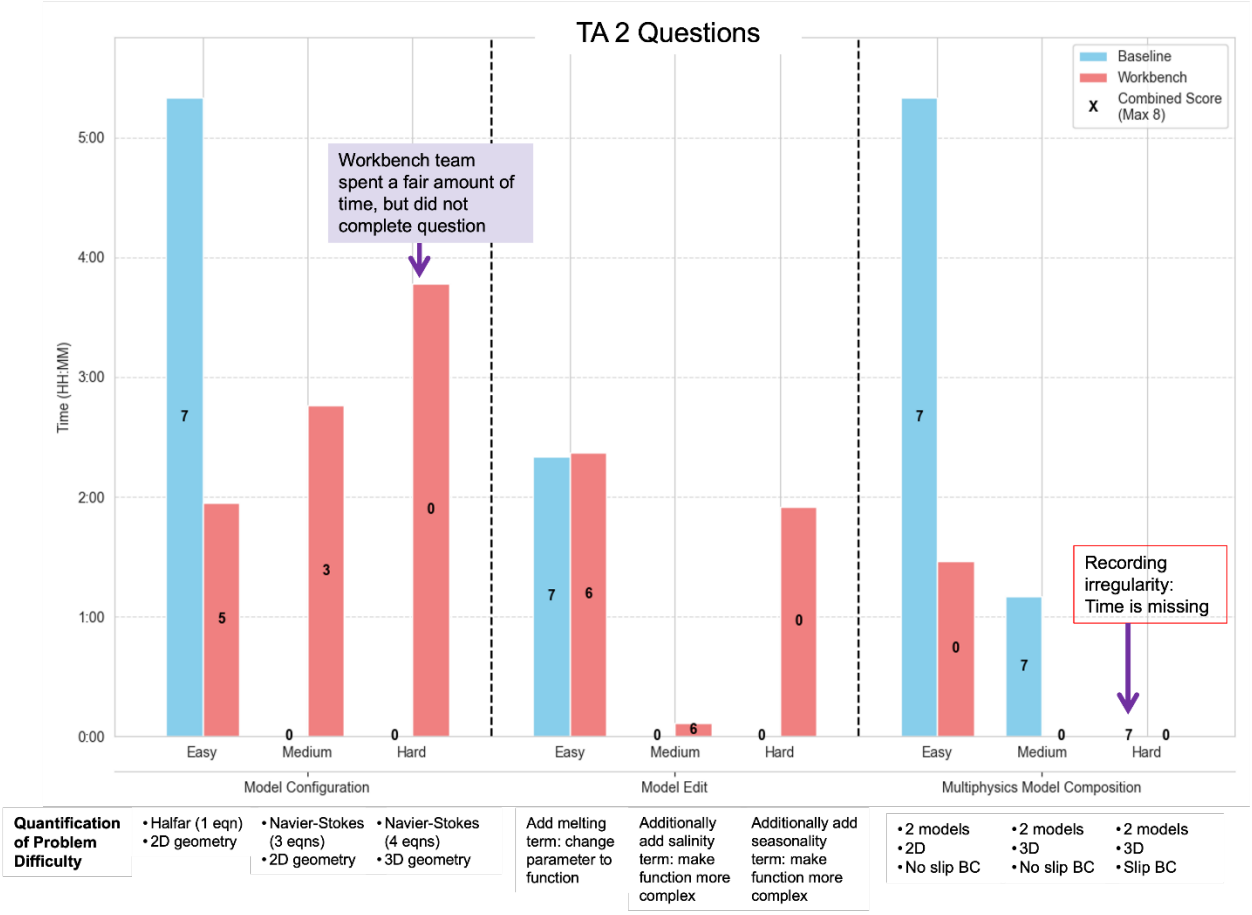


Figure 18. Timing and scores for Climate Independent Questions (Climate IQ), Section 1 (TA2) problems. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty.

The medium-level model configuration problem (Question 1.1.2) was only attempted by the Workbench team. This question asked the modelers to recreate a unit test for 2D turbulence, with the Baseline using the Oceananigans computational modeling package and the workbench using the equivalent Decapodes model. However, the result from Decapodes did not seem to demonstrate turbulent behavior, similar to the 18-month evaluation outcomes for a similar type of problem. Figure 19 shows the expected output while Figure 20 is the actual output from the Workbench team. Possible explanations for the lack of turbulence could be the current Decapodes solvers, evolution of error in the system, a misunderstanding of time steps, or some other user error. In addition, Decapodes has limitations on the types of topology it can support, requiring that some simulations be run on torus or spherical geometries; this was an additional challenge the Workbench team had to work through.

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The Workbench team spent a lot of time on the hard model configuration problem, but simply ran out of time, and did not produce any meaningful artifacts that were evaluable, resulting in a score of 0. Similarly, the Workbench team attempted, but ran out of time to accomplish much with the hard model editing question.

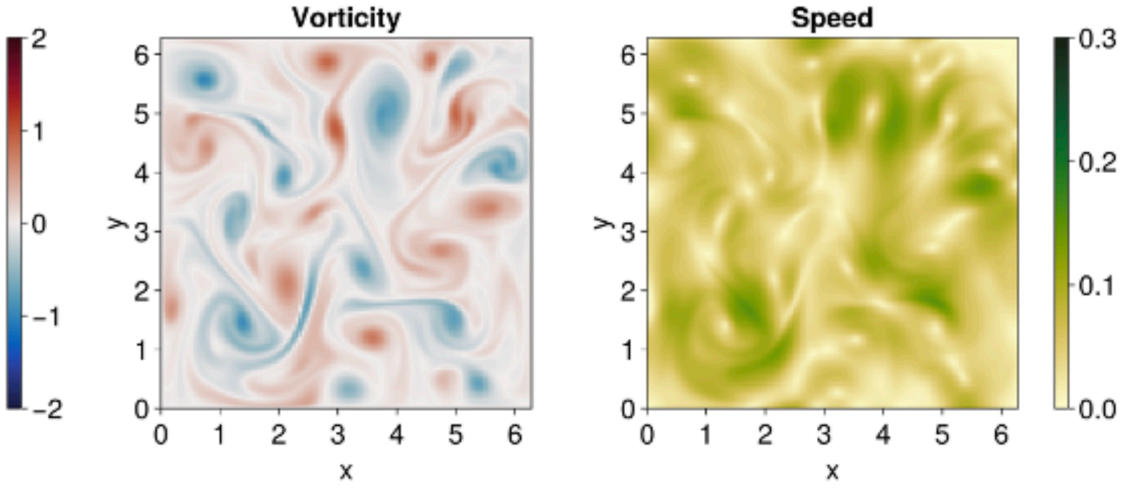


Figure 19. Expected output for Question 1.1.2, on turbulence with Oceananigans. Notice the structure of the vorticity field showing turbulent areas.

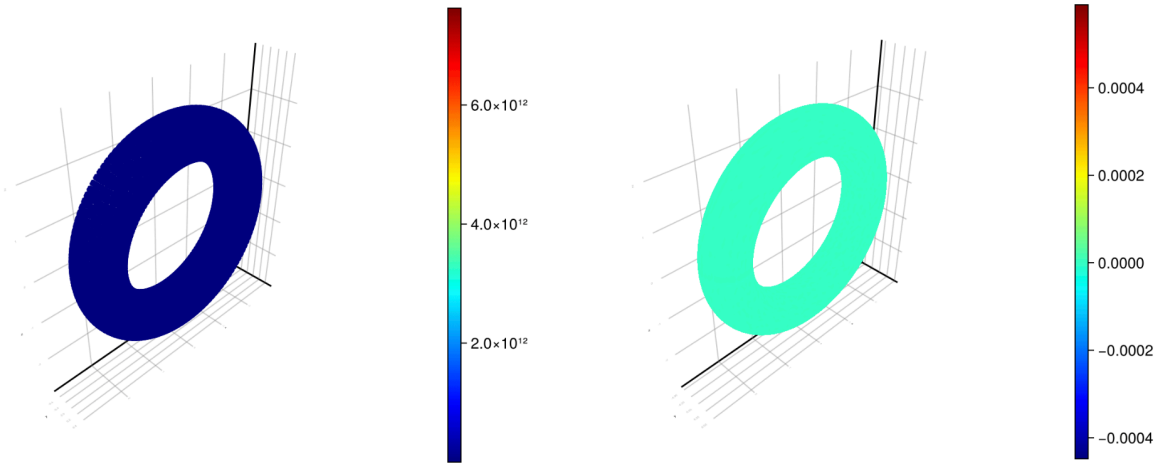


Figure 20. Workbench team's output for Question 1.1.2, on modeling turbulence. There does not seem to be any noticeable fine-grained structure or spatial variation for the vorticity or speed values.

Set 1.3 on multiphysics model composition, was only completed by the Baseline team using COMSOL. These questions required modelers to combine two to three different physics models into a multiphysics model. The Workbench team spent a few hours attempting to make a 3D pipe mesh that could be used with Decapodes (being able to create 2D and 3D meshes of this type was a requirement for all questions in this set). In the end, they were not able to create the

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required geometry to move forward with the medium and hard level questions in this section, but also didn't attempt the easy version of this question, as they were encouraged to move on to other question sets by the T&E team. The time spent on the easy multiphysics modeling question was focused on understanding how to do model composition in Decapodes, but there was no output for any questions in Set 1.3 that was evaluable. Based on this experience, the T&E team recommends that performer multiphysics tools include additional documentation and guidance (either in reference to their own tooling or other existing software libraries) to help new users of their tools to create geometries, a critical step in the modeling and simulation process with physics-based PDE models.

4.1.3.1.2 *Section 2 (TA3)*

Section 2 contained three sets of questions exercising TA3 functionality, all with the Halfar ice model: Set 2.1, forecasting, Set 2.2, sensitivity analysis, and Set 2.3, calibration. The full breakdown of timing and combined quality and completeness scores can be found in Figure 21. The Workbench team was able to complete more questions than the Baseline team.

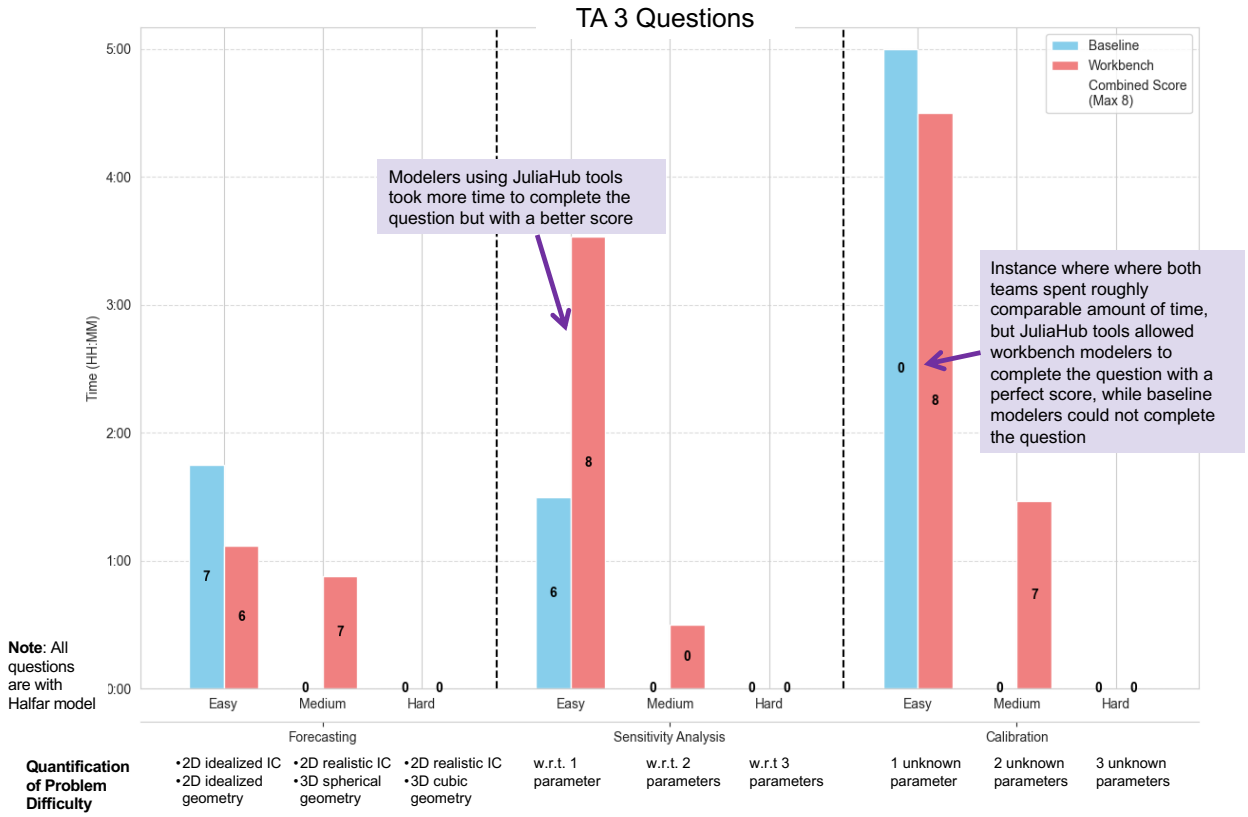


Figure 21. Timing and scores for Climate Independent Questions (Climate IQ), Section 2 (TA3) problems. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty.

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For this section, the Baseline team used pre-written MATLAB models that they needed to edit and simulate. Execution time could take potentially hours to days, depending on the simulation parameters and other model inputs; this is because the resolution of the geometry has a large effect on model computation time, making it difficult to iterate and run sufficiently large/complex models multiple times. In contrast, the Workbench team used pre-written Decapodes model for the base Halfar model, along with Julia tools and libraries, and were able to simulate the models in minutes (pure computation time and not the time taken for doing the model editing).

Another difficulty the Baseline team faced was applying the Halfar model to real data of the Grigoriev Ice Cap. In Decapodes, this is a simple exercise. All that is needed is to edit the code where the mesh is defined, no more difficult than applying a simplified mesh. However, for the Baseline modeling team using MATLAB, this task proved extremely difficult, and the team was unable to complete it. In general, making major changes to the geometry on which a PDE model is applied (i.e., transferring physics between different geometries), is a much easier task in Decapodes, compared to MATLAB or many similar general purpose programming tools/environments.

The integration of Decapodes with other Julia packages (including *Turing.jl*) provided huge benefits to the Workbench team. They were able to complete the easy and medium calibration questions with good quality and completeness by using pre-existing documentation. We anticipate that any future new documentation on using Decapodes with other Julia packages will make modeling and simulation tasks even easier to complete in the future. The Workbench team also utilized Beaker to explain how to use various Julia packages, since neither member of the team was familiar with *Turing.jl* or Julia as a programming language. The Beaker outputs were not completely free of errors but the team noted that it got them 60-70% towards a final solution.

Both teams spent time on questions that did not produce any evaluable output. For the Baseline team, this was the easy calibration question, which they spent a lot of time on but were unable to complete due to the difficulty of doing the problem in MATLAB. The Workbench team spent time on the medium-level sensitivity analysis question, but ran out of time to generate any meaningful output.

4.1.3.1.3 *Section 3 (TA4)*

Section 3, with six sets of questions, exercised TA4 functionality: Sets 3.1-3.4, on various aspects of Integrated Assessment Models (IAMs), and Sets 3.5-3.6, on data search and transformation. The full breakdown of timing and combined quality and completeness scores can be found in Figure 22. For the four sets of IAM questions, the Baseline team only attempted the first set on IAM model understanding (Set 3.1), and the Workbench team was able to attempt parts of all four sets of IAM questions. For the two IAM questions both teams attempted (easy and medium-level IAM understanding questions), the Workbench team was able to complete the questions much faster and with better quality and completeness.

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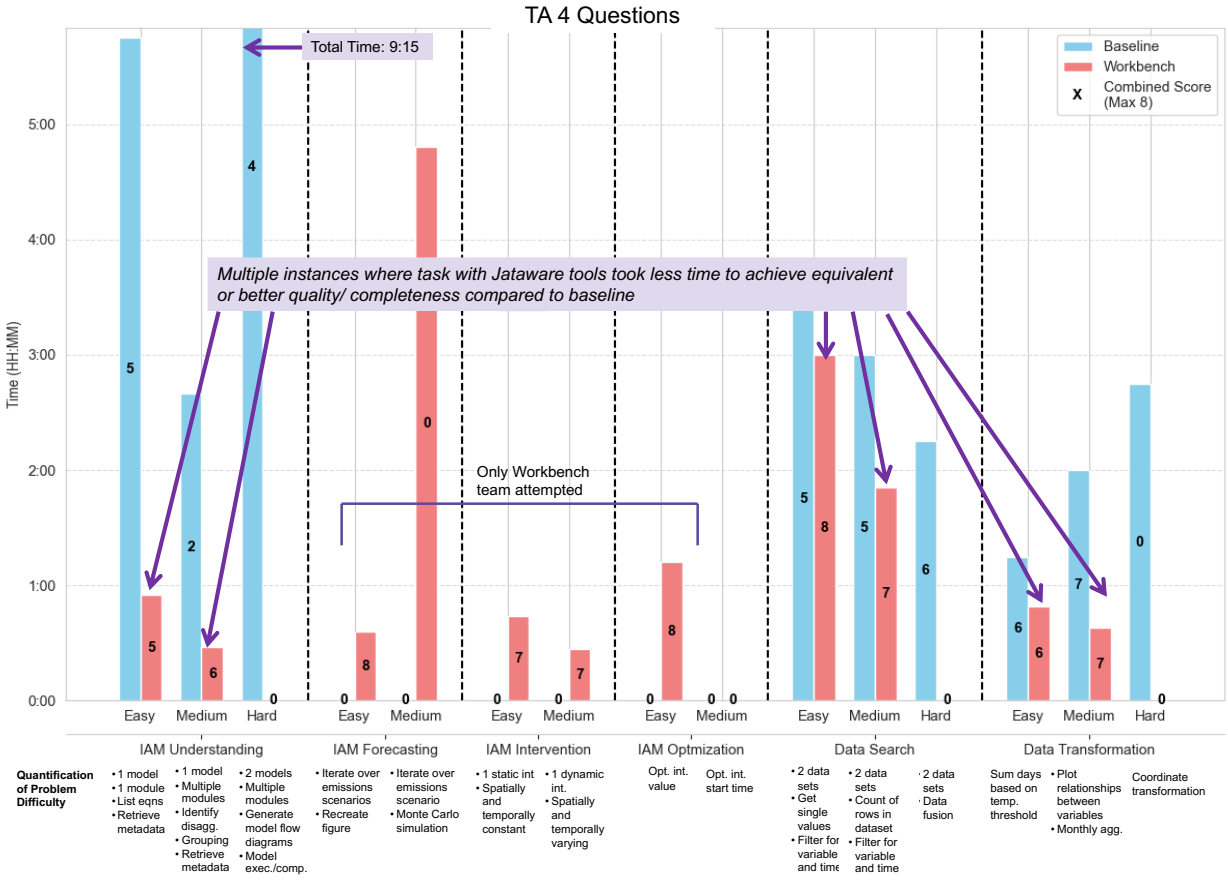


Figure 22. Timing and scores for Climate Independent Questions (Climate IQ), Section 3 (TA 4) problems.The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Below the x-axis is a high-level quantification of problem difficulty.

The Workbench modelers found that asking the Beaker LLM agent about the documentation for MimiFUND and MimiFAIR was extremely useful. For the same IAM questions, the Baseline team struggled to find the relevant information from the documentation.

Both modeler teams were equally inexperienced with Julia. The Workbench team was able to ask Beaker for help writing Julia code, which allowed them to answer the IAM forecasting, intervention, and optimization questions (Sets 3.2 to 3.4), but without this assistance, the Baseline team did not attempt these questions, most likely due to time constraints after the amount of time they spent on Set 3.1.

The data search questions (Set 3.5) required finding and downloading data from the Earth System Grid Federation (ESGF). The Baseline team manually went to the ESGF website and found the correct datasets to download. The intention was that the ESGF API could be accessed in Beaker but due to technical issues, this was not working at the time of the evaluation. Therefore, the Workbench team was able to use Beaker to get the name of the correct datasets, but had to go to the ESGF website and manually download the data as well. In spite of this

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challenge, the Workbench team was able to more quickly attain the data compared to the Baseline team, while also having a higher quality and completeness scores for the questions they attempted. The Workbench team did note that one of the dataset names Beaker returned seemed to not actually exist in ESGF, hinting at LLM hallucination.

4.1.3.2 Climate Independent Questions Results Summary

In summary, the Baseline team was unable to complete as many questions as the Workbench team. Table 14 shows the "critical path" total time spent by each team per section. To compute the critical path times, we excluded any time that was spent which did not directly contribute to addressing the problem, and any time spent learning the tools but not actually answering the question. If a team went down an exploratory path that did not lead to the final answer, then this time was subtracted from their total time. For example, the Workbench team tried to create a 3D pipe mesh in Blender that did not end up being compatible with Decapodes. Therefore, the time they spent installing, using, and exporting the pipe from Blender is considered non-critical path time and subtracted from the total time. Other instances of non-critical path time include system failure, attempts to debug packages that were not fruitful, and general discussions with the ASKEM performers that were about their tools. The biggest difference in time required between the two teams was in Section 3 where the Workbench team completed more questions (10/15 questions) in 17 hours less time than the time taken by the Baseline team to complete fewer questions (8/15 questions).

Table 14. Climate Independent Questions total times by section. All times are 'critical path' times except for the Additional Time category.

Part	Baseline Time In Person-Hours (Critical Path, hh:mm format)	Workbench Time In Person-Hours (Critical Path, hh:mm format)
Section 1 (TA 2)	14:10	14:22
Section 2 (TA 3)	8:15	12:00
Section 3 (TA 4)	32:25	15:28
Additional Time (Planning, Slides, Noncritical Path, Etc.)	7:10	15:41
Total Time Spent	**62:00**	**57:31**

Table 15 shows the average quality and completeness scores *across all completed questions,* by team and by section. The Workbench team was able to complete more questions in Section 2 than the Baseline, with higher average quality and completeness scores (see Table 15), but took longer (see Table 14). For Section 3, the Workbench team performed with better quality and completeness scores compared to the Baseline team (see Table 15) and was also faster (per Table 14).

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Table 15. Climate Independent Questions overall quality and completion scores

Part	Baseline Quality Score Avg	Baseline Complete-ness Score Avg	Baseline Questions Finished	Workbench Quality Score Avg	Workbench Complete-ness Score Avg	Workbench Questions Finished
Section 1	3.00	4.00	5/9	3.00	2.00	4/9
Section 2	3.00	3.50	2/9	3.20	4.00	5/9
Section 3	2.75	2.44	8/15	3.50	3.40	10/15
Average Scores Across All Completed Questions	**3.00**	**2.97**	15/33 Questions Finished	**3.32**	**3.26**	19/33 Questions Finished

4.1.4 Climate Decision-maker Scenario (Climate DM)

While the Climate Independent Questions were intended to get very direct comparisons of time and quality for particular aspects of climate and multiphysics modeling, the Climate Decision-maker Scenario was intended to support the decision-maker metrics with a more realistic and slightly more open-ended multi-part scenario.

4.1.4.1 The Scenario

See the Climate Decision-maker Scenario (Climate DM) section under Evaluation Design, Execution, and Post-Processing, for structure and summary of the scenario. Unlike the Epi DM experiment, the Climate DM scenario was sufficiently constrained that both the Workbench and Baseline modeling teams took generally similar approaches using different tool sets. Therefore, we discuss highlights organized by scenario part and task, comparing the team approaches and outputs along the way.

4.1.4.2 Part 1: Climate Data Search, Processing, and Plotting Data

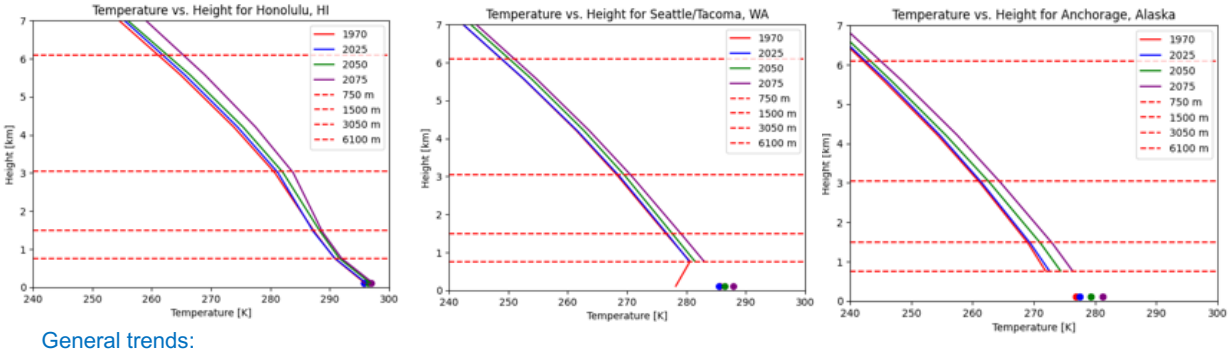
For the climate data search questions, the Baseline team found data by manually searching through ESGF; one modeler noted that they would have liked to use specific humidity but couldn't find it. The Workbench team found data using Beaker, and was able to find specific humidity. Both teams ended up using the same general model, and assumed the same SSP scenario (5.85), as shown in Table 16.

Table 16. Baseline and Workbench approaches for climate data search

Dataset Details	Baseline	Workbench
Search Strategy	Manual search through ESGF portal	Beaker
Model Selected	**GFDL-ESM4** Variant label r1i1f1p1	**GFDL-ESM4**
Scenario	SSP-5.85 (assumes most extreme changes in the future)	
Spatial resolution	100 km	

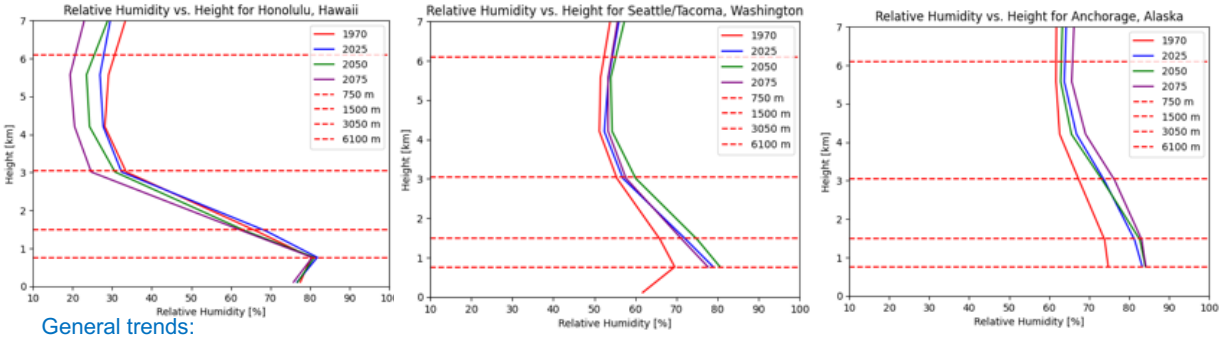
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In terms of processing the data and identifying trends, the Baseline team painstakingly processed data with a number of scripts. Data was in pressure levels (heights at specific pressure) and had to be interpolated to the altitudes indicated in the scenario description. To calculate height, the team used an equation from the National Weather Service (NWS) instead of the Barometric formula, because sea level temperature information was missing from the Alaska and Washington datasets. The team's results are shown in Figure 23, along with general trends they observed in their plots.



General trends:

- Temperature decreases with height regardless of location
- Temperature decreases with latitude
- Temperature increases with the year



General trends:

- Humidity generally decreases with altitude, but then slightly increases again at higher altitudes
- HI humidity drops significantly at higher altitudes
- HI humidity decreases as time passes for high altitudes, but for AK, it increases
- WA trends in between HI and AK

Figure 23. Baseline team's findings for climate data for the time periods, altitudes, and locations described in the scenario.

The Workbench team used Beaker to investigate what was in the climate data files they had found earlier in their search, and then followed up with very precise prompts asking the LLM agent to subset data in a certain way, and following a certain format, providing guardrails. Given the specificity of their prompts (see Figure 24), the team didn't have issues with hallucination and nondeterministic responses. Their data results are shown in Figure 25, along with observed trends. Although the format of their plots were different, both teams ended up observing similar trends in their plots (e.g. temperature decreasing with altitude, temperature increasing with year, moisture/humidity decreasing with altitude).

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from the files ta_Amon_GFDL-ESM4_historical_r1i1p1f1_gr1_195001-201412.nc and ta_Amon_GFDL-ESM4_ssp585_r1i1p1f1_gr1_201501-210012.nc for a region bounded by 10 to 30 degrees North latitude and -140 to -170 East longitude create independent subset netcdf files with dimensions of x and y for 1970, 2025, 2050, and 2075 each for 1000mb, 925mb, 850mb, 700mb and 500mb for mean temperature for December, January, and February and use a naming structure of Hawaii_Winter_{Year}_{level}_MeanTemp.nc

Figure 24. Example processing prompt for Beaker.

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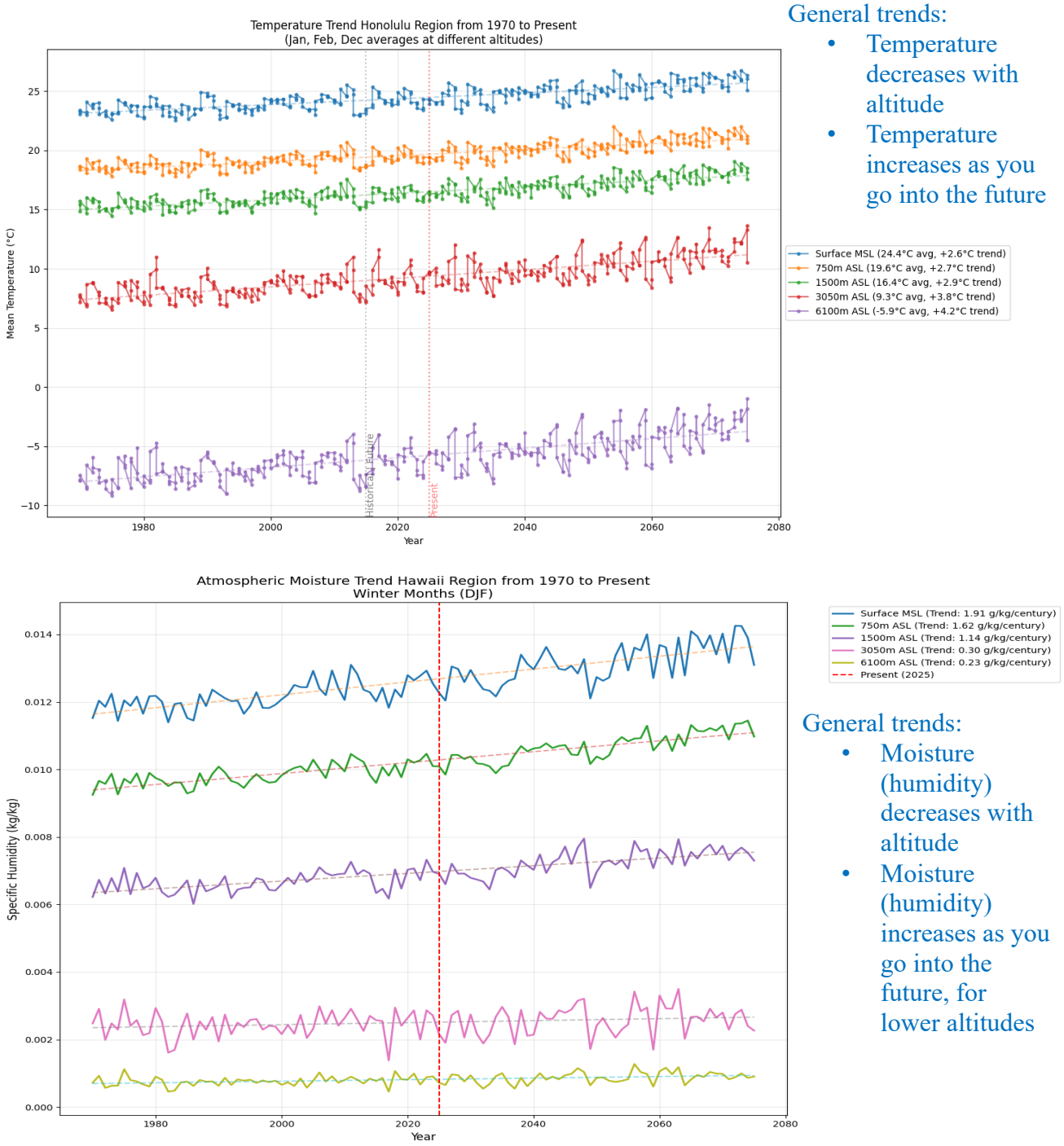


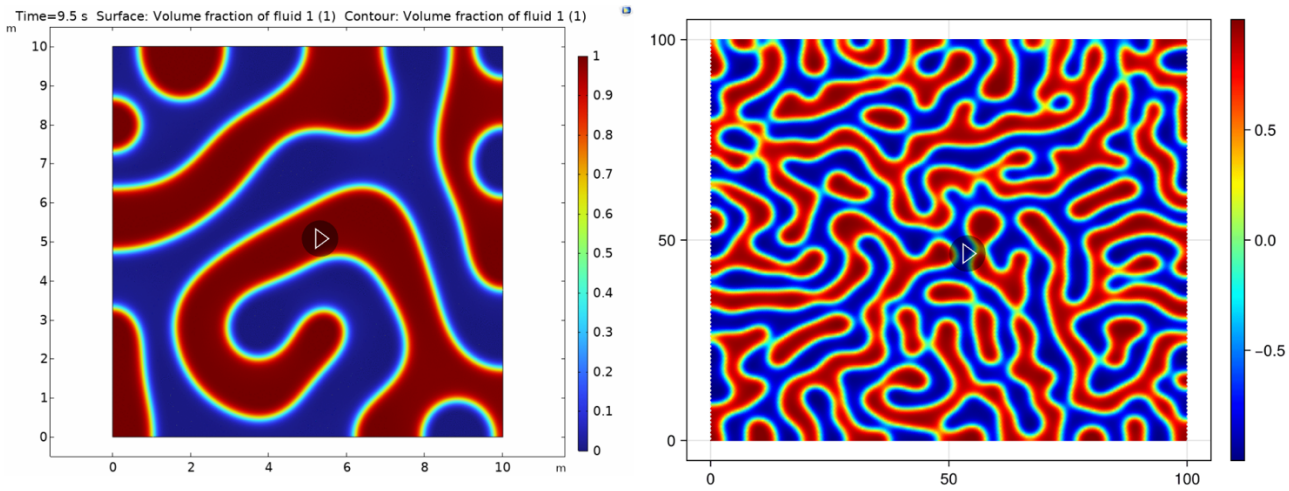
Figure 25. Workbench team's climate data visualizations, and observed trends.

4.1.4.3 Part 1: Component Multi-Physics Models

For all physics modeling in this scenario, the Baseline team used COMSOL (which the T&E team provided pre-installed on a loaner laptop), and the Workbench team used the Decapodes modeling library through the Beaker coding notebook interface.

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Both teams were able to successfully simulate the Cahn-Hilliard and Laminar Flow component physics models, with comparable outputs shown in and Figure 26 and Figure 27. Both teams generally met unit test expectations for these two models.



Baseline *Workbench*

Figure 26. Baseline and Workbench teams' simulation results for the Cahn-Hilliard model

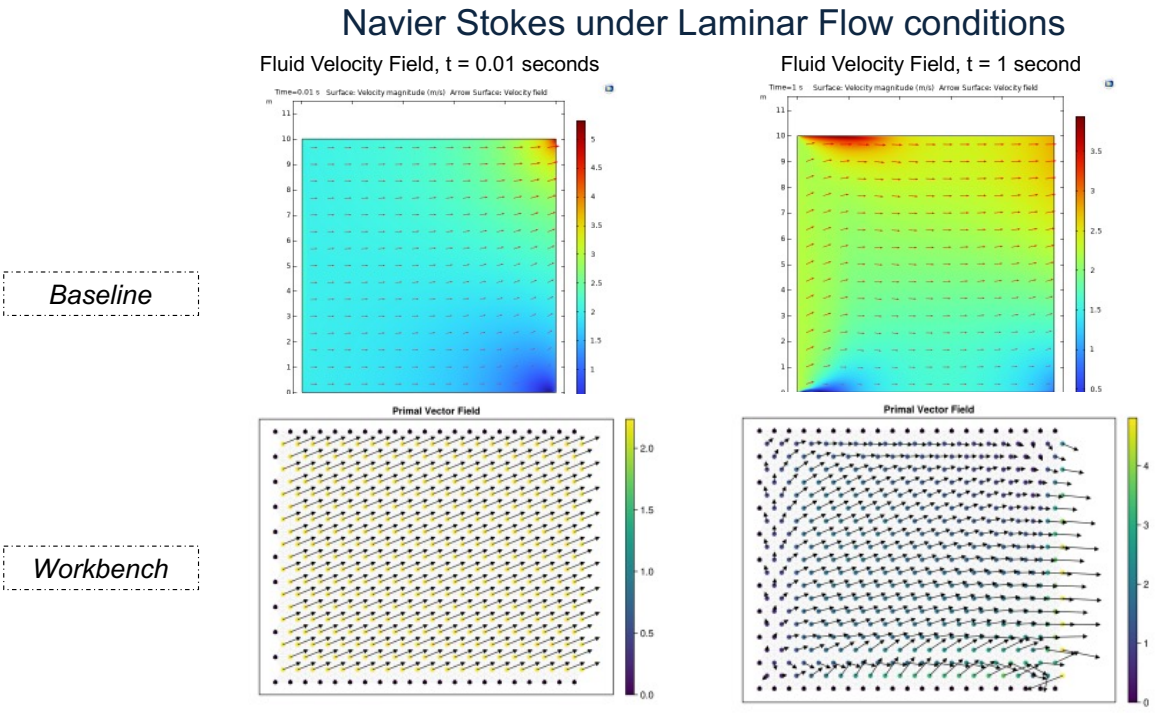


Figure 27. Baseline and Workbench simulation results for the Navier Stokes Laminar Flow model.

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4.1.4.4 Part 1: Cahn-Hilliard and Laminar Flow

Part 1 also asked modelers to do a model coupling, starting with the simplest models for this scenario – Cahn-Hilliard and Laminar Flow. While the teams simulated their results on different spatiotemporal scales, both sets of results (Figure 28) capture expected behavior, including 1) phase separation, and 2) general movement of both phases towards the right due to laminar flow. The Baseline team observed that their solution included something that looked like a small K-H wave trying to form in the bottom of their geometry. As explained by one of the Baseline modelers who is an expert in atmospheric science, K-H waves form where there is an abrupt change in wind direction in a portion of the atmosphere that is on the brink of cloud formation, or where a dry upper layer of air is passing over a moist layer. This simulation is similar to this physical phenomena, as there is a boundary (shear) that separates the upper extent of the fluid.

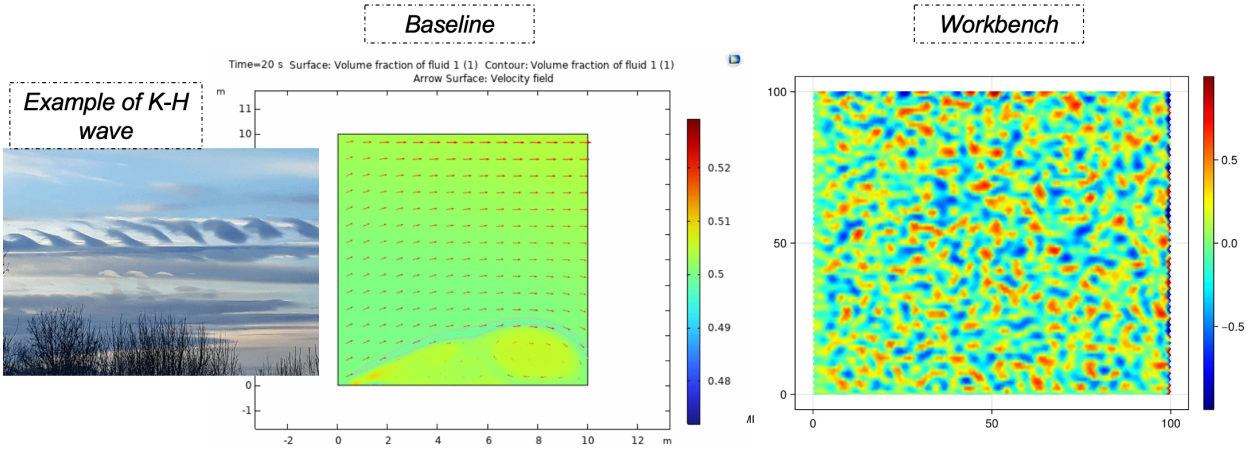


Figure 28. Snapshot of Baseline and Workbench team results for the Cahn-Hilliard and Laminar Flow coupling.

4.1.4.5 Part 2: Cahn-Hilliard and Turbulent Flow

In Part 2, modelers were asked to update their component physics models that they successfully simulated in Part 1, and add realism. Part of this process involved updating the Laminar Flow model to a Turbulent Flow model, which is a more realistic representation of atmospheric behavior. Simulating the Navier Stokes model under turbulent flow conditions was a challenge for the Workbench team (see Figure 29), and the inability to complete the unit test for this task detracted from decision-maker confidence later during the decision-maker panel. Model convergence was chaotic and failed to show expected output. The Workbench team explored multiple mesh sizes, solvers, time steps, boundary conditions, and vortex conditions, but was unable to complete this task successfully. In post-eval debrief sessions, the Workbench modelers suspected that it was handling of the boundary conditions with the circular obstacle in the center that was the biggest problem. It is unclear to what extent user error contributed to this issue, or whether modelers would have been able to successfully debug the situation with more time. Figure 29 shows snapshots of both team's results for the Turbulent Flow model.

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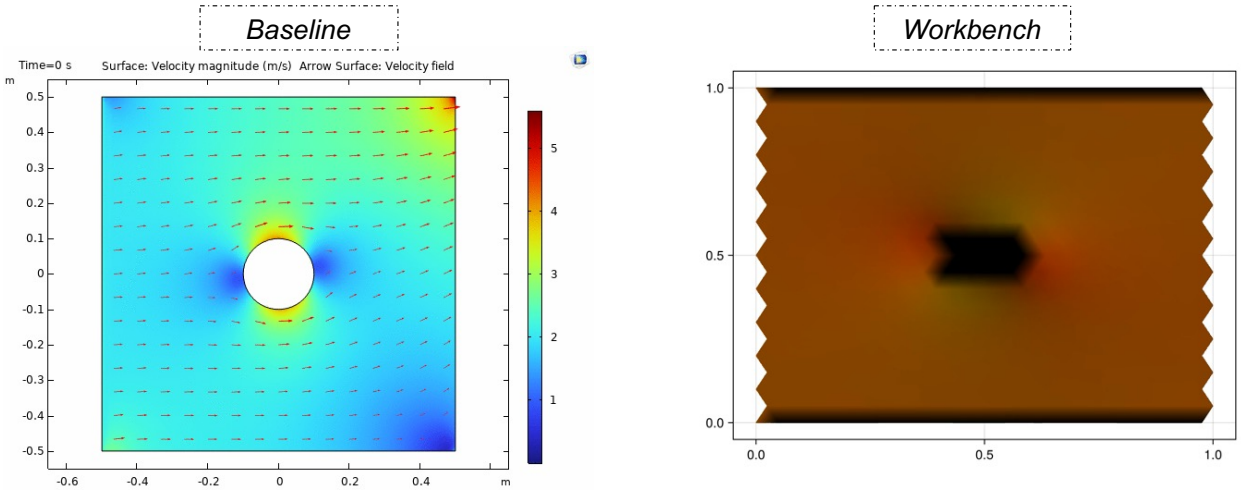


Figure 29. Snapshot of Baseline and Workbench teams results for the Turbulent Flow model.

The next task in Part 2 was to simulate the Cahn-Hilliard and Turbulent Flow models. The Baseline team was able to address this problem successfully, as shown in the example in Figure 30. The Workbench team was not able to complete this question due to their previous issues with simulating the Turbulent Flow model; they decided to move forward with rest of the scenario with the coupled Cahn-Hilliard and Laminar Flow models instead.

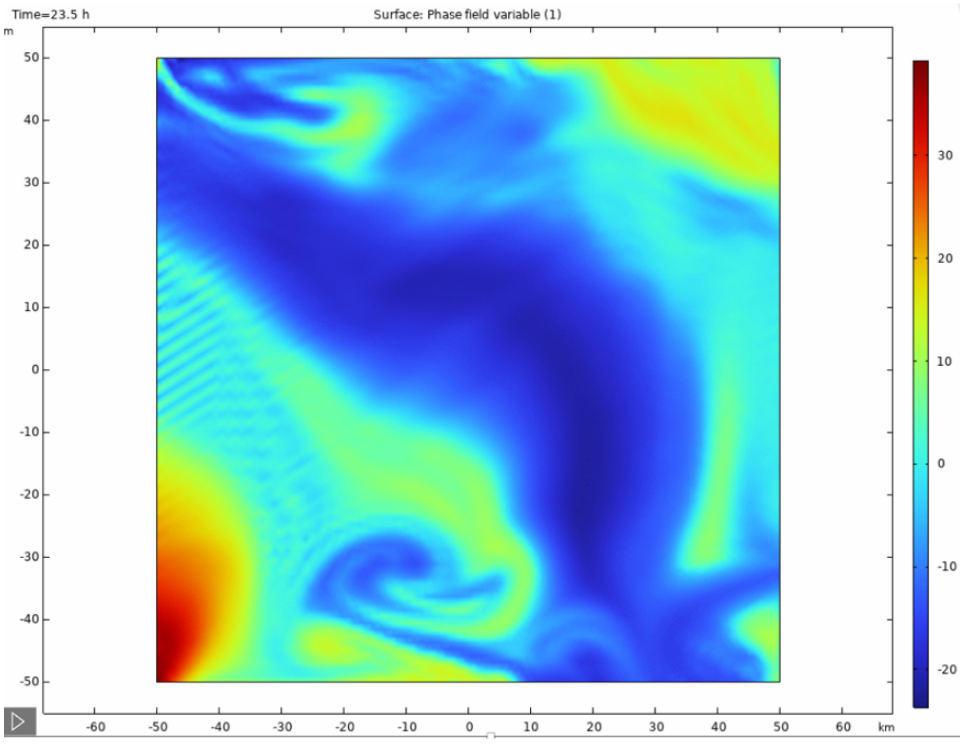


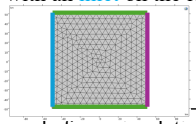
Figure 30. Example result from the Baseline team, for the coupled Cahn-Hilliard and Turbulent Flow models

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4.1.4.6 Part 3: Simulating Clouds

In Part 3, the modelers were asked to try and simulate realistic cloud patterns. Both teams had reasonable approaches to simulating clouds using realistic climate data (summarized in Table 17), although the Workbench team was constrained by the coupled models it was able to use, as explained in the previous section.

Table 17. Approaches taken by the Baseline and Workbench teams, for simulating clouds in Part 3 of the scenario.

Approach	Baseline	Workbench
Initializing Simulations	• Used humidity data to set average value of phase field variable (C) with random +/- 5% fluctuation around the mean humidity for the initial time step • Wind speed fixed to around 9 m/s at 35 degrees from the western boundary • Relative Humidity varied from 20-80% [20, 40, 60, 80] • Pressure varied from 0.4 atm to 1 atm [0.4, 0.6, 0.8, 1.0]	• Cahn-Hilliard diffusion parameter D set based on EPA • Wind direction was chosen based upon prevailing wind patterns • Average humidity data used set average of phase field variable (C) using a truncated normal distribution (0.0, 2μ) where μ is the local average humidity and σ=0.001 was scaled to reflect the local variation of humidities across similar grid cells
Model Used	Coupled Cahn-Hilliard and Turbulent Flow	Coupled Cahn-Hilliard and Laminar Flow
Geometry	• 100km x 100km 2D grid • Boundary Conditions are periodic on the top and bottom boundaries along with an inlet on the left face and open on the right face 	• 100km x 100km 2D grid • Models were run on a padded grid to avoid inclusion of computational artifacts from boundary conditions • Periodic boundary conditions were attempted, not successful
Simulation Output	Table of cloud length scale from simulations, as a function of pressure and humidity	Tables of C values for every point on the 2D grid from simulation

The Baseline team explored chaotic aspects of the simulations and tried to see if there were clear trends that emerged. To do this they ran three simulations for each configuration, each with an 8-hour simulation time with a 2-hour spin up time. Some example results are shown in Figure 31, demonstrating the non-deterministic nature of the simulations involved in this scenario. In Figure 32 however, the plot of Percentage Cloud Cover vs. Time shows pretty consistent trends even across multiple non-deterministic simulations; seeing analyses and plots like this positively impacted confidence for some of the decision-makers in the panel.

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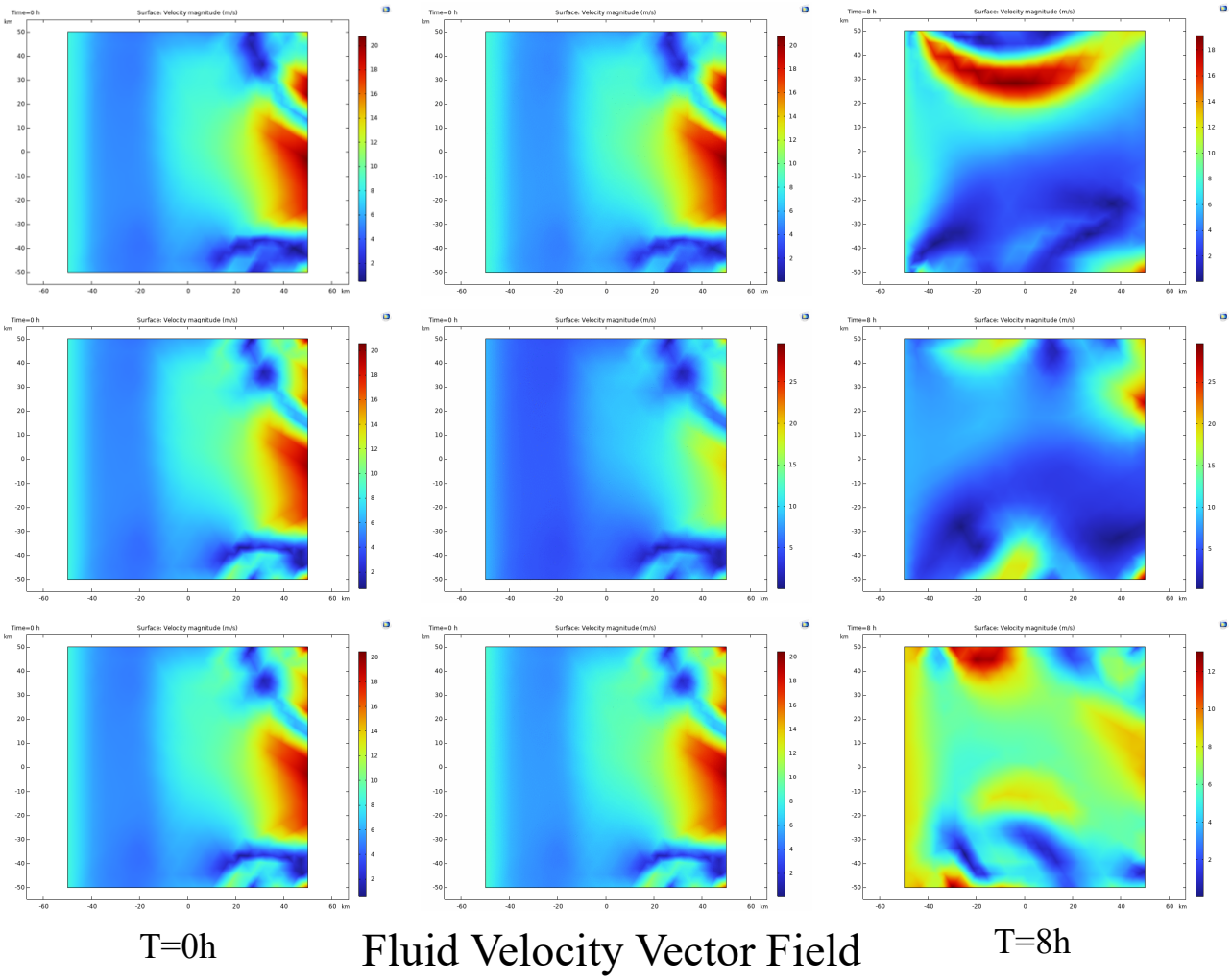


Figure 31. Plots demonstrating how the Baseline team explored the chaotic aspects of their simulations. Each row is a different simulation. The left column shows that all simulation began with the same initial conditions. The middle column shows how the simulation patterns are quite different between each simulation run, and the right column shows how the final simulation results at T = 8h are very different between the three runs, despite beginning from the same initial conditions.

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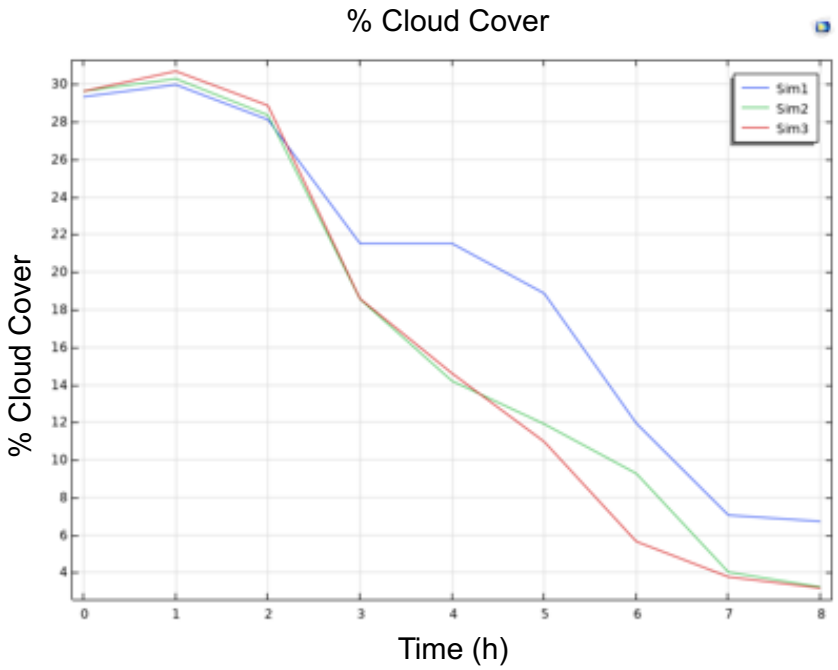


Figure 32. The Baseline team's exploration of the chaotic aspects of their simulations included plotting Percentage Cloud Cover over simulation time.

4.1.4.7 Part 3: Impact of Cloud Patterns on Radar Performance

For the final part of the scenario, modelers were asked to find the impact of cloud patterns (as simulated by their cloud simulations) on radar performance. Table 18 summarizes the approach that each team took. In general, the approaches were reasonable, although during scoring, both teams had points subtracted for either mistakes in their process (a quality issue), or incompleteness of their results. In implementing the radar equations, it should be noted that the Workbench team encountered a valuable lesson on not over-relying on the Beaker LLM agent's ability to generate code, without checking it for scientific reasonableness. In this specific scenario, the LLM was provided a Matlab documentation page with examples, inputs, and outputs. It generated Python code that upon first glance seemed complete and equivalent, but upon further inspection did not generate sensible results. The Workbench team found that the original Matlab source was needed before the LLM could generate the correct and equivalent Python code.

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Table 18. Summary of the approach taken by both teams in connecting their cloud simulations to radar performance.

<table>
<tr><th>Approach</th><th>Baseline</th><th>Workbench</th></tr>
<tr><td>Cloud Simulation Outputs → Radar Equations</td><td>
• Generated cloud length scale table from simulations
<table>
<tr><th colspan="2">Cloud Length Scale [km]</th><th>Pressure [atm]</th><th></th><th></th><th></th></tr>
<tr><th></th><th></th><th>0.4</th><th>0.6</th><th>0.8</th><th>1</th></tr>
<tr><td rowspan="4">Humidity [%]</td><td>20</td><td>20</td><td>100</td><td>60</td><td>100</td></tr>
<tr><td>40</td><td>100</td><td>100</td><td>100</td><td>100</td></tr>
<tr><td>60</td><td>30</td><td>60</td><td>100</td><td>100</td></tr>
<tr><td>80</td><td>80</td><td>50</td><td>30</td><td>90</td></tr>
</table>
• Set liquid water density to constant 0.5 g/m^3, for cumulus clouds
• Detection range <u>family of curves</u> established as a function of temperature, cloud length scale, RCS, SNR
• For a given location, year, altitude, and temperature:
  ○ From RCS select plot from detection range family of curves
  ○ Pick appropriate pressure/humidity combination in table, select cloud length scale
  ○ From cloud length scale, select curve on plot
  ○ From temperature, find intersection point on curves
</td><td>
• Take output C values from every grid point in cloud simulations, interpret as concentration and convert to cloud liquid water density
• Input each liquid water density value into python fogpl function, and get distributions of atmospheric loss
• From atmospheric loss distributions, calculate radar range distributions
• Did not vary path length based on cloud size
</td></tr>
<tr><td>Radar Equations</td><td>
• Matlab function fogpl() (<u>input</u>: path length, frequency, temperature, cloud liquid water density; <u>output</u>: atmospheric loss)
• Matlab function radareqrng() (<u>input</u>: loss from fogpl; <u>output</u>: max radar detection range)
</td><td>
• Generated Python equivalent of Matlab function fogpl() and hand-coded detection range formula
• Initially gave Beaker LLM agent the fogpl Matlab documentation page with examples, inputs and outputs → results were nonsensical → Matlab source code needed before correct Python code could be generated by LLM
</td></tr>
</table>

Figure 33 shows one set of radar performance results from the Baseline team (they also generated similar sets of curves for other radar cross sections and detection targets described in the scenario). Figure 34 shows a subset of results from the Workbench team. A summary of the different trends observed by each team (including similarities), is found in Table 19. The Workbench team generated distributions while the Baseline team did not incorporate uncertainty into their plots. However, it should be noted in Figure 34 that for the Workbench team's results, the amount of change in detection range and atmospheric loss is very small, suggesting very little sensitivity to many of the factors that were varied, which is unrealistic. Reasons for this could include the fact that they had to use Laminar Flow instead of the Turbulent Flow models, leading to lack of realistic cloud formation and variation in signal path length. The T&E team also noted unit conversion mistakes during post-evaluation scoring of the results, which may have contributed to this issue.

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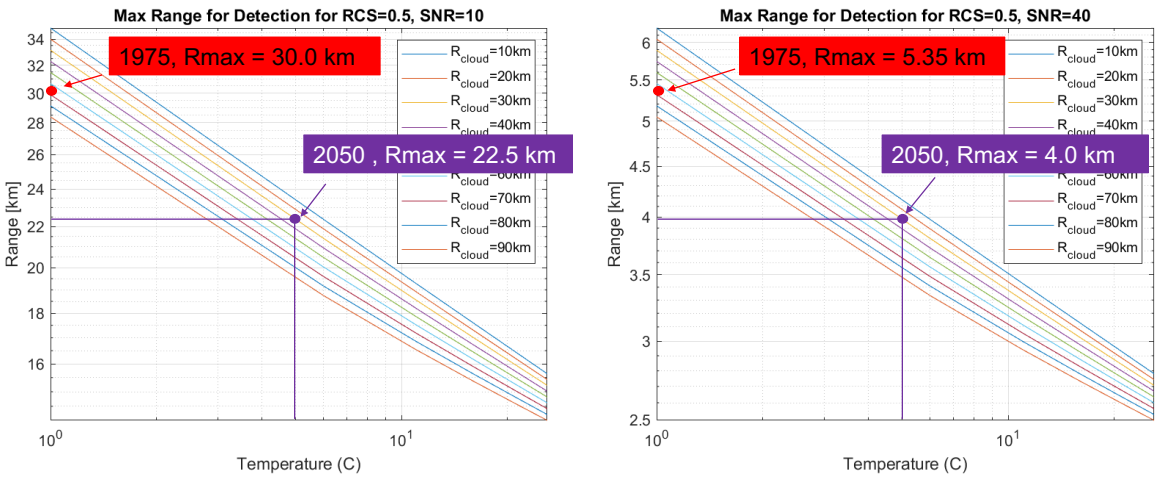


Figure 33. One set of radar performance results from the Baseline team. These plots show max range for detection of a Tomahawk, in Anchorage, AK.

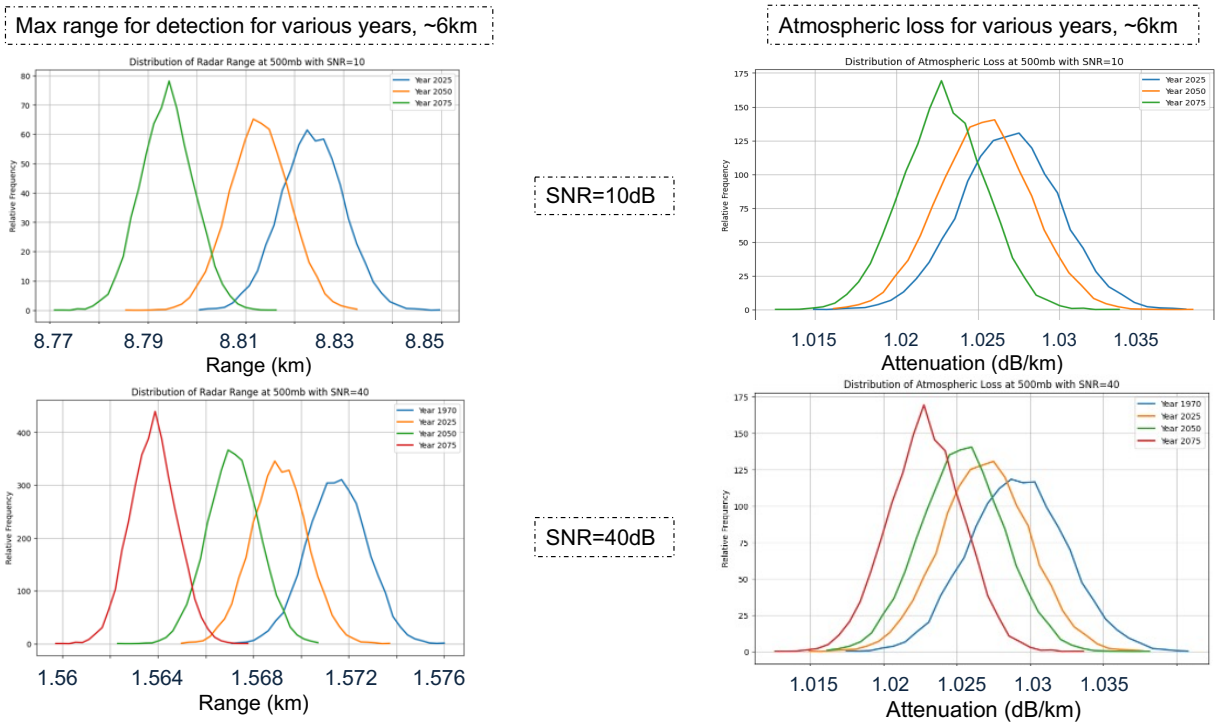


Figure 34. A subset of radar performance results from the Workbench team. The team only had time to calculate results for Honolulu, HI.

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Table 19. A summary of trends found in the radar performance analyses completed by both teams. Similar trends are bolded. For other trends, it was difficult to do a crosswalk because outputs from both teams were quite different in format, etc.

Baseline	Workbench
• For a given cloud length scale and SNR, detection range decreases as temperature increases • For a given cloud length scale, **detection range decreases significantly as required SNR increases** • For a given temperature and SNR, detection range decreases as cloud length scale increases • **For given SNR, detection range decreases as you go into the future (and ambient temperature also increases as years increase)**	• For a given year, detection range increases as altitude increases • For a given year, atmospheric loss decreases as altitude increases (density decreases as altitude increases); in other words, atmospheric loss decreases as density decreases • **For a given altitude and SNR, detection range decreases as you go into the future (ambient temperature also increases)** • For a given altitude and SNR, atmospheric loss decreases as you go into the future (as ambient temperature and humidity increase) • For a given altitude and year, **detection range decreases significantly as required SNR increases** • For a given altitude and year, attenuation doesn't change much as required SNR increases

4.1.4.8 Decision-maker Panel Outcomes

Similar to the Epi DM experiment, after four days of work on the Climate DM Scenario, on the final day of the evaluation event, the decision-maker panel (consisting of 3 panelists) heard presentations from the Baseline team in the morning session and the Workbench team in the afternoon session. The panel was asked to judge the two teams independently rather than comparatively. They were asked to score Parts 1-2 and Part 3 on a scale from 1 (very low) to 7 (very high) indicating their confidence in the answers and outputs for that part.

For each part, the panelists were given a high-level confidence question and sub questions to support their assessment, as described below.

Parts 1-2: Setup with Multiphysics Modeling and Updating Models to Add Realism

How confident are you that the modeling team appropriately modified and developed a model or set of models, to sufficiently explore the scenario/problem?

Metric Explanation: This part involves updating or modifying physics models. Decision makers will assess whether this was approached in sensible way and whether the final model(s) is likely to support decision-maker priorities for this scenario.

The decision-maker confidence score should be supported by the answers to the following questions:

- How confident are you that individual models and physics were correctly set up, implemented, and simulated?
- Were parameters chosen and updated appropriately for your context?

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Part 3: Addressing Decision-maker Priorities

With what degree of confidence do you feel that you understand the modelers' results?

Metric Explanation: Based on the information presented to you by the modelers, determine your confidence in model results, and (if relevant) how uncertainty factors in.

The decision-maker confidence score should be supported by the answers to the following questions:

- Did simulation results make sense? If results did not match your intuition or expectations, did modelers explain the reason(s)? Did their explanations make sense? (Please focus on general trends rather than exact numbers).
- Is it clear how to interpret any uncertainty in the results? Do you understand the key drivers of uncertainty in the results?
- Are you confident in making decisions based on the modeling process, model results, and recommendations from the modelers?

Figure 35 shows the score distributions and mean scores for each of the graded sections (Parts 1-2, and Part 3). For all parts scored, the mean score was higher for the Baseline team. As with the epidemiology use case, the order of the presentations (Baseline first) could have influenced scores – ideally, we would want to have two different panels see the presentations in different orders, although this was not logistically feasible. Also similar to the epidemiology use case, there was a lot of variation in scores and across panelists (with one exception being the score for the Workbench team for Parts 1-2, which was graded 'Somewhat Low' by all panelists); panelists struggled to score consistently across both teams, changing their minds several times during the panels. Table 20 and Table 21 summarize the panel outcomes. Based on the panel outcomes and debrief sessions, most likely the biggest detractors of confidence for the Workbench team trace back to the challenges with the Turbulent Flow model, conversion to the right geometry (scenario asked about a 2D rectangle where documentation involved torus and sphere), and not being able to implement period boundary conditions as the modelers would have liked to; some of this traces back to lack of sufficient Decapodes documentation examples with different types of geometries and commonly-used boundary conditions.

During the debrief session with modelers and decision makers, one topic of discussion that arose was the potential advantage of using different tools for different types of physics. After going through the evaluation, the modelers were of the opinion that for well-known, established physics, it may generally be easier to use well-known and established multiphysics tools like COMSOL, which have the advantage of decades of development and funding and many resources for learning how to use the tools (classes, tutorials, other learning materials). However, the Workbench modelers commented that Decapodes was designed for the 'researcher mentality', and has the potential to demonstrate meaningful advantage over baseline tools for developing and implementing new physics models and equations. For example, if the scenario goal was instead to develop new equations for cloud microphysics and formation, it may be easier to do that in Decapodes, compared to COMSOL and other baseline tools. However, the T&E team is of the opinion that this is a difficult aspect of research to evaluate in a structured and constrained evaluation event, and was beyond the scope of the final program evaluation, with the potential to be better communicated through further unstructured exploration and performer research publications.

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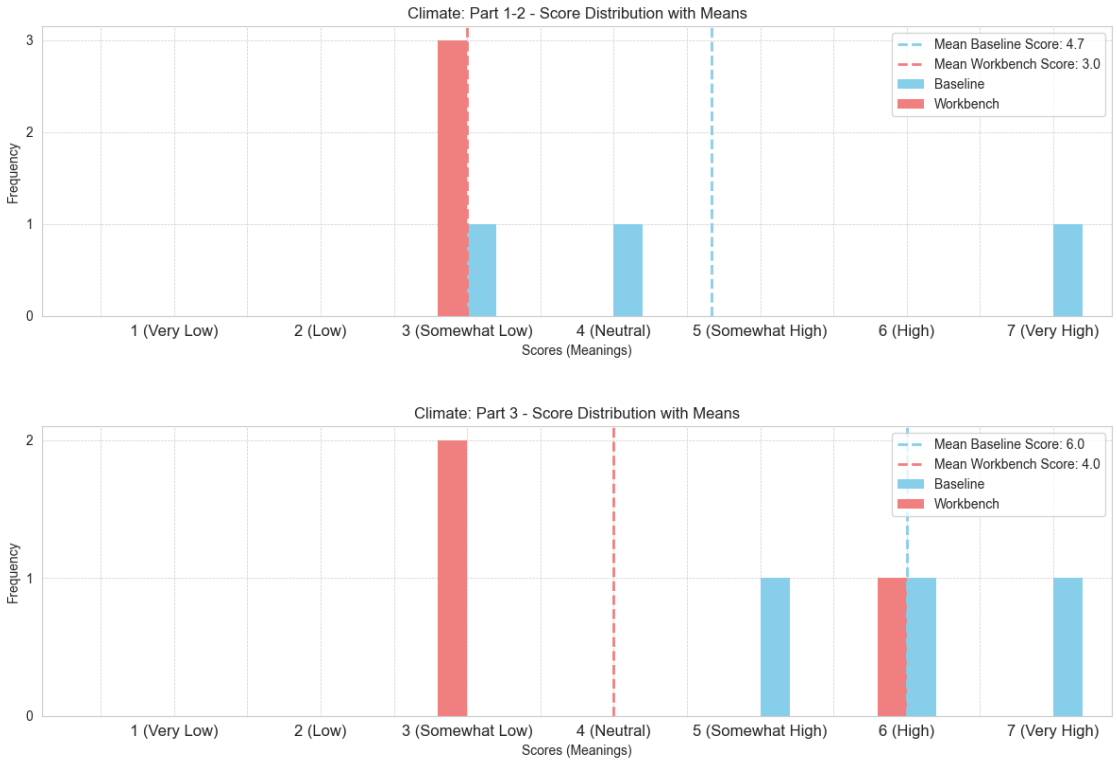


Figure 35. Climate Decision-maker Panel scores.

Table 20. Summary of Climate Decision-maker Panel comments and outcomes.

Part	Baseline Avg Score	Baseline Summary	Work-bench Avg Score	Workbench Summary
Part 1: Setup with Multiphysics Modeling **Part 2:** Updating to Add Realism	4.67	• Plots produced generally match unit tests **(+)** • Modelers able to describe activity in all plots **(+)** • Comp. limits that made modelers go with linear instead of quadratic discretization, and raise error tolerance for convergence **(-)** • Modelers had to do a lot of math to get info into COMSOL acceptable format. Mistakes were unlikely, but it is an error path **(-)** • Wanted to see more on how temperature and relative humidity relate to Cahn-Hilliard • Things DMs wanted to see to increase confidence (validation with historical data, seasonality, decreases in variances), were difficult with time constraints. Within time constraints, team balanced concerns appropriately. • Modelers had clear command of limitations of their process **(+)** • Really impressive output given time/compute restraints! • Really great work tying into intuition w/ real clouds **(+)**	3	• Difficult to understand if the tools work • Understand fluid viscosities, boundaries were difficult • Additional validation steps needed; lack of validations significantly contributed to score **(-)** • Clearly need experts to run the ASKEM tools • Did not find tool output visualization compelling **(-)** • Comparing graphs to unit test outputs (COMSOL images) was more challenging. Would've preferred heat map **(-)** • **Lack of turbulence validation w/ unit test output really made the difference here (-)** • **Not being able to implement periodic boundary conditions lowered confidence (-)** • Ability to input equations directly **(+)** • Liked Decapodes model composition diagram that showed building blocks **(+)**, even though DMs could not digest it all during the presentation
Part 3: Addressing Decision-maker Priorities	6	• COMSOL graphics/gifs are good, helped DMs understand variations **(+)** • Modelers didn't have enough time to run additional experiments to increase confidence **(-)** • Cloud cover graph with 3 runs boosts confidence, as it showed % cloud cover isn't too dependent on chaotic processes **(+)** • Lack of precipitation as input data? **(-)** • Modelers had clear understanding of the limitations of their approach, which lends confidence in next steps for further areas of exploration **(+)** but also means their lack of confidence in their own results leads DMs to not be confident either **(-)**	4	• Lots of QC for tools needed • Not confident results are reproducible **(-)** • The relative frequency plots were informative (+), but confusing units **(-)** • The trend analysis graph of humidity and temperature data **(+)** • The conclusions section was helpful **(+)** • Too little info on uncertainties **(-)** • Less focus on validation **(-)**

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Table 21. Summary of Climate Decision-maker Panel outcomes, in terms of aspects that raised or lowered decision-maker confidence.

	Aspects that raised confidence (+)	Aspects that detracted from confidence (-)
Baseline	• Plots produced generally match unit tests • COMSOL graphics are good, helped DMs understand variations/chaotic output • Modelers able to describe/explain activity in all plots • Tying into intuition by connecting back to real cloud images/descriptions • Cloud cover graph with 3 runs and seeing trends emerge despite chaotic process • Modelers understood limitations of process → lends confidence to next steps (further research, exploration)	• Computational limits that made modelers go with linear discretization rather than quadratic, and raise error tolerance for convergence • Having to do a lot of math to get info into COMSOL acceptable format. (Doesn't think modelers made mistakes, but this is an error path) • Lack of validation with historical data, seasonality data, and more experimentation • Modelers understood limitations of process but their lack of confidence in their own results leads DMs to not be confident in results
Workbench	• Ability to input equations directly • Model composition diagram that showed building blocks (even though panel couldn't digest it all during the presentation) • Trend analysis graph of humidity and temperature data	• Lack of/less focus on validation (partly due to lack of time to do more iteration) • **Lack of turbulence validation w/ unit test** • **Not being able to implement periodic boundary conditions** • Output visualizations not compelling; more difficult to compare to unit test outputs (*T&E note: this is in some respects an unfair comparison of visualization quality because unit test plots were created in COMSOL, and Decapodes visuals are constrained by standards in Julia-style visualization ecosystem*) • Confusing units in relative frequency plots

4.1.4.9 Climate Decision-maker Scenario Summary Times and Scores

Although the Climate DM experiment was not used for program timing metrics, because of the generally comparable approaches prescribed to both modeling teams, the T&E team did score and time the results, with a summary in Table 22, and Figure 36. In general the Baseline team spent less time on most questions with equal or better quality and completeness scores; the only exception is the simulating clouds question in Part 3. Please note that Workbench times for Part 3 may include some untracked performer help not reflected in total time.

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Table 22. Timing and outcomes summary for the Climate Decision-maker experiment.All times are 'critical path' times except for the Additional Time category. Please note that the Workbench times for Part 3 may include some untracked performer help not reflected in total time.

Part	Baseline Time Person-Hours (hh:mm)	Baseline Summary	Workbench Time Person-Hours (hh:mm)	Workbench Summary
Part 1: Data and Multiphysics Model Setup	11:53	• Data search and wrangling was handled without issue, required a fair amount of custom coding • COMSOL usage despite pre-eval training/exposure was still challenging, but successful with a learning curve	26:21	• Getting the right prompts required some iteration but once identified, modelers found Beaker to be very helpful with ESGF data search, processing of data, and creation of plots • Cahn-Hilliard and Laminar Flow models successfully implemented
Part 2: Updating to Add Realism	7:17	• Successful in implementing turbulence and periodic boundary conditions, • Fair amount of time spent trying to make simulations computationally tractable	11:21	• Unable to simulate turbulence and periodic boundary conditions
Part 3: Cloud Simulations and Impact on Radar Performance	28:20	• Explored chaotic aspects and emerging trends • Reasonable approaches for cloud simulation and radar calculations; used family of curves approach	32:30 * May include some untracked performer help not reflected in total time	• Was able to use Beaker to generate significant amount of the Python code for radar calculations • Reasonable approach for cloud simulation and radar calculations; outputs included uncertainty
Additional Time (Planning, Slides, Noncritical Path, Etc.)	11:48		3:40	
Total time spent	59:18		73:52	* May include some untracked performer help not reflected in total time

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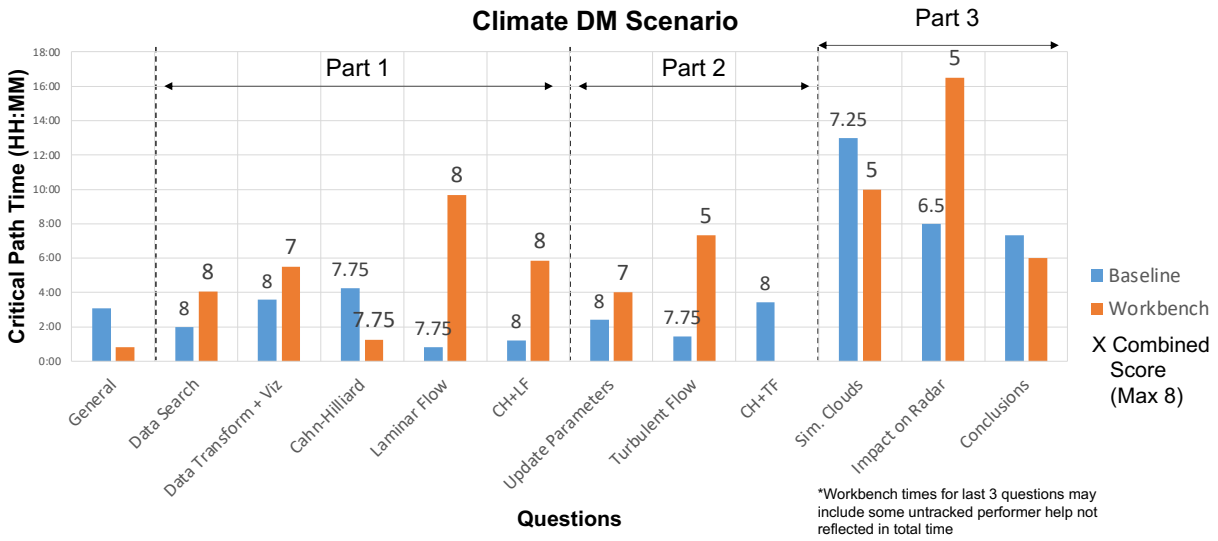


Figure 36. Timing and scores for the Climate Decision-maker (Climate DM) experiment.The heights of the bars represent 'critical path' time, and the numbers beside the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology. Although the quality and completeness scores should be interpreted separately, we sum them here only for the sake of concise visualization and ease of comparison in a single image. Please note that the Workbench times for the last three questions may include some untracked performer help not reflected in total time.

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4.2 Challenges and Successes (Technical Aspects and Modelers' Experiences)

In the following sections, we describe the biggest challenges and highlights, focusing on the experience of the modelers during the evaluation. Content reflects technical and other aspects related to teaming approach, usability of tools, etc. Table 23 covers the Epi IQ experiment, Table 24 covers the Epi DM experiment, Table 25 covers the Climate IQ experiment, and Table 26 covers the Climate DM experiment.

Table 23. Challenges and successes for the Epidemiology Independent Questions (Epi IQ)

Team	Biggest Challenges	Highlights, What Worked Well
Epi Baseline IQ	• **Calibration**: Modelers found calibration with data challenging. This may be partially due to time constraints. • **Stratification**: Modelers found stratification a challenging topic, either opting for the easiest approach of using different configurations, or running out of time. Modelers mentioned they understood the concept and the goal of the questions, but did not have the right R tools in their "toolbox" offhand and struggled with implementation. • **Ensembles**: Modelers found ensemble problems difficult and did not explore a fine-grained parameter space when searching for ensemble model weights. This may be partially due to time constraints. • **Limited Preparation Time**: Modelers commented they felt like more time with the training materials and types would have been helpful, especially given that the Workbench team had more training sessions in order to learn the ASKEM tools.	• Modelers were highly efficient in conducting basic modeling tasks and maintained inspectable, replicable R code. • **Bounded Questions**: Modelers liked the bounded nature of questions • **Unit Tests**: Modelers incorporated many unit tests along the way to check their work, which led to high quality on many questions • **Teamwork and Flexibility**: Modelers opted to pair-program for a majority of the evaluation, and both enjoyed working through problems of their choice as a team • **Limited Context-Switching**: The Baseline team did not have to do much context-switching between scenarios – they were able to focus on one question at a time and make meaningful progress on each one before moving on to the next
Epi Workbench IQ	• **Model Ingestion**: Modelers found ingesting the models to be a bit cumbersome; in general, model ingestion took more time in the workbench than it did for the Baseline team, except for the hard model ingestion problems. • **Lack of Context**: Since the independent questions were detached from a decision-making context, modelers noted it was hard for them to assess the workbench's usefulness in a realistic setting, since they were primarily doing functionality checks. • **Validation**: While doing functionality checks, modelers wanted assurances that they were getting the right output, and found it hard to validate their results when no expected output was provided in the question	• **Independent Working**: Unlike the Baseline team, the Workbench modelers generally worked on separate questions individually. Modelers commented that the workbench lends itself to independent working; working on the same question simultaneously was difficult. Modelers seemed to enjoy this method. • Modelers generally found the tools and workflows to be useful for completing the tasks in the evaluation. This was especially notable for TA3 tasks where there were substantial reductions in time needed to complete certain tasks (forecasting, ensembling, etc.) relative to the baseline.

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Table 24. Challenges and successes for the Epidemiology Decision-maker Scenario (Epi DM)

Team	Biggest Challenges	Highlights, What Worked Well
Epi Baseline DM	• Modelers "fought the scenario" a little bit in the beginning, stating that re-creating unit tests from academic papers would not be their normal workflow • **Complex Model**: The model they eventually developed was very complicated (576 state variables and multiple transitions per state variable) and required time on an HPC to run • **Time Constraints**: The modelers felt constrained by the time given to complete the work, often working extended hours in the evenings in their hotel rooms, and even then, they didn't quite complete everything • **Finding Parameters, Data Munging**: The process of finding parameters and munging data into the right probability formats required by the model, was challenging • **Literature Search**: Modelers chose not to use off-the-shelf LLM models for literature search due to lack of familiarity (it did not occur to them to use such resources), but later upon reflection in the debrief sessions, mentioned that they would have appreciated AI assistance and thought AI tools could have utility for this topic.	• **Efficient Approach**: Modelers were highly efficient in building their model and maintained inspectable, replicable Python code. • Created a flexible, comprehensive model • **Reasonable Outputs**: The results obtained were reasonable (especially the differences between antagonistic/synergistic models), though not perfect
Epi Workbench DM	• **Model Ingestion**: Model ingestion took substantially longer than expected, and one of the models (B) posed challenges that the performers could not explain • **Transition Names**: The need to know the arbitrary name of a transition (e.g., t67) in order to change it required an added step to attempt to find the transition and its name before doing an editing step. This is harder with larger models. • **Stratification**: Modelers started with a model that partially stratified by disease and then extended it to be fully the same across all diseases, creating a stratified model that Terarium did not recognize as such, leading to slower performance. • **Stratification**: Age stratification of their complex model resulted in a model that they could not use to run simulations. • **From Scratch vs. Modify**: Modelers did not understand that it would likely have been easier and better Terarium practice to start	• **Metadata Extraction**: Modelers appreciated extraction of parameter descriptions and units from papers • **Model Ingestion**: To the extent that model ingestion worked, it felt "like magic" to the modelers • **Debugging**: Modelers found having multiple ways to view/verify the model very helpful for debugging • **Parameter Stratification**: Gridded/matrix view of stratified parameters was helpful in quickly highlighting certain classes of model errors • **Adding Model Components**: Modelers found adding new compartments and transitions to models very easy in the workbench • **Model Modification and Validation**: Modelers appreciated the possibility to directly modify the human-readable JSON

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Team	Biggest Challenges	Highlights, What Worked Well
	from a very simple model and build it up using inspiration from the more complex provided models versus trying to modify a complex model • **Complex Rate Laws**: Modelers hand-generated very complex rate laws (for Exposed compartments where the exposure could have come from an infected person in any one of several different Asymptomatic or Infected compartments) that Terarium had a hard time with. (Performers provided an outside "simplify" operation that unfortunately introduced other errors that the modelers didn't have the time or confidence to wait for performers to debug.) • **Implausible Forecasts**: Ultimately forecasts were not plausible, and modelers were unable to debug this during the evaluation timeframe • **Visualizations**: Default graphs were hard to read • **AI Overreliance**: Modelers may have over-relied on AI in using parameter values ingested from papers that were not appropriate for the scenario details/timeframe • **Logistics**: The inability to work simultaneously in a single project caused logistical challenges	representation of the model, and the tool provided a validity check before importing a modified model into the workbench • **Clear Communication**: Terarium supported clear communication of model differences, model edits, and assumptions • Despite all the challenges, the modelers really enjoyed using the software and one has plans to use it for an outside project

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Table 25. Challenges and successes for the Climate Independent Questions (Climate IQ)

Experiment	Biggest Challenges	What Worked Well
Climate Baseline IQ	• **Software Setup**: Software installation (e.g., MATLAB and COMSOL) was more challenging than expected. This required several rounds of help to finalize. • **Julia**: Modelers were unfamiliar with Julia programming language and COMSOL software, which required some support and a learning curve (more so with COMSOL) • **Question Clarifications**: Question descriptions required additional minor clarification • **Lack of Strategy**: Modelers admitted to not spending enough time to strategize a plan and simply worked independently • Ignored questions that seemed too challenging to solve initially and made progress where possible	• **Technology of Choice**: Modelers were mostly unconstrained by programming language or IDE environment, freely using Python in Jupyter notebooks or MATLAB. For problems requiring Julia, modelers initially required support but seemed to adapt quickly. • **Context Switching**: Although working independently, the Baseline team was able to switch between questions when waiting on support to make continued progress.
Climate Workbench IQ	• **Model Coupling**: Reconfiguring problems for coupling different physics was a challenge; this could be due to inexperience with Julia or not understanding Decapodes documentation • **Creating Geometrical Mesh**: Making a 3D pipe in Blender had triangulated elements, and Decapodes requires tetrahedral elements. Modelers spent a lot of time making the pipe out of tetrahedrals. Used *tetgen* to construct rectangular prism and converted to cylinder using coordinate transformations. It was surprisingly difficult to make a cylinder out of tetrahedrals. (*T&E Note: Meshing tools are outside the scope of Decapodes, which relies on external meshing tools that the modelers didn't have much prior experience with, so this challenge actually reflects tasks that needed to be done with Blender/tetgen, as a prerequisite for using Decapodes to solve the evaluation problems. However, Decapodes would benefit from additional documentation on how to generate compatible tetrahedral meshes in some of these external tools*). • **LLM Stuck in Cycle**: There were instances where the LLM agent in Beaker got really close to the right answer, but then got stuck in a cycle of fixing errors and losing sight of the original ask	• **Time Saved Writing Code**: Modelers think Beaker saved them a lot of time in writing code for the questions • **Tool Utility**: Modelers think Beaker would be a useful tool in conjunction with other tools; some modelers were really excited to use it with other projects • **Arbitrary Geometry**: Time was saved in applying physics models to arbitrary geometries, with Decapodes

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Table 26. Challenges and successes for the Climate Decision-maker Scenario (Climate DM)

Experiment	Biggest Challenges	What Worked Well
Climate Baseline DM	• **COMSOL Convergence**: Modelers didn't expect to spend so much time fighting with COMSOL to get their solution to converge. Point at which calculations would diverge was inconsistent, sometimes it would get to 95% and then crash. • **Lack of Adaptation/Automation**: Had expected COMSOL would be more adaptive, but there was a lot more manual tweaking needed • **COMSOL Plots**: Tedious at first, but made more sense by end of the evaluation week • **Model Selection**: Overwhelming to find/select the right model; there were too many choices. Went through wizard to try to find differential equations that were similar to question. Phase field was under general math, which confused and lowered modelers' confidence that they were doing the right thing. Without Cahn-Hilliard tutorial (found by Googling), would've taken a lot more to convince modelers they were selecting the right model • **Scenario Length**: Thought the scenario could be shorter. Too many combinations for Part 3, had to loop over different locations, altitudes, etc. • **Strategy**: Next time would focus more on the end goal/expected output of scenario, then focus on developing MVP for overall scenario	• **Boundary Conditions, Initial Conditions**: Expected these to be tricky but wasn't as bad as expected • **MATLAB**: MATLAB Radar Toolbox was easy to use, even though one modeler last used MATLAB > 10 years ago • **COMSOL Animations**: Weren't very difficult to use, look nice, and definitely helped with sanity checks. Was expecting this to be tricky, but was easier than plotting in some cases • **COMSOL Documentation/Forum**: Was able to readily find information online, from COMSOL website and independent sources/contributors. It did take some sifting through the information, but helpful content could be found.
Climate Workbench DM	• **More BC Examples**: Would have liked more examples of Decapodes models with boundary conditions, and variety of boundary conditions (especially the typical ones you use very often with PDE models). The torus and sphere examples currently in documentation have no boundary conditions • **Turbulent Flow**: Modeler thinks the challenges with turbulent flow (Q7) largely came down to conversion to the right geometry (scenario asked for a 2D rectangle and documentation had examples for torus and sphere), and BCs. Thinks many problems could've been resolved if they had access to more examples • **Generating Meshes**: Would have liked to look at additional Julia libraries for this, but ran out of time	• **Documentation**: Found Decapodes documentation to be quite useful • **Exterior Calculus**: Liked coherent separation between physical objects and the forces that operate on them. This is a topic that's more challenging with traditional finite element modeling, especially for understanding what's happening when things go wrong in the modeling and simulation process. As a mathematician, modeler appreciated this separation with exterior calculus. • **Decapodes Modular Model Composition Diagrams**: Liked the cleanly separated modular components for multiphysics coupling

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Experiment	Biggest Challenges	What Worked Well
	• **Visualization**: Modelers commented that Decapodes visualization capabilities are consistent with standard Julia-style visualizations, which are fine for academic purposes, but not as compelling for industry or government audiences. • **Units**: Would have liked more unit tracking throughout; when pulling in someone else's model, would like to make sure their units line up properly. Also, would be good to have functionality to help physicists' understanding of units. • **File Management**: Early issues with Beaker not knowing about files, but this was quickly resolved with more specific prompt engineering. • **Kernel Restart**: Modelers would have liked kernel restart options (similar to regular Jupyter notebooks), as it is a general best practice and needed quite often • **Questionable/Incorrect Outputs From LLM Highlights Need for QC**: Sometimes Beaker would output something, and the modeler had to question it as a scientist. For one question, the modeler asked Beaker to create a Python version of a radar function, giving it reference to the MATLAB documentation page and example. It confidently gave an answer, but upon further inspection, the code was incorrect. Once the MATLAB source code for the radar function was provided, Beaker generated the correct Python version. This highlights the need for quality control and checking from the modeler, and not blind reliance on the veracity of the output provided by the LLM agents, not matter how confident they seem in their responses.	in Decapodes; the Graphviz composition diagrams helped in the debugging process • **Finding ESGF data**: Was able to find the most relevant data more easily than the Baseline team. With the ESGF website, not all filters are helpful, it's hard to parse what you want. Modeler was able to relay what he wanted into Beaker (e.g. 'I want data that meets xyz criteria'), and it returned list of files and where they could be downloaded. • **File Management**: Once modeler figured out the general approach to let Beaker know about certain file names and that they were in the working directory, could do pretty much anything with those files, including having the AI agent clean up files no longer needed, etc. • **Complex Geospatial Tasks**: One modeler used Beaker to help with a complex task (unrelated to the evaluation), on embedding lines from a shapefile onto a geospatial tiff with a given resolution and boundaries. Beaker was able to do this fairly easily where other common SOTA tools (ArcGIS) were not able to (or it was difficult to know how exactly to do it in those specialized programs). • **Beaker as Onboarding Tool**: Beaker was helpful for onboarding with Decapodes, and becoming acquainted with the documentation. However, this became less useful once the modelers delved deeper into the tech stack and questions.

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4.3 Forecasting Challenge Results

The forecasting challenge compared the results from the ASKEM performers to the results from the FluSight forecasters, which collectively represent the non-ASKEM Baseline capabilities. The FluSight forecast hub model output page notes that FluSight forecasters were asked to make forecasts for horizons ranging from 0 weeks ahead (a nowcast for the week of submission) to 3 weeks ahead (which is essentially 4 weeks ahead.) Since the ASKEM performers labeled their forecasts as being 1 to 4 weeks ahead, the T&E team aligned the forecasts between the ASKEM performers and FluSight forecasters, comparing the ASKEM 1 week ahead forecasts, to the FluSight 0 week ahead forecasts. All results below include this shift.

The ASKEM performer's single model forecast and the ensemble model forecast performed relatively well for both Pennsylvania and Florida forecasts.

Figure 37 presents the weighted interval scores (WIS) for the state of Florida for all FluSight forecasters (dotted lines) and the performer (bolded tan lines.) As mentioned earlier in Section 3.2.5, a lower WIS score indicates better performance. Both the single model performer forecast and the ensemble model performer forecast start out with a low WIS and decrease as time goes on. For the last two timepoints of the forecast period, their forecasts had the *lowest WIS scores of all forecasts*, indicating that their forecasts were generally accurate and contained narrow uncertainty intervals.

In Figure 38 we also plot the median forecast of all forecasters versus the actual hospitalization numbers for Florida (ground truth is bold purple line). We see that the ground truth hospitalizations are closer to the performer's ensemble model for the first two weeks of the forecast period, but follow the performer's single model forecast very closely for the last two weeks. Other FluSight forecasts were generally underestimating the number of hospitalizations, and also not trending in an upward direction to match with the ground truth data. Thus, ***both of the ASKEM performer forecasts outperformed all Baseline forecasts for the last two timepoints of the forecast period, for the state of Florida***.

Figure 39 displays the WIS scores for the state of Pennsylvania for all the FluSight forecasters (dotted lines) as well as the ASKEM performer (bold blue lines). The ASKEM performer forecasts compare well to the forecasts from the FluSight teams. ***For the middle two timepoints of the forecast period, the performer's ensemble forecast outperforms all the other Baseline forecasts as measured by WIS***; however, the performer's single model forecast doesn't perform as well, falling right in middle of the neighborhood of scores from the FluSight Baseline forecasters. This instance supports the general claim that ensemble forecasts perform better than single model forecasts.

In Figure 40 we plot the median forecast of all forecasters versus the actual hospitalization numbers for Pennsylvania. We see that the ASKEM performer's single model forecast underestimated the true incident hospitalizations for all four weeks of the forecast timeframe. The performer's ensemble forecast, however, had two forecasts timepoints which were close to the ground truth, for December 23rd and December 30th (tracking with the low WIS scores for these middle two timepoints of the forecast period), before significantly overestimating the actual hospitalizations on January 6th.

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While previous forecasting challenges results in performer forecasts being comparable to the general neighborhood of Baseline forecast scores, with this challenge, there are clearly several timepoints and circumstances where one or more of the ASKEM performer forecasts outperform all other Baseline forecasts, which is a meaningful achievement.

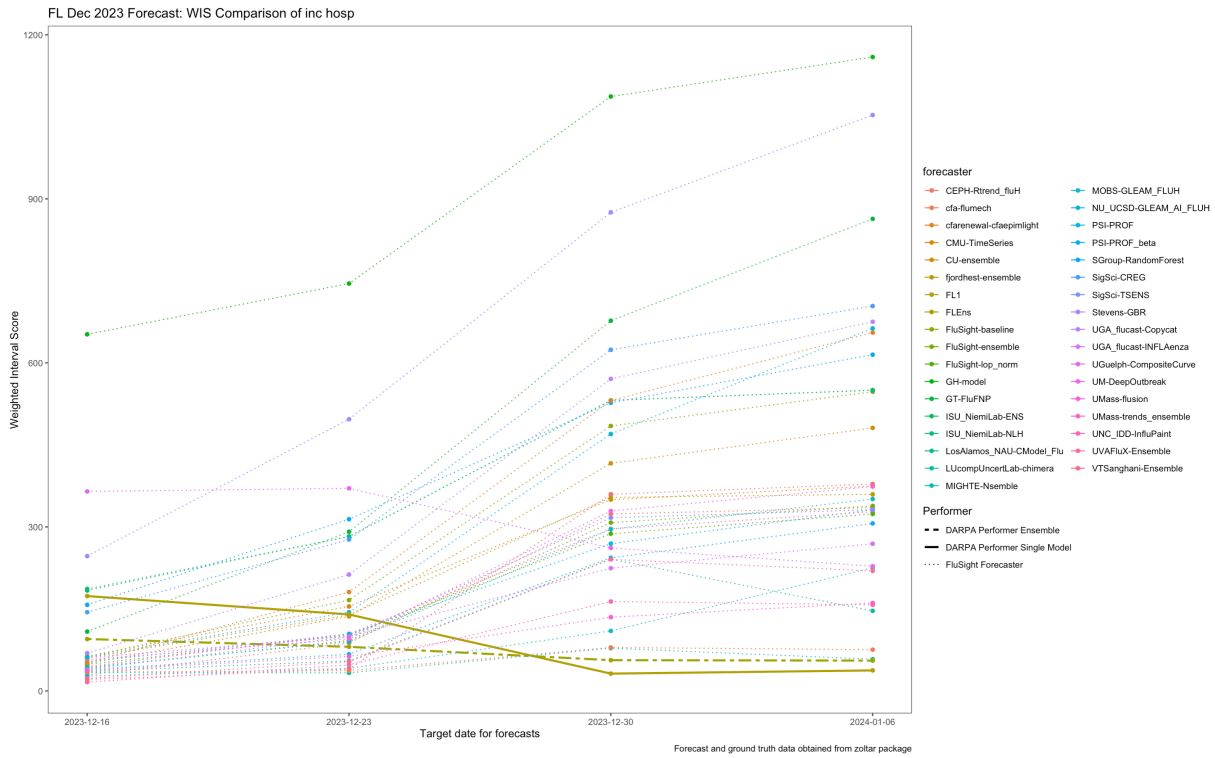


Figure 37. WIS scores for ASKEM performer (bold tan lines) and FluSight forecasts (thin dotted lines in various colors), for the state of Florida

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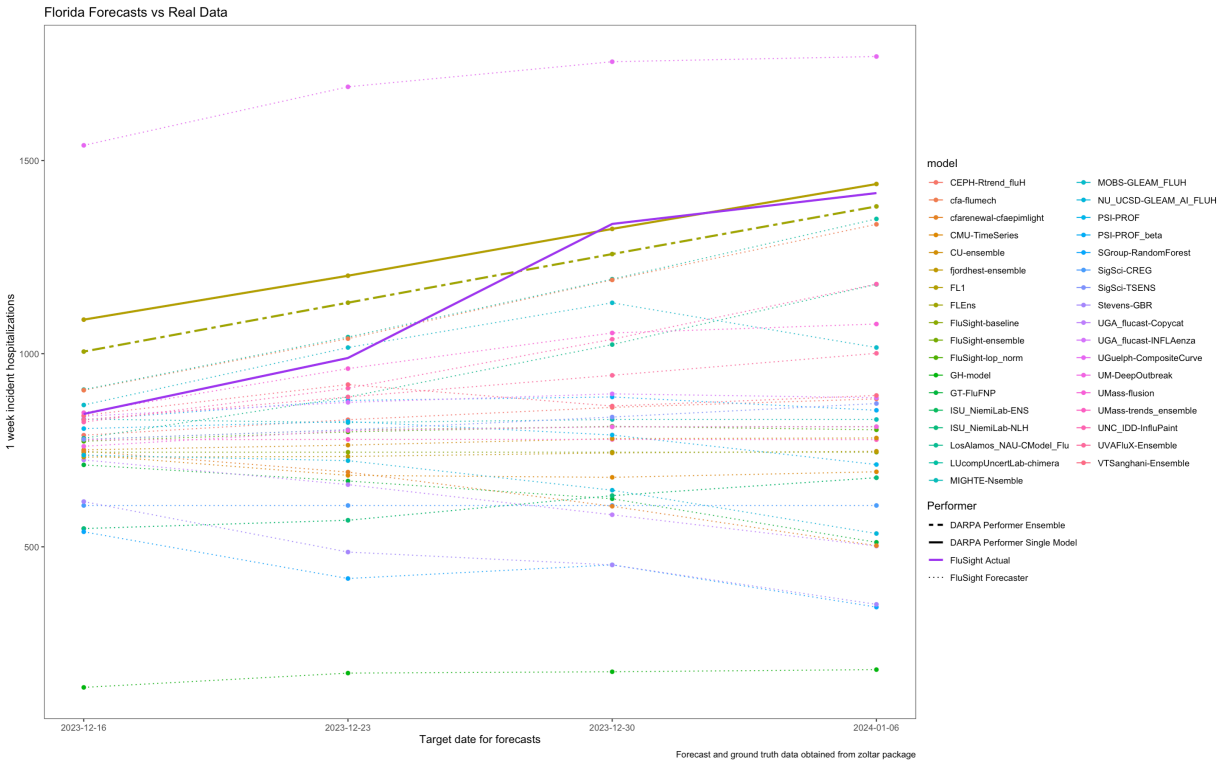


Figure 38. Florida Forecasts vs Ground Truth Data (bold purple line), for ASKEM performer team (bold tan lines) and FluSight forecasters (thin dotted lines in various colors)

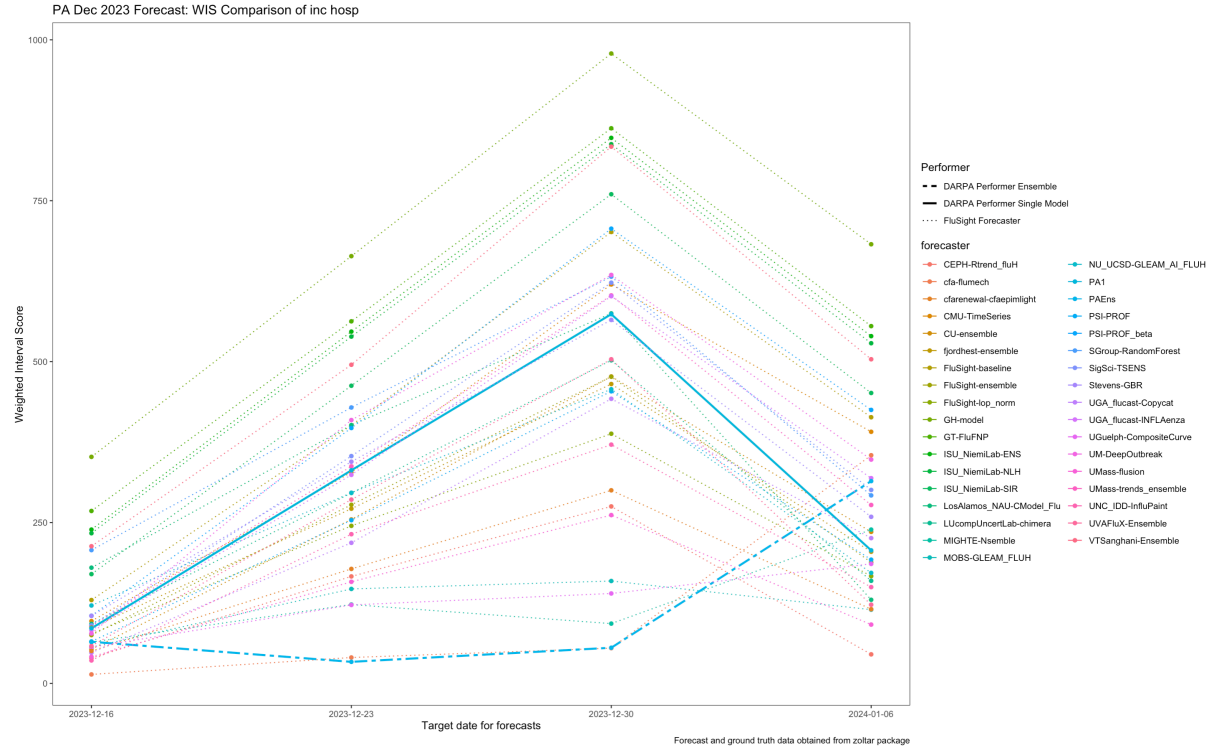


Figure 39. WIS scores for ASKEM performer (bold blue lines) and FluSight forecasts (thin dotted lines in various colors), for the state of Pennsylvania

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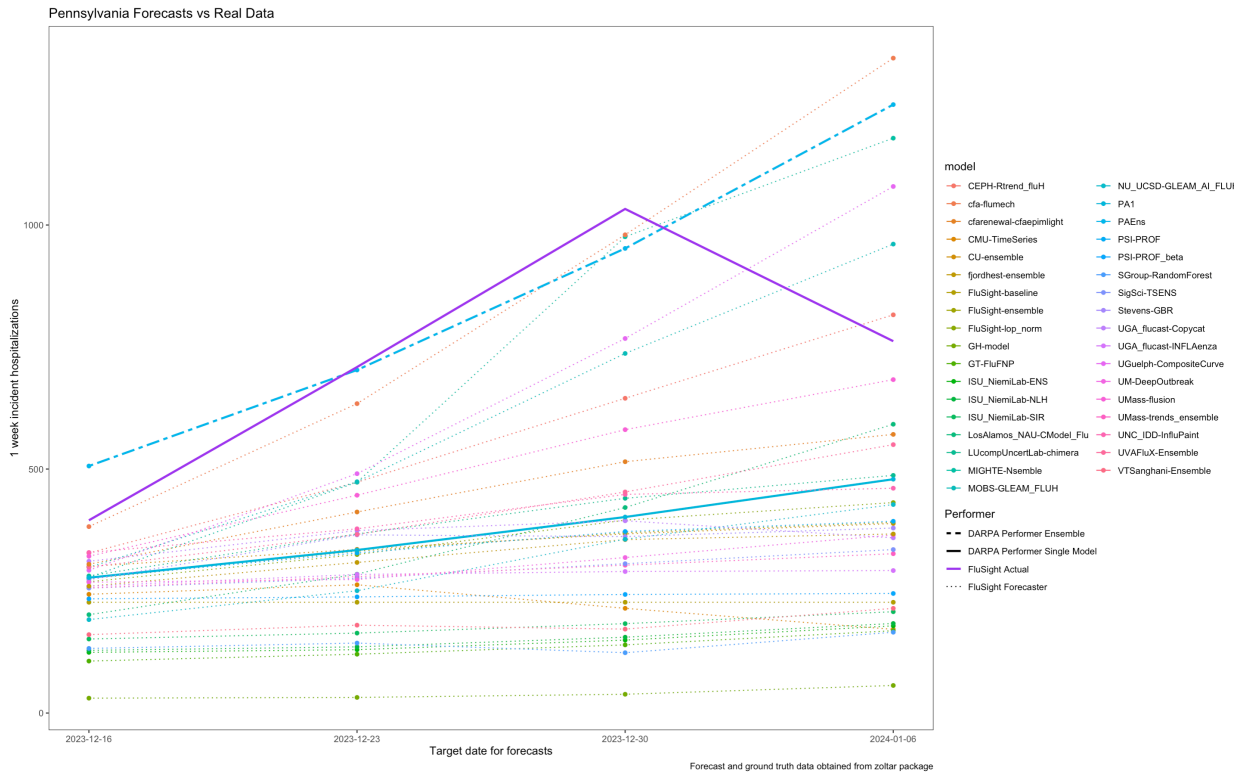


Figure 40. Pennsylvania Forecasts vs Ground Truth Data (bold purple line), for ASKEM performer team (bold blue lines) and FluSight forecasters (thin dotted lines in various colors).

4.4 Lessons Learned on Evaluation Structure and Design

Here we note several lessons learned regarding the February 2025 evaluation structure and design. Although this was the final evaluation for the ASKEM program, we note aspects that worked well and we would want to replicate for future T&E events for other programs with similar types of evaluation structures, and aspects that would need to be modified or could be expanded.

- **Positive:** In-person, teamed approach was successful. Several teams worked very well together with complementary skills from one modeling expert and one subject-matter expert.
- **Positive:** Having an on-site team of performers to support and help with debugging was critical. Modelers found the performers to be responsive to their questions and needs and appreciated having people available to support them in using novel research tools.
- **Positive**: For tasks that do not focus on data search, have data preprocessed, accessible, and ready to use, with clear descriptions to avoid confusion in interpretation.
- **Positive**: There were comparable 'tool learning curves'. For the climate use case, both Baseline and Workbench modeling teams were not familiar with the tool sets, and this ended up providing a fair comparison, as each team had a little bit of a 'tool learning curve' to work through.
- **Modify:** For the Baseline teams, incorporate more opportunities for going through hands-on exercises together as a group during pre-evaluation training, as opposed to expecting

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individuals to study on their own. This would provide a more comparable experience to the Workbench teams, who went through several training sessions together.

- **Modify:** Some of the Baseline teams chose not to use state-of-the-art AI models for their tasks as they were not part of their normal workflow or state-of-practice. At the time of the evaluation, this was reasonable as the latest models were not efficient at many different types of modeling tasks. However, since the evaluation, commercial and open-source AI models have progressed significantly such that they are much more integrated into the state-of-practice for many disciplines; therefore future evaluations of new AI technology should consider requiring the use of state-of-the-art AI models by Baseline teams (or recruiting users who are familiar enough with them to use them in their daily work).
- **Modify/Expand**: Consider designing a scenario around models and analyses that the T&E team know can be completed with performer tools, and see whether state-of-the-art baseline tools can achieve the same functionality.

5 Features and Capabilities Not Evaluated

Here we note features and capabilities that were in development around the time of the evaluation, or brought up during post-eval debrief sessions, but were not formally evaluated during the February 2025 evaluation for various reasons.

5.1 Epidemiology Use Case

With the Modeling with an Intelligent Research Assistant (MIRA) library, the Northeastern team has been working on an 'automatic stratification detection' ability to recognize when a model is stratified but the stratification had not been done using the Terarium stratification operator, and can re-represent the model as if it had been stratified with the operator, to make model diagrams and other representations in Terarium more streamlined. They have also implemented a capability for automated simplification of models, which allows further processing with the model more efficient. Additionally, MIRA's LLM-based approach for automatically grounding models was not exercised during the evaluation.

The Smart Information Flow Technologies (SIFT) team's Functional Model and Analysis (FUNMAN) tools have been integrated into the Terarium workbench and are used for model checks and parameter exploration. Although we encouraged modelers to use model checks in the Epi Decision-maker Scenario, they did not end up using this functionality. Model checks were also not explicitly asked about during the Epi Independent Questions scenario. Therefore, FUNMAN was not explicitly evaluated during this final program evaluation, although there are some metrics documented with post-evaluation activities.

Over the course of the ASKEM program, the Causal Inference Engine for Multi-scale Simulations (CIEMSS) team developed the ChiRho library for causal reasoning. This library provides functionality to ask and answer causal questions around interventions, counterfactuals, explaining causal impact, mediation analysis, and more. The ChiRho library also includes functionality for building hierarchical Bayesian models with dynamical systems, which allows users to build multilevel Bayesian SIR-type models. Although some of these features were

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exercised in some capacity in notebook format previously, these capabilities were not integrated into the Terarium workbench and therefore not evaluated during this final program evaluation.

Several enhancements and features have also been added to the Terarium workbench since the conclusion of the final evaluation period:

- The workbench now includes enhancements to the model ingestion process, such as more advanced model checks for transition validity and automatically generated information about the parameters, transitions, and state variables to include in the model metadata.
- The workbench now includes the ability to produce sensitivity analyses as a check-boxed option within every simulation. These analyses offer a few plots showing how sensitive model variables are to a parameter of choice.
- The workbench also includes additional scenario templates that were not assessed by the T&E team during the final evaluation. These include templates named "Decision making", "Calibrate an ensemble model", "Value of information", "Reproduce models from literature", "Necessary or sufficient analysis", and "Policy design."
- And finally, within the context of scenario configurations and parameter space validation, the performers have improved the insights that can be gained when trying to assess what parameter ranges to use within an optimization context.

Additionally, see Appendix B: Performer-Specific Metrics for more information on specific performer capabilities assessed and documented post-evaluation. These include the SIFT team's parameter synthesis scale-up for FUNMAN and exploration of optimization questions to understand the impact of FUNMAN. Material also includes plots of the speedup in CIEMSS simulation times with and without Julia backend solvers, reflecting an integration win for both CIEMSS and JuliaHub.

5.2 Climate Use Case

Shortly before the evaluation, the University of Florida team finished developing multigrid capabilities with Decapodes. Multigrid methods increase computation speed by using a hierarchy of discretization, solving equations on a coarse grid except where finer resolution is required. This was a new, relatively untested feature at the time of the evaluation and was not exercised by the Workbench modelers.

Post-evaluation, the University of Florida team also demonstrated that due to integration of Decapodes with Julia's auto-differentiation capabilities, they could now take advantage of Julia's implicit solvers (something that was not possible earlier in the program, and was not exercised during the evaluation). In their demonstration, they showed that a Brusselator Decapodes model (used to model autocatalytic reactions with two different chemicals) with a stiff model configuration using an implicit Julia solver was faster and more memory efficient (with shorter execution runtime, significantly smaller memory usage due to sparse Jacobians, significantly fewer function evaluations, and significantly fewer overall timesteps), compared to an explicit solver. This reflects an integration win for both University of Florida and JuliaHub teams.

In Phase 2, the JuliaHub team developed sparse auto-differentiation capabilities that speed up execution time compared to non-sparse approaches; this speedup scales very well with simulation mesh density. Sparse auto-differentiation was not quite ready at the time of the

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evaluation, but post-evaluation, the JuliaHub team completed this development and documented their findings; more details can be found in Appendix B: Performer-Specific Metrics.

One of the features of Beaker is the search and retrieval of datasets through ESGF's API. However, due to changes in the ESGF API right before the evaluation, the T&E team was only able to evaluate the data search component (and not retrieval) with Beaker. The Workbench team instead had to take the dataset name given by Beaker and manually download the data from ESGF's website.

Another aspect of Beaker that was not evaluated was the ability to stand up new custom contexts. A Beaker context consists of the extra information provided to Beaker on top of the LLM itself. For example, for creating an ESGF-relevant context, Beaker was provided documentation on how to use the ESGF API, including examples of successful API calls. This allows Beaker to more successfully perform specialized tasks than if it relied solely on the LLM agent. The Beaker contexts used during the evaluation had been created beforehand, not by the modeling teams. For standing up custom contexts, it is difficult to imagine what an equivalent task would be for a Baseline team. Therefore, we did not design scenarios around this feature of Beaker.

6 Potential for Future Impact

6.1 Epidemiology Use Case

The Terarium workbench was built with epidemiology in mind and has been demoed to a number of stakeholders in the epidemiology and public health spaces (e.g. the Centers for Disease Control and Prevention's InsightNet and its academic and public government partners including the University of California, San Diego's Resilient Shield team, and the California Department of Public Health). Since the February 2025 evaluation, the workbench has been used to show how interventions could reduce the health care costs of diabetes patients in Canada, demonstrating the ability to extend disease modeling to include economic cost components. Additionally, Terarium has been used to model critical mineral supply chain risks, and develop forecasts for the West Nile Virus in collaboration with the California Department of Public Health, further demonstrating its generalizability beyond just respiratory disease modeling, which has been the main focus of the epidemiology use case for almost the entirety of the ASKEM program. In addition, there are other domains where its modeling and simulation capabilities have the potential to be applied. For example, population growth, ecology and predator-prey systems, game theory models, migration flows, some biological signaling systems, and economic models all have the potential to be represented by ODEs and modeled in the workbench. The wide array of potential use cases leads to a large set of possible transition partners, ranging from organizations interested in disease spread, like the Centers for Disease Control and Prevention (CDC), the National Institute for Allergy and Infectious Diseases (NIAID) at National Institutes of Health (NIH), and state and local public health departments, to government entities that monitor supply chains.

There have also been several adaptations of ASKEM technology outside the workbench. For example, the causal identification algorithms in y0 (supported in part by ASKEM funding) have been extended to hierarchical models, allowing for the identification of causal queries over hierarchical models specified in ChiRho. Work done on ChiRho has also been used in 1) the US

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Department of Energy's Biosystems Design program to simulate the dynamics of gene regulatory networks, 2) in a project internally funded by the PNNL Predictive Phenomics Initiative to simulate the dynamics of microbial interactions, and 3) in the Defense Threat Reduction Agency (DTRA)-funded RAPTER program, to apply counterfactual transportability algorithms to estimate vaccine efficacy across different hosts, pathogens, and vaccine platforms.

6.2 Climate Use Case

As was discussed in the Decision-maker Panel Outcomes section for the Climate DM experiment, Decapodes has the potential to demonstrate advantage over state-of-the-art baseline software tools for developing and implementing new physics models and equations through its multiphysics model coupling capabilities and ability to easily apply complex models to arbitrary geometries. Post-evaluation, the University of Florida is transitioning and expanding Decapodes' capabilities for space weather, magnetohydrodynamics (MHD), and the chemistry and physics of lightning with transition partner US Army Combat Capabilities Development Command Army Research Laboratory (DEVCOM ARL). In addition, the University of Florida team will continue to build upon the theoretical developments made during ASKEM in the areas of category theory and with the Algebraic Julia software ecosystem, through the establishment of the multi-institute Designing Integrated, Generic, and Interoperable Compositional Analyses of Modern Systems (DIGICAMS) center, in collaboration with Georgia Tech, Duke University, Topos Institute, and transition partner Air Force Office of Scientific Research (AFOSR)/Air Force Research Laboratory (AFRL). DIGICAMs will develop unified mathematical frameworks for decision-making processes across optimization, dynamical systems, and control.

Throughout the ASKEM program, JuliaHub has partnered with various academic organizations and commercial companies, in a variety of domains. Particularly for climate, it has collaborated with the research team behind the ODINN glacier evolution model. And as of February 2025, JuliaHub has become a corporate partner of the National Science Foundation (NSF) Colorado-Wyoming (CO-WY) Climate Resilience Engine. JuliaHub will support the CO-WY Engine's R&D projects and ecosystem, through its technical expertise in scientific modeling and simulation, high-performance computing, scientific machine learning, and digital twin technologies, with focus areas on predicting and preparing for the impacts of wildfire and drought.

Beaker has also been used to demonstrate impact beyond the climate use case. The transition value in Beaker lies in the fact that a custom context can be created quickly for any use case, making the technology stack generalizable. This has already been demonstrated in the wildfire, critical minerals, and geospatial domains, with demos developed for the final PI meeting and other post-evaluation stakeholder discussions. At the March 2025 ASKEM PI Meeting the NSF CO-WY Engine expressed interest and alignment with Beaker's capabilities relating to wildfire data analysis. Using Beaker code notebooks, it is also possible to create other analytic products, such as bespoke, data-driven web dashboards, as was shown with the wildfire demo at the PI meeting; this type of workflow demonstrates how AI tools can assist in quickly gathering data, developing analyses with the data, and visualizing results in a custom-built interface, to support stakeholders and decision makers. The US Air Force 16th Weather Squadron is another possible transition partner who has expressed interest in leveraging Beaker for their analytic and computational workflows.

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Appendix A: Metrics Tables

Appendix A contains full Technical Area (TA)-level metrics tables for both the epidemiology and climate use cases, and includes definitions, targets, and evaluation outcomes.

Table 27. TA-level program metric definitions, targets, and outcomes for the epidemiology use case. Please note that TA1 performers were not part of Phase 2 of the program; at the time of the final evaluation, almost all of the TA1 model extraction technology stack was updated, extended, or replaced by TA4 capabilities (with contributions from TA2 performers). Targets are indicated as fully met (✅), partially met (◪), or not met (❌).

Goals	Details	TA1	TA2	TA3	TA4
Accuracy	Definition	N/A for this TA	A1: accuracy of single model	A2: multi-model ensemble accuracy	N/A for this TA
	Targets (Phase 1, Phase 2)		Specific Targets N/A. Show improvement compared to Baseline	Specific Targets N/A. Show improvement compared to Baseline	
	Evaluation		• Accuracy results were obtained from the Flu Forecast Challenge done in March 2025, and are quantified in terms of the weighted interval score (WIS), commonly used in many forecast challenges. • For the state of Florida, the ASKEM performer's single model forecast compares very well with all FluSight forecasters (representing the Baseline) – ***it had the lowest WIS score for the last two timepoints of the forecast period***, and performed well for the other two timepoints (falling in the middle of the general neighborhood of scores among all other forecasters). The forecast tracks very well with the ground truth trajectory, matching it in terms of general upward trend, where most of the FluSight forecasters did not capture this trend. • For the state of Pennsylvania, the ASKEM performer's single model forecast falls in the middle of the neighborhood of scores among all other forecasters.	• Accuracy results were obtained from the Flu Forecast Challenge done in March 2025, and are quantified in terms of the weighted interval score (WIS), commonly used in many forecast challenges. • For the state of Florida, the ASKEM performer's ensemble model (created from 3 component models) forecast compares very well with all FluSight forecasters (representing the Baseline) – ***it had the second-lowest WIS score for the last two timepoints (only outperformed by its own single-model forecast)***, and performed well for the other two timepoints (falling in the middle of the general neighborhood of scores among all other forecasters, and outperforming its own single model forecast). • The ensemble forecast tracks very well with the ground truth trajectory, matching it in terms of general upward trend, where most of the FluSight forecasters did not capture this trend.	

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Goals	Details	TA1	TA2	TA3	TA4
			• For more details and plots, see the Forecasting Challenge Results ✅**Target Met** (ASKEM forecast demonstrated improvement over Baseline forecasts)	• For the state of Pennsylvania, the ***ASKEM performer's ensemble model forecast had the lowest WIS score for the middle two timepoints*** and is in the middle of the neighborhood for the other two timepoints. In terms of WIS score, it outperforms the single model forecast for this state for all timepoints except the last. The ensemble forecast trajectory matches the ground truth trajectory quite well for the first three timepoints of the forecast period, but diverges for the last timepoint because the ground truth data trend changes direction. • For more details and plots, see the Forecasting Challenge Results ✅**Target Met** (ASKEM forecast demonstrated improvement over Baseline forecasts)	

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<table>
<tr><th>Goals</th><th>Details</th><th>TA1</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr><td rowspan="2">Timeliness</td><td>Definition</td><td>T1: Time for accurate model extraction</td><td>T2: Time to extend/modify/compose model</td><td>T3: Time to create simulators for validation and prediction</td><td>Although this is not part of the official program TA metrics, TA4 functionality for this use case and evaluation included:
High-level (non-structural) model comparison
Complex visualization
Scenario templates</td></tr>
<tr><td>Targets (Phase 1, Phase 2)</td><td>Targets derived from modeler survey, based on operational needs of decision makers.
<table><tr><th>TA Task</th><th>Phase 1 Target</th><th>Phase 2 Target (50% reduction)</th></tr><tr><td>TA1.1 (Find Relevant Models)</td><td>2 days</td><td>1 day</td></tr><tr><td>TA1.2 (Reproduce Findings)</td><td>2 days</td><td>1 day</td></tr><tr><td>TA1.3 (Find Parameter Values)</td><td>2 days</td><td>1 day</td></tr><tr><td>TA1.4 (Find Data)</td><td>1.5 days (12 hrs)</td><td>6 hrs</td></tr></table></td><td>Targets derived from modeler survey, based on operational needs of decision makers.
<table><tr><th>TA Task</th><th>Phase 1 Target</th><th>Phase 2 Target (50% reduction)</th></tr><tr><td>TA2.1 (Stratification)</td><td>5 days</td><td>2.5 days</td></tr><tr><td>TA2.2 (Add/Remove Compartments and Transition Pathways)</td><td>1 day</td><td>4 hrs</td></tr><tr><td>TA2.3 (Incorporate New Policy Interventions)</td><td>5 days</td><td>2.5 days</td></tr><tr><td>TA2.4 (Compose Multiple Models)</td><td>2 days</td><td>1 day</td></tr></table></td><td>Targets derived from modeler survey, based on operational needs of decision makers.
<table><tr><th>TA Task</th><th>Phase 1 Target</th><th>Phase 2 Target (50% reduction)</th></tr><tr><td>TA3.1 (Calibration)</td><td>6 hours</td><td>3 hrs</td></tr><tr><td>TA3.2 (Quantifying Uncertainty)</td><td>6 hours</td><td>3 hrs</td></tr><tr><td>TA3.3 (Forecasting)</td><td>3 hours</td><td>1.5 hrs</td></tr><tr><td>TA3.4 (Optimization)</td><td>3 days</td><td>1.5 days (12 hrs)</td></tr><tr><td>TA3.5 (Interventions)</td><td>7 days</td><td>3.5 days</td></tr><tr><td>TA3.6 (Sensitivity Analysis)</td><td>1.5 days (12 hrs)</td><td>6 hrs</td></tr><tr><td>TA3.7 (Ensembles</td><td>1 day</td><td>4 hrs</td></tr></table></td><td>N/A</td></tr>
</table>

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	Evaluation	For this evaluation, the only functionality aligning with TA1 was model extraction coupled with simple unit tests. (This was ***Task TA1.2 ,'reproducing findings' from a paper.***) See Epi IQ *Section 1 (TA1)* for plots and discussion on TA1 outcomes. **Summary**: • **TA1.2 (Reproduce Findings)**: As all attempted questions were completed in less than 4 hrs, both Baseline and Workbench teams *exceeded Phase 2 timing target* (1 day). The largest speedup of the Workbench over the Baseline for this task was 4.15x with equivalent outcomes. • **TA1.1 (Find Relevant Models), TA1.3 (Find Parameter Values), and TA1.4 (Find Data)**: These tasks were not explicitly exercised in this evaluation due the fact that TA1 performers were not part of Phase 2. • For a couple questions, the Workbench team took longer but achieved better outcomes. • For one question (hard Covid model extraction), the Workbench team took significantly less time with same level of quality/ completeness in outcome. • Workbench team was able to complete one more question than the Baseline team	See Epi IQ *Section 2 (TA2)* for plots and discussion on TA2 outcomes. **Summary:** • **Task TA2.1 (Stratification)**: As all attempted questions were completed in less than 5 hrs, both Baseline and Workbench teams *exceeded Phase 2 timing target* (2.5 days). • **Task TA2.2 (Add/Remove Compartments and Transition Pathways or Model Edits)**: As all questions were completed in less than 2 hrs, both Baseline and Workbench teams *met Phase 2 timing target* (4 hrs). Both teams achieved perfect scores for all 3 questions, with Workbench speedups for 2 of them (max 1.56x speedup with comparable outcomes). • **Task TA2.3 (Incorporate New Policy Interventions)**: Modifying models to incorporate new interventions was part of the Epi Decision-maker Scenario but this task was not designed be carefully timed or compared between Workbench and Baseline. • **Task TA2.4 (Compose Multiple Models)** was not exercised for the epidemiology use case. • For a couple questions, the Workbench team took longer but achieved better outcomes • **Structural Model Comparison**: While was not a formal task with defined metrics targets, the Workbench had speedups of a factor of 5-9x over the Baseline, depending on the question.	See Epi IQ *Section 3 (TA3)* for plots and discussion on TA3 outcomes. **Summary:** • **Task TA3.1 (Calibration)**: As all attempted questions were completed in less than 2 hrs, both Baseline and Workbench teams *met Phase 2 timing target* (3 hrs). Max Workbench speedup is 9.7x over the Baseline, with better Workbench outcomes. • **Task TA3.2 (Quantifying Uncertainty)**: This was an aspect of the Epi Decision-maker Scenario but this task was not designed to be carefully timed or compared between Workbench and Baseline. • **Task TA3.3 (Forecasting)**: As all attempted questions were completed in less than 90 mins, both Baseline and Workbench teams *met Phase 2 timing target* (90 min). For all questions, the Workbench demonstrated speedups with better or similar outcomes, compared to the Baseline (max speedup was 5.7x. with better outcomes for Workbench) • **Task TA3.4 (Optimization)**: As all attempted questions were completed in less than 3 hrs, both Baseline and Workbench teams *exceeded Phase 2 timing target* (12 hrs). For all questions, the Workbench demonstrated speedups, with comparable outcomes (one exception being the hard optimization problem). Max speedup was 3.5x with equal outcomes. • **Task TA3.5 (Interventions)**: As all questions were completed in less than 2 hrs, both Baseline and Workbench	See Epi IQ *Section 4 (TA4)* for plots and discussion on TA4 outcomes. **Summary:** • **Model Comparison**: Workbench team completed all questions, Baseline only completed one. Workbench speedup was 1.8x. • **Viz . and Data Comparison**: Workbench achieved max 2.9x speedup over Baseline. • **Scenario Templates**: Workbench completed all questions, Baseline only completed 2/3. • **All questions:** Workbench team finished all questions, Baseline team finished 6/9 questions • Both teams achieved perfect scores for almost all questions they completed • For questions both teams completed, Workbench spent less or roughly equal time

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- The Workbench team did better with the hard extraction problems, compared to the Baseline team
- In general, time for extraction increased with problem complexity (with one exception, the hard Covid model extraction)

***B** indicates the Baseline,*
***W** indicates Workbench*

Task	Phase 2 Target	Outcome
TA1.1	1 day	N/A for this eval
TA1.2 (Rep. Findings)	1 day	✅**Target Met (W,B)** W max 4.15x speedup
TA1.3	1 day	N/A for this eval
TA1.4	6 hrs	N/A for this eval

***B** indicates the Baseline,*
***W** indicates Workbench*

Task	Phase 2 Target	Outcome
TA2.1 (Strat.)	2.5 days	✅**Target Met (W,B)**
TA2.2 (Edits)	4 hrs	✅**Target Met (W,B)** W max 1.56x speedup
TA2.3 (Edits for Policy Interv.)	2.5 days	Questions not designed to support timing metric for this evaluation
TA2.4	1 day	N/A for this eval

teams *exceeded Phase 2 timing target* (3.5 days). The Workbench was able to complete all questions, and the Baseline was only able to complete 2/3. The largest Workbench speedup over the Baseline was 12.7x with comparable outcomes.

- **Task TA3.6 (Sensitivity Analysis)**: This functionality was met with a combination of TA3 simulation tools and TA4's Sensitivity Analysis scenario template. Both Workbench and Baseline completed the question in less than 2 hrs, and therefore *met the Phase 2 target* (6 hrs). Both teams spent equal amount of time on the question.
- **Task TA3.7 (Ensembles)**: The Workbench team *met Phase 2 timing target* (4 hrs), while the Baseline did not meet this target for one of the questions. The largest Workbench speedup over the Baseline was 3.9x with comparable outcomes.
- The Workbench team completed all questions, and the Baseline team completed 12/14.

***B** indicates the Baseline,*
***W** indicates Workbench*

Task	Phase 2 Target	Outcomes
TA3.1 (Calib.)	3 hrs	✅**Target Met (W,B)** Workbench max 9.7x speedup
TA3.2 (Quant. Uncert.)	3 hrs	Questions not designed to support timing metric for this evaluation

***B** indicates the Baseline,*
***W** indicates Workbench*

Task	Outcomes
Model Comp.	W has 1.8x speedup over B
Viz. and Data Comparison	W has 2.9x speedup over B
Scenario Templates	W completed more questions

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				TA3.3 (Forecast.)	1.5 hrs	✅**Target Met (W,B)** Workbench max 5.7x speedup
				TA3.4 (Opt.)	1.5 days (12 hrs)	✅**Target Met (W,B)** W max 3.5x speedup
				TA3.5 (Interv.)	3.5 days	✅**Target Met (W,B)** W max 12.7x speedup
				TA3.6 (Sensitivity Analysis)	6 hrs	✅**Target Met (W,B)** No meaningful Workbench speedup
				TA3.7 (Ensembles	4 hrs	✅**Target Met by Workbench** ❌**Target Not Met by Baseline for 1 question** W max 3.9x speedup

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<table>
<tr><th>Goals</th><th>Details</th><th>TA1</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr><td rowspan="3">Decision Making Confidence</td><td>Definition</td><td>Confidence in finding and selecting an appropriate starting point for the scenario/problem</td><td>Confidence in developing an appropriate model and associated parameter space to sufficiently explore the scenario/problem</td><td>Confidence in understanding model results and tradeoffs between potential interventions</td><td rowspan="3">N/A for this TA</td></tr>
<tr><td>Targets (Phase 1, Phase 2)</td><td>Confidence scores on 7-point scale
1 Very Low
2 Low
3 Somewhat Low
4 Neutral
5 Somewhat High
6 High
7 Very High

Target – Average Score: (4.8, 5.2)
Target – Proportion of Low Scores (4 or below): (25%, 10%)
<table><tr><th>Category</th><th>Phase 1 Target</th><th>Phase 2 Target</th></tr><tr><td>Average Score</td><td>4.8</td><td>5.2 (~10% increase)</td></tr><tr><td>Proportion of Low Scores</td><td>25%</td><td>10% (60% reduction)</td></tr></table></td><td>Confidence scores on 7-point scale
1 Very Low
2 Low
3 Somewhat Low
4 Neutral
5 Somewhat High
6 High
7 Very High

Target – Average Score: (5.3, 5.7)
Target – Proportion of Low Scores (4 or below): (22%, 9%)
<table><tr><th>Category</th><th>Phase 1 Target</th><th>Phase 2 Target</th></tr><tr><td>Average Score</td><td>5.3</td><td>5.7 (~10% increase)</td></tr><tr><td>Proportion of Low Scores</td><td>22%</td><td>9% (60% reduction)</td></tr></table></td><td>Confidence scores on 7-point scale
1 Very Low
2 Low
3 Somewhat Low
4 Neutral
5 Somewhat High
6 High
7 Very High

Target – Average Score: (4.4, 4.8)
Target – Proportion of Low Scores (4 or below): (30%, 12%)
<table><tr><th>Category</th><th>Phase 1 Target</th><th>Phase 2 Target</th></tr><tr><td>Average Score</td><td>4.4</td><td>4.8 (~10% increase)</td></tr><tr><td>Proportion of Low Scores</td><td>30%</td><td>12% (60% reduction)</td></tr></table></td></tr>
<tr><td>Explana-tion</td><td>Decision maker evaluates whether material includes an appropriate base model, with relevant parameters, starting values, etc.</td><td>The scenario involves updating or modifying the model, and decision makers will evaluate whether this was done in a sensible way and whether final model can support all the questions asked in the scenario.</td><td>Confidence in being able to assess effectiveness of all interventions considered in the scenario and understand how uncertainty factors into model results and their implications.</td></tr>
</table>

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<table>
<tr><th>Goals</th><th>Details</th><th>TA1</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr><td></td><td>Evaluation</td><td>
See Epidemiology Decision-maker Scenario (Epi DM) section for more detailed discussion on both team's approaches, example outputs, and decision-maker panel outcomes related to Part 1 of the scenario (aligned with TA1 functionality).

Decision-maker confidence scores for scenario components relevant to TA1, on 7-point scale.
• B indicates the decision-maker score for the Baseline
• W indicates the decision-maker score for the Workbench
<table>
<tr><th>Panelist</th><th>B</th><th>W</th></tr>
<tr><td>#1</td><td>Somewhat Low (3)</td><td>Somewhat High (5)</td></tr>
<tr><td>#2</td><td>Neutral (4)</td><td>Neutral/ Somewhat High (4.5)</td></tr>
<tr><td>#3</td><td>High (6)</td><td>High (6)</td></tr>
<tr><td>#4</td><td>Somewhat High (5)</td><td>Somewhat High (5)</td></tr>
<tr><td>#5</td><td>Somewhat Low (3)</td><td>Somewhat High (5)</td></tr>
<tr><td>Average Score</td><td>4.2</td><td>5.1</td></tr>
<tr><td>Proportion of Low Scores (4 or below)</td><td>20%</td><td>0%</td></tr>
</table>
Alternative baseline from decision-maker surveys, on TA1-aligned tasks:
Average Score: 4.36
Proportion of Low Scores: 50%
</td><td>
See Epidemiology Decision-maker Scenario (Epi DM) section for more detailed discussion on both team's approaches, example outputs, and decision-maker panel outcomes related to Part 2 of the scenario (aligned with TA2 functionality).

Decision-maker confidence scores for scenario components relevant to TA2, on 7-point scale.
• B indicates the decision-maker score for the Baseline
• W indicates the decision-maker score for the Workbench
<table>
<tr><th>Panelist</th><th>B</th><th>W</th></tr>
<tr><td>#1</td><td>Somewhat Low (3)</td><td>Neutral/ Somewhat High (4.5)</td></tr>
<tr><td>#2</td><td>Somewhat High (5)</td><td>Somewhat Low/ Neutral (3.5)</td></tr>
<tr><td>#3</td><td>Somewhat High (5)</td><td>Somewhat High (5)</td></tr>
<tr><td>#4</td><td>Somewhat High/ High (5.5)</td><td>Neutral/ Somewhat High (4.5)</td></tr>
<tr><td>#5</td><td>Somewhat High (5)</td><td>Somewhat High (5)</td></tr>
<tr><td>Average Score</td><td>4.7</td><td>4.5</td></tr>
<tr><td>Proportion of Low Scores (4 or below)</td><td>20%</td><td>20%</td></tr>
</table>
Alternative baseline from decision-maker surveys, on TA2-aligned tasks:
Average Score: 4.79
Proportion of Low Scores: 43%
</td><td>
See Epidemiology Decision-maker Scenario (Epi DM) section for more detailed discussion on both team's approaches, example outputs, and decision-maker panel outcomes related to Part 3 of the scenario (aligned with TA3 functionality).

Decision-maker confidence scores for scenario components relevant to TA3, on 7-point scale.
• B indicates the decision-maker score for the Baseline
• W indicates the decision-maker score for the Workbench
<table>
<tr><th>Panelist</th><th>B</th><th>W</th></tr>
<tr><td>#1</td><td>Low (2)</td><td>Low (2)</td></tr>
<tr><td>#2</td><td>Low/ Somewhat Low (2.5)</td><td>Low (2)</td></tr>
<tr><td>#3</td><td>Neutral (4)</td><td>Somewhat Low (3)</td></tr>
<tr><td>#4</td><td>High (6)</td><td>Somewhat Low (3)</td></tr>
<tr><td>#5</td><td>Somewhat Low (3)</td><td>High (6)</td></tr>
<tr><td>Average Score</td><td>3.5</td><td>3.2</td></tr>
<tr><td>Proportion of Low Scores (4 or below)</td><td>60%</td><td>80%</td></tr>
</table>
Alternative baseline from decision-maker surveys, on TA3-aligned tasks:
Average Score: 4
Proportion of Low Scores: 60%
</td><td></td></tr>
</table>

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<table>
<tr><th>Goals</th><th>Details</th><th>TA1</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr><td></td><td></td><td>Conclusions:
Workbench scores were an improvement over the Baseline for both average score and proportion of low scores.
Workbench team’s average score (5.1) was very close to but did not quite meet Phase 2 target (5.2)
Proportion of low scores for Workbench team (0%), exceeded Phase 2 target (10%)
Given the small sample size of the decision-maker panel (N = 5), these are not necessarily robust or generalizable conclusions
<table><tr><th>Category</th><th>Phase 2 Target</th><th>Outcome</th></tr><tr><td>Average Score</td><td>5.2</td><td>❌Target Not Met by either Workbench or Baseline</td></tr><tr><td>Proportion of Low Scores</td><td>10%</td><td>✅Target Met by Workbench
❌Target Not Met by Baseline</td></tr></table></td><td>Conclusions:
Workbench scores were slightly worse than the Baseline for average score, and equal for proportion of low scores.
Both Baseline and Workbench scores fell significantly short of the Phase 2 targets
Given the small sample size of the decision-maker panel (N = 5), these are not necessarily robust or generalizable conclusions
<table><tr><th>Category</th><th>Phase 2 Target</th><th>Outcome</th></tr><tr><td>Average Score</td><td>5.7</td><td>❌Target Not Met by either Workbench or Baseline</td></tr><tr><td>Proportion of Low Scores</td><td>9%</td><td>❌Target Not Met by either Workbench or Baseline</td></tr></table></td><td>Conclusions:
For Part 3, both teams had poor scores and fell significantly short of the Phase 2 targets, likely due to running out of time.
Given the small sample size of the decision-maker panel (N = 5), these are not necessarily robust or generalizable conclusions
<table><tr><th>Category</th><th>Phase 2 Target</th><th>Outcome</th></tr><tr><td>Average Score</td><td>4.8</td><td>❌Target Not Met by either Workbench or Baseline</td></tr><tr><td>Proportion of Low Scores</td><td>12%</td><td>❌Target Not Met by either Workbench or Baseline</td></tr></table></td><td></td></tr>
<tr><td rowspan="2">Generalizability</td><td>Definition</td><td>Semantic domains supported (applies to TA1/TA2)</td><td>Mathematical frameworks supported (applies to TA1/TA2/TA3/TA4)</td><td>Simulator components supported</td><td rowspan="2">N/A for this TA</td></tr>
<tr><td>Targets (Phase 1, Phase 2)</td><td>(4, NA)</td><td>(4, 4)</td><td>(10, 25)</td></tr>
</table>

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<table>
<tr><th>Goals</th><th>Details</th><th>TA1</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr>
<td></td>
<td>Evaluation</td>
<td>Because TA1 was not a part of Phase 2, this metric was out-of-scope for the final evaluation.</td>
<td>In Phase 1, the target goal of 4 frameworks supported in the workbench, was partially met, and this metric has not been a focus in Phase 2. There have some updates to the capabilities supported across multiple frameworks, compared to Phase 1, as indicated in the following table

Conclusion: ◩Partially met Phase 2 target of 4 frameworks supported through TA4 interfacing tools.
<table>
<tr><th></th><th>TA 1</th><th>TA2</th><th>TA3</th><th>TA 4</th></tr>
<tr><td>Petri Net</td><td>Y</td><td>Y</td><td>Y</td><td>Y</td></tr>
<tr><td>Stock & Flow</td><td>Y</td><td>Y</td><td>N (no change since Phase 1)</td><td>Y</td></tr>
<tr><td>Regnet</td><td>Y</td><td>Y</td><td>N (no change since Phase 1)</td><td>Y</td></tr>
<tr><td>Decapodes</td><td>N/A</td><td>Y (see climate table for Phase 2 improv ements)</td><td>Y (improve ment since Phase 1 – can use more optimal TA3 solvers)</td><td>N/A</td></tr>
</table>
</td>
<td>At the time of the final program evaluation, the workbench supports the following simulator components:

Running single models (5 blocks):
• Simulate
• Calibrate
• Optimize intervention policy
• Simulate ensemble
• Calibrate ensemble

Additionally, the ‘Validate configuration’ block incorporates FUNMAN model check functionality, which is a cross TA2-3 area.

Additionally, the TA3 functionality of implementing interventions through parameters, is supported by the ‘Configure model’ and the ‘Create intervention policy’ blocks.

In addition, the workbench now supports 9 new canvas templates which vastly simplify planning and executing several common workflows:
• Situational awareness
• Sensitivity analysis
• Decision making
• Horizon scanning
• Value of information
• Reproduce models from literature
• Calibrate an ensemble model
• Policy design
• Necessary or sufficient analysis

Conclusion: ◩Phase 2 target has been partially met. There is a total of 8 TA3 simulator components supported by the workbench, and an additional 9 templates</td>
<td></td>
</tr>
</table>

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Goals	Details	TA1	TA2	TA3	TA4
				which were not anticipated by the original metric but nonetheless substantially improve the usability of the workbench for TA3 simulations in useful research and decision-making contexts. There are some additional pieces of TA3 functionality available outside the workbench: • JuliaHub's EasyModelAnalysis.jl functionality that provides an alternative approach to CIEMSS functionality in workbench, including: • Sensitivity analysis • Model calibration • Interventions • Optimization • Ensemble modeling • CIEMSS's ChiRho library on causal modeling, which includes: • Causal effect estimation • Mediation analysis • Counterfactual analysis • Functionality for building hierarchical Bayesian models with dynamical systems **Total Integrated in Workbench:** 8 simulator components and 9 simulation template workflows **Additional Components Outside Workbench**: At least 9 pieces of key functionality	

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Goals	Details	TA1	TA2	TA3	TA4
Scalability	Definition	Extracted model size (# of variables)	Number of components that can be (de)composed	Sim configurations automatically explored	N/A for this TA
	Targets (Phase 1, Phase 2)	(100s, NA)	(5, 8)	(100, 1000)	
	Evaluation	Because TA1 was not a part of Phase 2, this metric was out-of-scope for the final evaluation.	Stratification with 5 dimensions was demonstrated in the workbench in Phase 1 (meeting the Phase 1 target). In Phase 2, the following stratified model was attempted but could not be executed in the workbench due to the level of complexity and number of total strata: • Disease dynamics (4 states) • Vaccination (2 groups) • Age groups (3 groups) • Sex (2 groups) • Location (16 groups) Final model was designed to have 192 strata in total. In this final program evaluation, the above model with the first 4 dimensions (and 12 strata in total) was successfully executed. **Conclusion**: ❌Phase 2 target has not been met.	TA3 capabilities with probabilistic simulation include sampling when there is an uncertainty range in model configurations (e.g. when parameters have a range of values they could take). In Phase 1, the workbench demonstrated the ability to explore 1000 configurations automatically through sampling of parameter space (uncertainty range), simulating, and visualizing results in a single graph; sampling and simulation did not take long to run, but visualization was very slow and memory-intensive for the browser. This worked for configurations with multiple parameters having uncertainty ranges. At the end of the program, we repeated the exploration with 1000 configurations and found that the sampling and simulation was quick to run and generate structured data (~ 15 seconds for a model with 8 variables and 8 parameters for 100 timesteps), and visualization of variable outputs was still slow (2+ minutes to generate 5 state variable plots) **Conclusion**: ✅Phase 2 target was met during both Phase 1 and Phase 2, albeit with some issues with the speed of visualization.	

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Table 28. TA-level program metric definitions for the climate use case. Explicit targets were not developed for this use case. TA1 performers left the program before they were able to develop meaningful capabilities for the climate use case, and therefore the metrics table does not include TA1 in any capacity.

Goals	Details	TA2	TA3	TA4
Accuracy	Definition	N/A		
	Targets			
	Evaluation			
Timeliness	Definition	T2: Time for physics model configuration/execution, model edits, model composition	T3: Time to create simulators for validation and prediction	Although this is not part of the official program TA metrics, TA4 functionality for this use case and evaluation included: • Integrated Assessment Model (IAM) understanding • IAM forecasting • IAM interventions • IAM optimization • Data search • Data processing and transformation
	Targets	N/A		
	Evaluation	See Climate IQ *Section 1 (TA2)* for plots and discussion on TA2 outcomes.	See Climate IQ *Section 2 (TA3)* for plots and discussion on TA3 outcomes.	See Climate IQ *Section 3 (TA4)* for plots and discussion on TA4 outcomes.
Decision Making Confidence	Definition	Confidence in having/developing an appropriate model (or set of models) and associated parameter space to sufficiently explore the scenario/problem	Confidence in understanding model results	N/A for this TA
	Targets	N/A for this use case	N/A for this use case	
	Explanation	The scenario involves updating or modifying physics models. Decision makers will assess whether this was approached in sensible way and whether the final model(s) is likely to support decision-maker priorities for this scenario.	Decision makers will be asked to determine their confidence in model results, and (if relevant) how uncertainty factors in.	Although this is not part of the official program TA metrics, TA4 functionality involved in the climate decision-maker scenario included: • Data search, processing and wrangling • Visualization/ plotting

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<table>
<tr><th>Goals</th><th>Details</th><th>TA2</th><th>TA3</th><th>TA4</th></tr>
<tr><td></td><td>Evaluation</td><td colspan="3">
The Decision-making Confidence metrics were supported by the Climate Decision-maker Scenario. Although Parts 1-2 of the scenario and corresponding decision-maker panel scores were meant to largely support Decision-maker Confidence Metric 1 (aligning with TA2 capabilities), TA4 functions relating to data search, transformation, and visualization, also contributed to the work and were reflected in the panelist scores. Part 3 of the scenario was largely meant to support Decision-maker Confidence Metric 2 (aligning with TA3 capabilities), but TA2, TA3, and TA4 functions contributed to the work and were reflected in the panelist scores. For these reasons, it is difficult to cleanly separate out the decision-maker confidence results by TA. Therefore, we discuss all scenario parts and results here.
See Climate Decision-maker Scenario (Climate DM) section for detailed discussion on the various approaches by Baseline and Workbench teams, organized into the scenario Parts 1, 2, and 3. See Decision-maker Panel Outcomes for discussion on panelist feedback and scores.
All decision-maker confidence scores were given on 7-point scale:
<table>
<tr><td>1</td><td>Very Low</td></tr>
<tr><td>2</td><td>Low</td></tr>
<tr><td>3</td><td>Somewhat Low</td></tr>
<tr><td>4</td><td>Neutral</td></tr>
<tr><td>5</td><td>Somewhat High</td></tr>
<tr><td>6</td><td>High</td></tr>
<tr><td>7</td><td>Very High</td></tr>
</table>
Decision-maker confidence scores for Parts 1-2 (align most strongly with TA2 capabilities but with TA4 contributions as well)

B indicates the decision-maker score for the Baseline
W indicates the decision-maker score for the Workbench

<table>
<tr><th>Panelist</th><th>B</th><th>W</th></tr>
<tr><td>#1</td><td>Somewhat Low (3)</td><td>Somewhat Low (3)</td></tr>
<tr><td>#2</td><td>Neutral (4)</td><td>Somewhat Low (3)</td></tr>
<tr><td>#3</td><td>Very High (7)</td><td>Somewhat Low (3)</td></tr>
<tr><td>Average Score</td><td>4.67</td><td>3</td></tr>
<tr><td>Proportion of Low Scores (4 or below)</td><td>66%</td><td>100%</td></tr>
</table>
Summary for Parts 1-2:

Workbench scores were overall worse, with a lower average and greater proportion of low scores
Lot of variation in scores for Baseline, but not for Workbench where all panelists agreed on low score; in general panelists struggled to score consistently across both teams, changing their minds several times during the panels.
Most likely the biggest detractors of confidence for the Workbench team trace back to the challenges with the Turbulent Flow model, conversion to the right geometry (scenario asked about a 2D rectangle where documentation involved torus and sphere), and not being able to implement periodic boundary conditions as the modelers would have liked to
Aspects that boosted panelists' confidence included modelers ability to explain all their results as well as limitations of their approach (Baseline team), ability to input model equations rather than search for it in a wizard (Workbench bench), and visualize model building blocks through composition diagram (Workbench team).
Aspects that detracted from panelists' confidence, included having to do a lot of math to get into the COMSOL acceptable format (Baseline), computational limits that caused modelers to raise error tolerance for convergence (Baseline), lack of validation steps (Workbench), and challenges with the Turbulent Flow model described above (Workbench).

</td></tr>
</table>

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Goals	Details	TA2	TA3	TA4

Decision-maker confidence scores for Part 3 (TA2, TA3, and TA4 functionality contributed to this part)

- ***B*** *indicates the decision-maker score for the Baseline*
- ***W*** *indicates the decision-maker score for the Workbench*

Panelist	B	W
#1	Very High (7)	Somewhat Low (3)
#2	Somewhat High (5)	High (6)
#3	High (6)	Somewhat Low (3)
Average Score	6	4
Proportion of Low Scores (4 or below)	0%	66%

Summary for Part 3:

- Workbench scores were overall worse, with a lower average and greater proportion of low scores
- Lot of variation in scores and across panelists
- Aspects that boosted panelists' confidence included running processes multiple times and finding average trends boosted confidence (Baseline team), and trend analysis graphs of humidity and temperature data (Workbench team).
- Aspects that detracted from panelists' confidence, included lack of time for running additional experiments and/or validation (for both teams), lack of specific data in analysis (for Baseline), or lack of confidence that results were reproducible (Workbench)

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Goals	Details	TA2	TA3	TA4
Generalizability	Definition	Mathematical frameworks supported (applies to TA1/TA2/TA3/TA4)	N/A for this use case	N/A for this TA
	Targets	(4, 4)		
	Evaluation	Decapodes was represented and simulated for physical climate models (PDEs), contributing to 1/4 mathematical frameworks. See epidemiology metrics table for more details on the 4 mathematical frameworks. Decapodes improvements since Phase 1 include: • Multigrid capabilities that increase computation speed • Better integration with TA3 capabilities, that allows Decapodes to leverage Julia's implicit solvers, resulting in significant efficiency gains when solving stiff systems • Many more types of physics supported in Decapodes form, including a robust set of canonical models in physics, chemistry, biology, and environmental science • Support for GPU simulations and parallel processing to significantly accelerate simulation execution time		
Scalability	Definition	Number of physics models that can be composed	N/A for this use case	N/A for this TA
	Targets	N/A for this use case		
	Evaluation			

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Appendix B: Performer-Specific Metrics

Appendix B contains performer-specific metrics for Technical Areas (TAs) 2-4, and spanning both the epidemiology and climate use cases.

TA	Performer Team	Metrics
2	**Northeastern University**	**See Figure 41 for Epi IQ Section 2 timing and quality scores for multi-model structural comparison, model editing, and model stratification**, which utilize Northeastern's Mira library, exercised through Terarium integrated workbench interface (developed by TA4 performer Uncharted) • There do not seem to be any clear trends between completion time and problem difficulty • The Workbench team had a significant time advantage for the structural comparison questions, although their answers were not always better than the Baseline. • All questions are independent, except for the stratification questions, which build upon each other sequentially • See Epi IQ Section 2 (TA2) and Appendix A: Metrics Tables for more detailed discussion and summary on how Mira functionality in Terarium met TA2 targets for stratification and model edits

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TA	Performer Team	Metrics

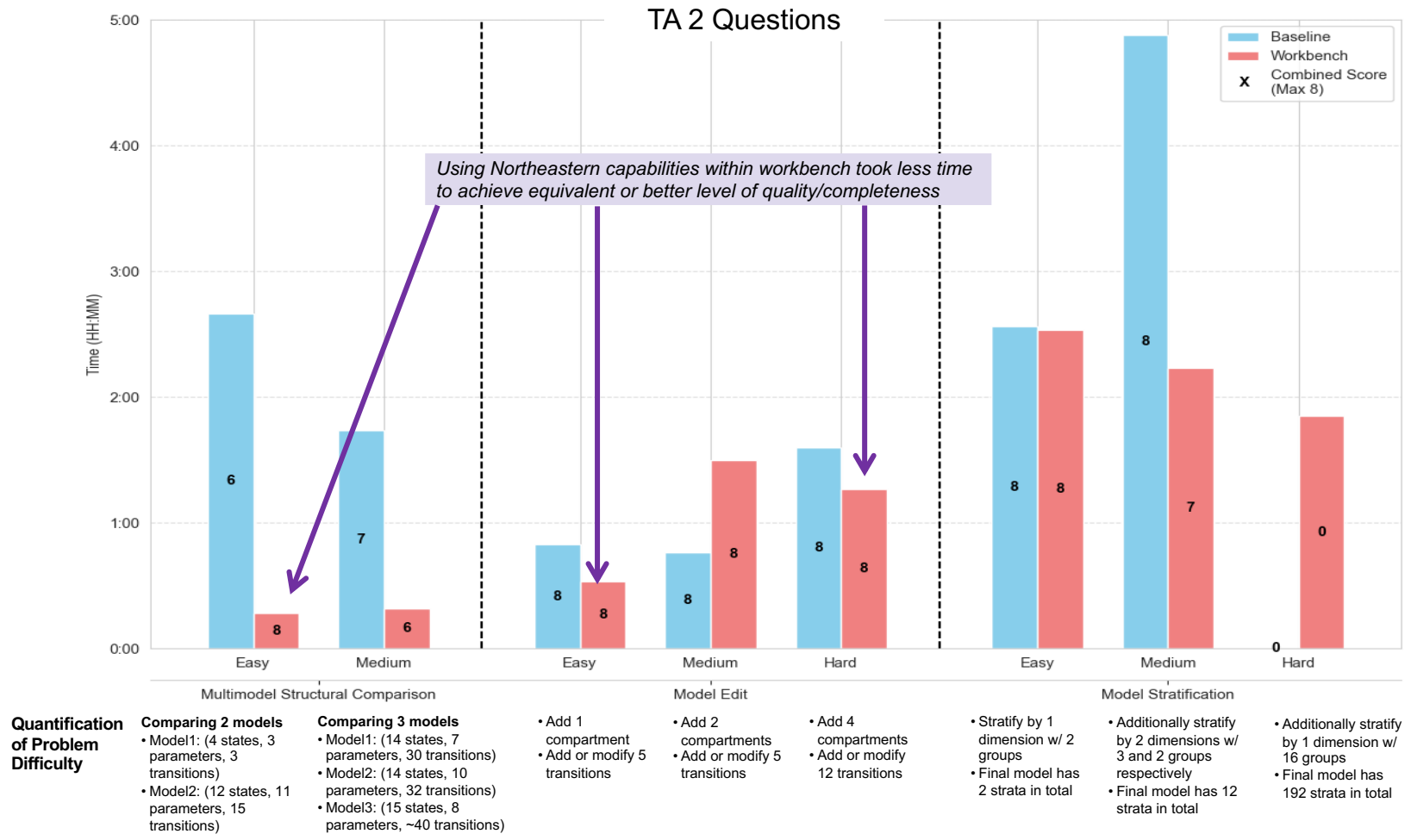


Figure 41. Epidemiology Independent Questions (Epi IQ) Section 2 timing and quality scores for multi-model structural comparison, model editing, and model stratification, which utilize Northeastern's Mira library, exercised through TA4's Terarium workbench interface. Copy of Figure 7, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
2	**Smart Information Flow Technologies (SIFT)**	**FUNMAN demonstrated improved parameter space exploration between program Phase 1 and Phase 2** • Figure 42 left plot shows that for equivalent parameter space size (indicated by volume), capabilities developed in Phase 2 resulted in improved or comparable runtime • Figure 42 right plot shows an example of Phase 1 parameter space coverage for a volume of 2500x, and Phase 2 parameter space coverage for a volume of 30625x volume taking a similar order magnitude of runtime, demonstrating significant scaling. • Additional commentary from SIFT team: '…[we] adjust resolution of the methods so that as we considered a larger space, we continued to refine it to the same level (allowing for an apples-to-apples comparison). The greatest volume parameter space we could solve within our time limit was 2500x the original volume (1x) for the Phase 1 method. The greatest volume we could solve in comparable time with our Phase 2 was 30625x the original volume, which is 30625/2500 = 12.25 times (1225%) that of Phase 1. That is a one order of magnitude (10x) improvement.' • In the Figure 42 right plot, the SIFT team configures the setup so that the parameter space was centered around a decision boundary (blue data points). The larger the volume solved, the more of the decision boundary is considered, and this is the area where the algorithms need to do the most work

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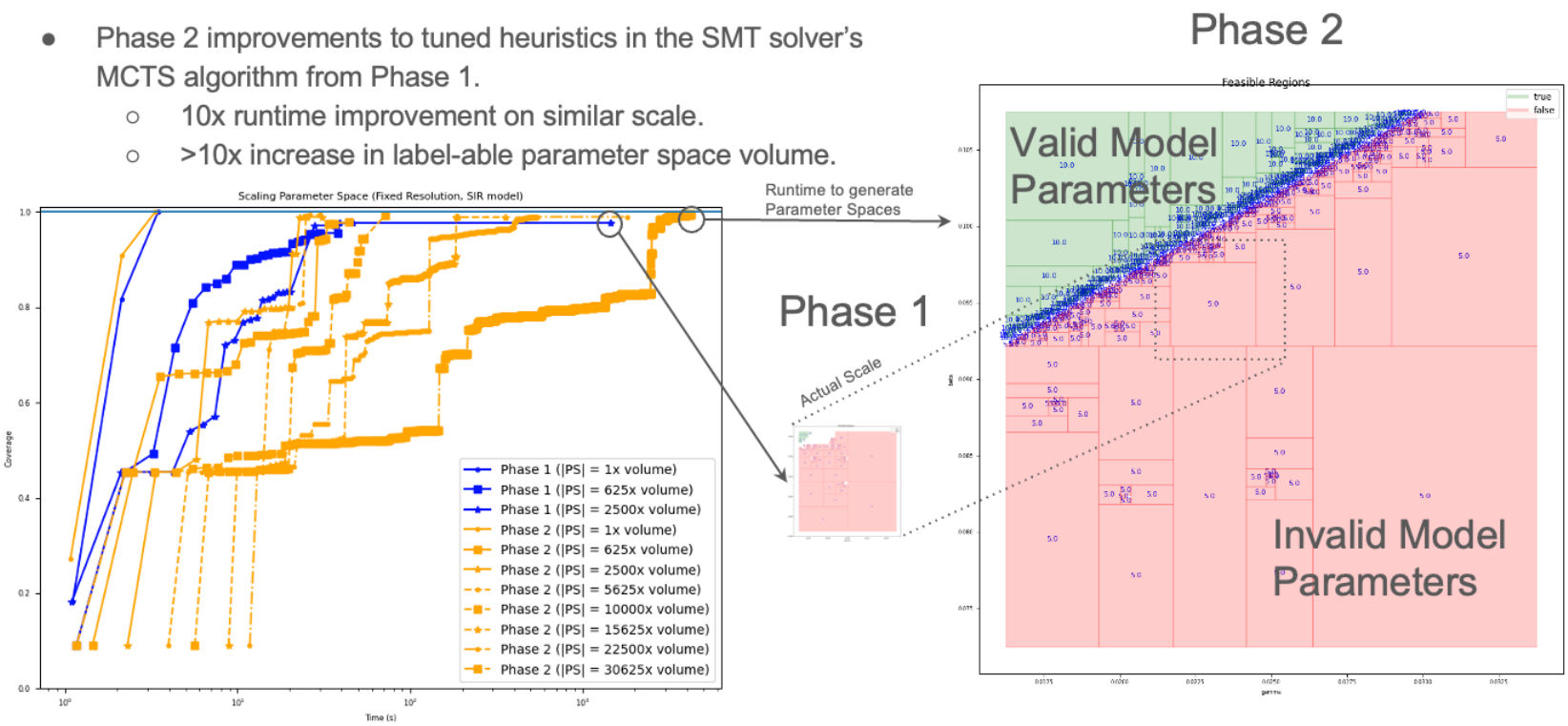


Figure 42. Demonstration of FUNMAN improvements in parameter space exploration between Phase 1 and Phase 2. Analysis and image provided by the SIFT team.

Evaluating impact of FUNMAN on optimization: The SIFT team completed the optimization set of equations from the final program evaluation, with and without the use of FUNMAN to inform the parameter bounds used in the optimization problem. The results are described in Table 29.

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TA	Performer Team	Metrics

Table 29. FUNMAN evaluation with final evaluation's optimization questions.

Problem	Time to converge to a result without FUNMAN (Baseline) in mm:ss	Time to converge to a result with FUNMAN in mm:ss	Comments
Q3.4.1 - Optimizing 1 intervention parameter value (single objective)	2:32, with optimal p_T value of 0.315	1:32 with optimal p_T value of 0.333	• Optimization with FUNMAN output informing parameter space exploration resulted in shorter time to converge to a solution • Both the Baseline and Funman approaches resulted in somewhat similar optimization results for parameter value, but this was not the minimum value that satisfied the constraints (p_T = 0.268)
Q3.4.2 - Optimizing 1 intervention parameter start time (single objective)	3:06 with latest start time for p_T being 17 days from start of simulation	2:21 with latest start time for p_T being 20 days from start of simulation	• Optimization with FUNMAN output informing parameter space exploration resulted in shorter time to converge to a solution • Both the Baseline and Funman approaches resulted in somewhat similar optimization results for parameter start time (17 vs 20 days from start of simulation), but this was not the latest start time that satisfied the constraints ($37\ days$)
Q3.4.3 - Optimizing 2 intervention parameter start times and 1 intervention parameter value (multiple objective)	1:28 but failed to converge to a solution satisfying constraints	3:44 • Minimum p_T value of 0.7 • Latest start time for p_T being 5 days into simulation • Latest start time for quarantine intervention parameter λ is 0 days into simulation	• Optimization with FUNMAN output informing parameter space exploration resulted in successful converge to a solution meeting constraints • However, results found with FUNMAN seem to be quite different from true solutions (minimum $p_T = 0.3$, latest start time for p_T is 10 days, and latest start time for λ is 13 days). We hypothesize that this is due to the difficulties in interpreting the Validate Configuration block in the workbench, which collapses FUNMAN's high dimensional parameter space representation into a 1D bar chart with differently colored regions which is very confusing to interpret for a question at this level of difficulty and complexity (but might be sufficient for an easier question).

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TA	Performer Team	Metrics
2	**University of Florida (UFL)**	**See Figure 43 for Climate IQ Section 1 (TA2) timing and quality scores for model configuration and execution, model edits, and multiphysics model composition** with UFL's Decapodes models and computational library, which were exercised through Beaker interface (developed by TA4 performer Jataware). • See Climate IQ Section 1 (TA2) for more detailed discussion on outcomes • The only clear trend is increasing computation time for the Workbench team as model config/execution problems became more difficult. • For this set of questions, there were not any instances *where both teams could be directly compared on a question **and*** the Workbench team had an advantage over the Baseline team. The only question in which the Workbench team spent less time compared to the Baseline team was the first model configuration/execution question, but the results were worse. • However, using Decapodes, the Workbench team was able to attempt and complete more questions in this set, compared to the Baseline team. • All questions were independent except for the model edit questions, which built upon each other to some extent.

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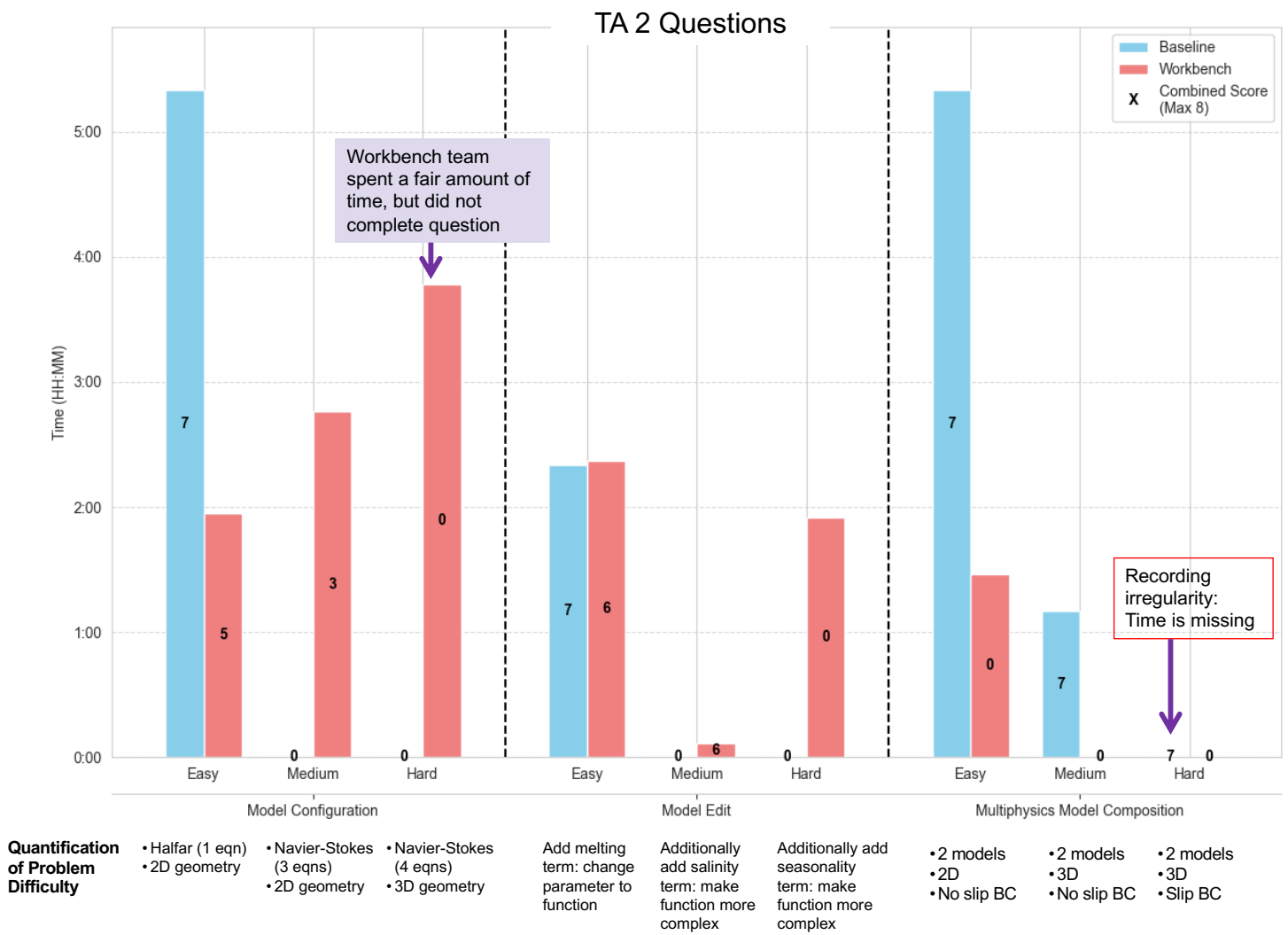


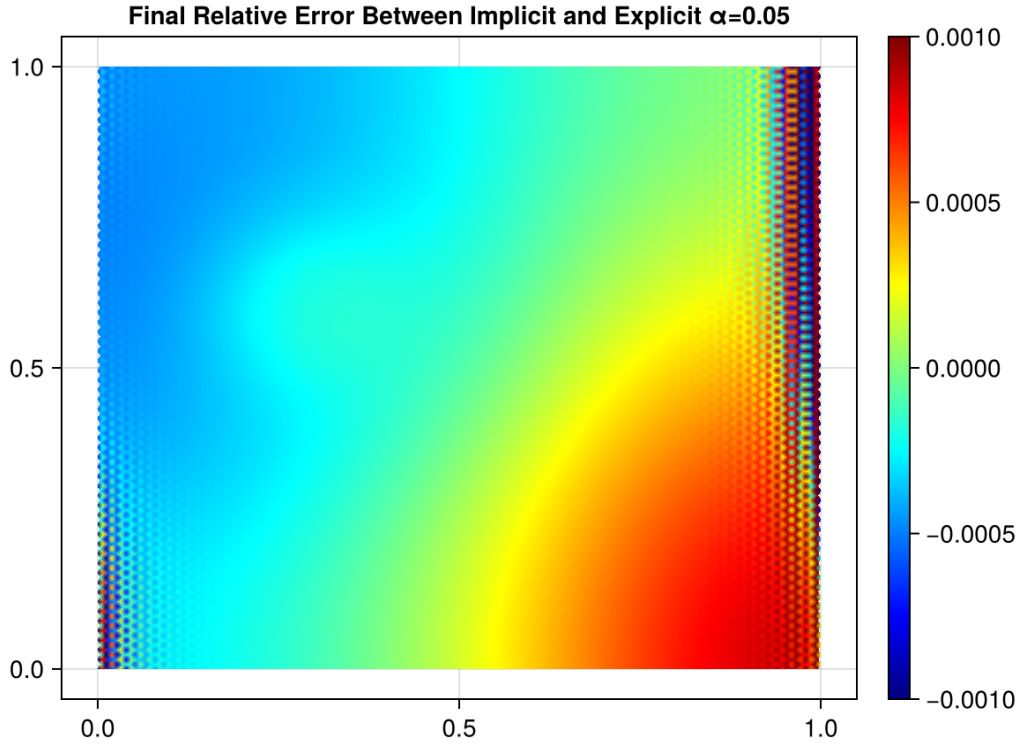
Figure 43. Climate Independent Questions (Climate IQ) Section 1 timing and quality scores for model configuration and execution, model edits, and multiphysics model composition with UFL's Decapodes models and computational library, which were exercised through TA4's Beaker interface. Copy of Figure 18, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
		Post-evaluation Demonstration of Brusselator w/ Julia's Implicit Solvers • Post-evaluation, the UFL team also demonstrated that due to integration of Decapodes with Julia's auto-differentiation capabilities, they could now take advantage of Julia's implicit solvers, something that was not possible earlier in the program, and was not exercised during the final evaluation. • Running a Brusselator Decapodes model (used to model autocatalytic reactions with two different chemicals) with a stiff model configuration (i.e. using a diffusion rate 20x the norm) and ***using an implicit Julia solver was faster and more memory efficient*** (with shorter execution runtime, significantly smaller memory usage due to sparse Jacobians, significantly fewer function evaluations, and significantly fewer overall timesteps), ***compared to an explicit solver***. Simulation was run for 8 timesteps. See Figure 44 for detailed quantitative results. • Because this is a very stiff system, the explicit solver needs to take very small timesteps to be stable; therefore, the explicit solution is likely more accurate due to small time steps taken. However, final simulation results at the end of the simulation period seem quite close between the two solvers. See Figure 45 for error between explicit and implicit solutions, at the end of the simulation period. For most of the rectangular domain, the error appears low and increases a little towards the right and left boundaries. • This demonstration reflects an integration win for both University of Florida and JuliaHub. **Figure 44. UFL results running a stiffened Brusselator model with explicit and implicit Julia solvers. Runtime was 8 time units. Analysis and image provided by the UFL team.**

Solver Type	Runtime (s)	Memory Usage (GiB)	Function 1 Evals	Accepted Steps
Explicit	90.78	91.719	57,831	9,637
Implicit	67.39	1.669	620	144

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TA	Performer Team	Metrics
		 **Figure 45. Error between implicit and explicit solutions, at the end of the simulation time frame. General direction of error is top-left to bottom-right, aligning with the dynamics of the Brusselator model itself. UFL hypothesizes that moving concentrations hitting the boundaries may be causing the higher error in those regions. Analysis and image provided by the UFL team.**

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TA	Performer Team	Metrics
3	**Causal Inference Engine for Multi-scale Simulations (CIEMSS)**	**See Figure 46 for Epi IQ Section 3 timing and quality scores for parameter calibration, forecasting, interventions, optimizing interventions, and ensembles**, largely utilizing CIEMSS capabilities exercised through Terarium integrated workbench interface (developed by TA4 performer Uncharted). • CIEMSS capabilities in the Terarium workbench utilize Mira (developed by TA2 performer Northeastern) for model ingestion, and also use Julia backend solvers (developed by TA3 performer JuliaHub) for simulation. The optimization capabilities also can leverage FUNMAN (developed by TA2 performer SIFT) output. • There are many instances where Workbench team using CIEMSS capabilities had a significant time advantage over the Baseline, with equal or better quality/completeness of results. • There do not seem to be any clear trends between execution time and problem difficulty, except for ensembles, where both the Baseline and Workbench team took less time to complete the problems as difficulty increased. • ***See Final Report Epi IQ Section 3 (TA3) and Appendix metrics tables for more detailed discussion and summary on how CIEMSS functionality in Terarium meets TA3 targets***

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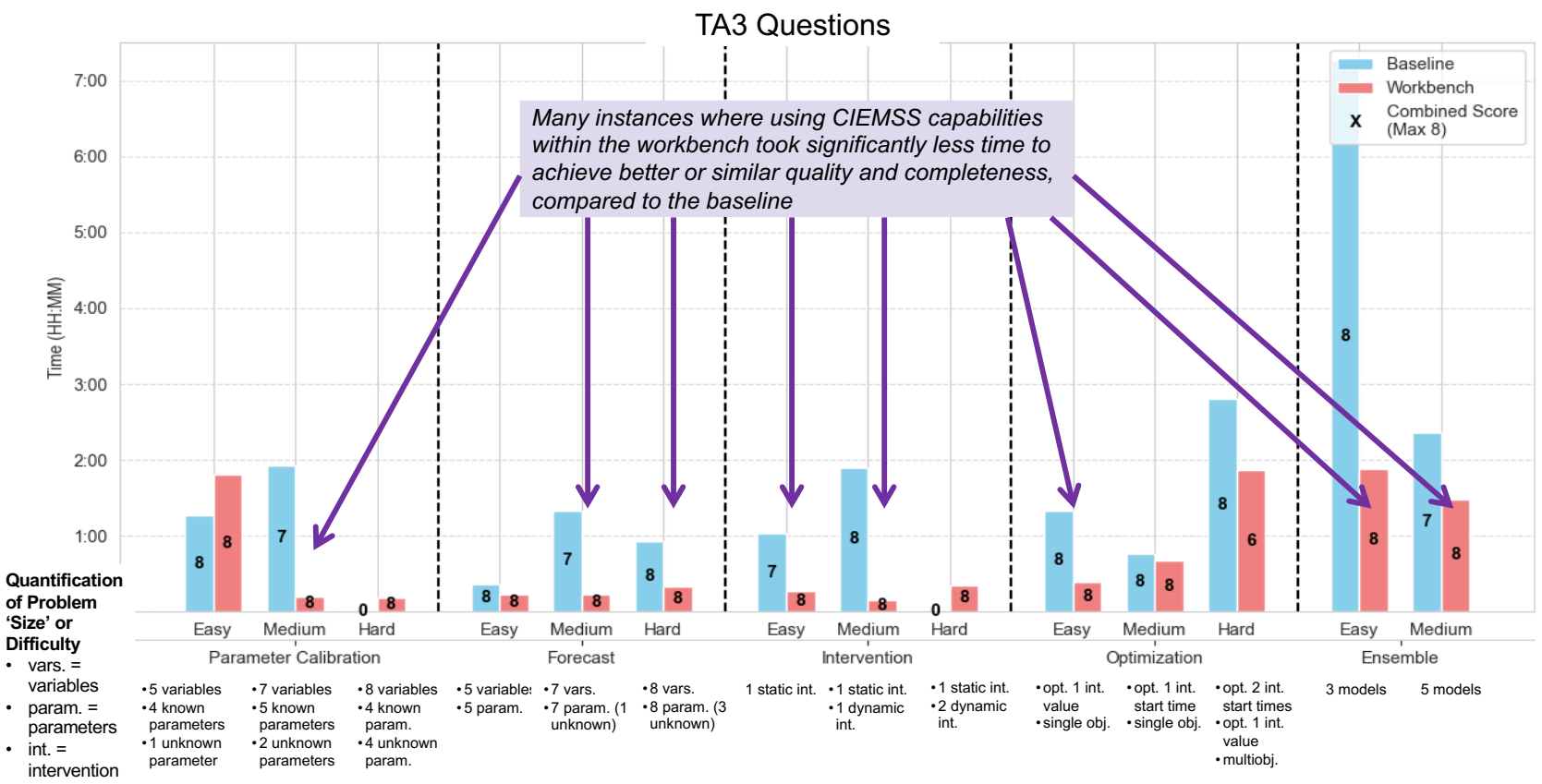


Figure 46. Epidemiology Independent Questions (Epi IQ) Section 3 timing and quality scores for parameter calibration, forecasting, interventions, optimizing interventions, and ensembles, largely utilizing CIEMSS features exercised through TA4 workbench, as well as some TA2 and TA3 capabilities from other performers. Copy of Figure 9, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
		Speedup in simulation when using Julia backend solvers (reflecting an integration win for both CIEMSS and JuliaHub): • Figure 47 shows difference in solve time for a simple SIR system • Figure 48 shows difference in solve time for a simple predator-prey Lotka-Volterra model. • For both Figure 47 and Figure 48, 'DiffEqPy' is JuliaHub's backend solver. • Figure 47 and Figure 48 demonstrate at least an order-of-magnitude solver time speedup with JuliaHub solvers. • Figure 47 and Figure 48 are excerpts from CIEMSS' code notebooks[1][2]

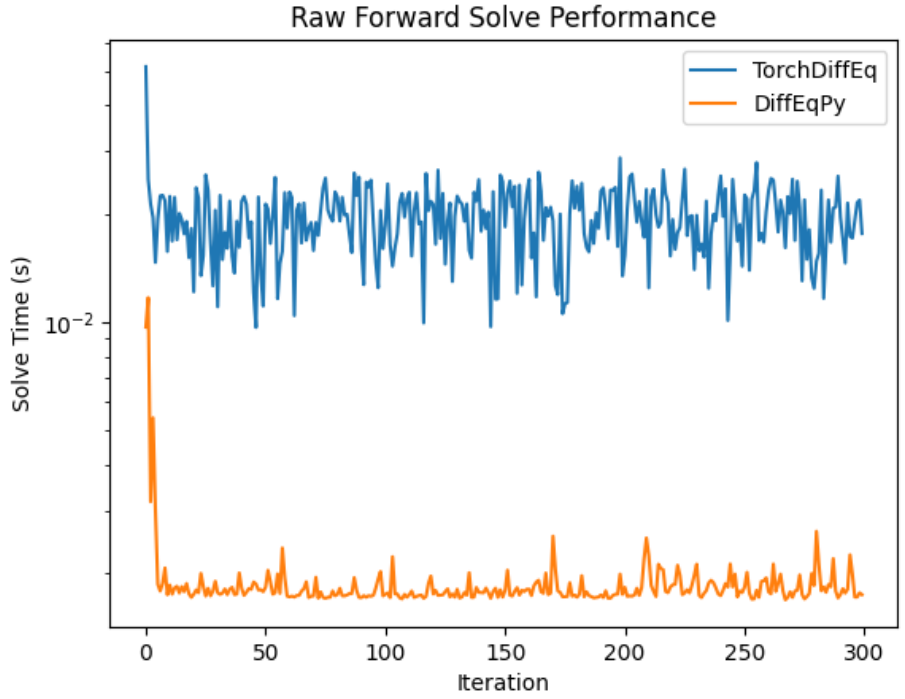


Figure 47. Difference in solve time with and without JuliaHub's backend solvers (DiffEqPy), for a simple SIR system of equations.

[1] https://github.com/ciemss/chirho_diffeqpy/blob/1-create-performance-benchmark-using-sir-model-not-lotka-volterra/docs/source/sir_performance_comparison.ipynb

[2] https://github.com/ciemss/chirho_diffeqpy/blob/1-create-performance-benchmark-using-sir-model-not-lotka-volterra/docs/source/performance_comparison.ipynb

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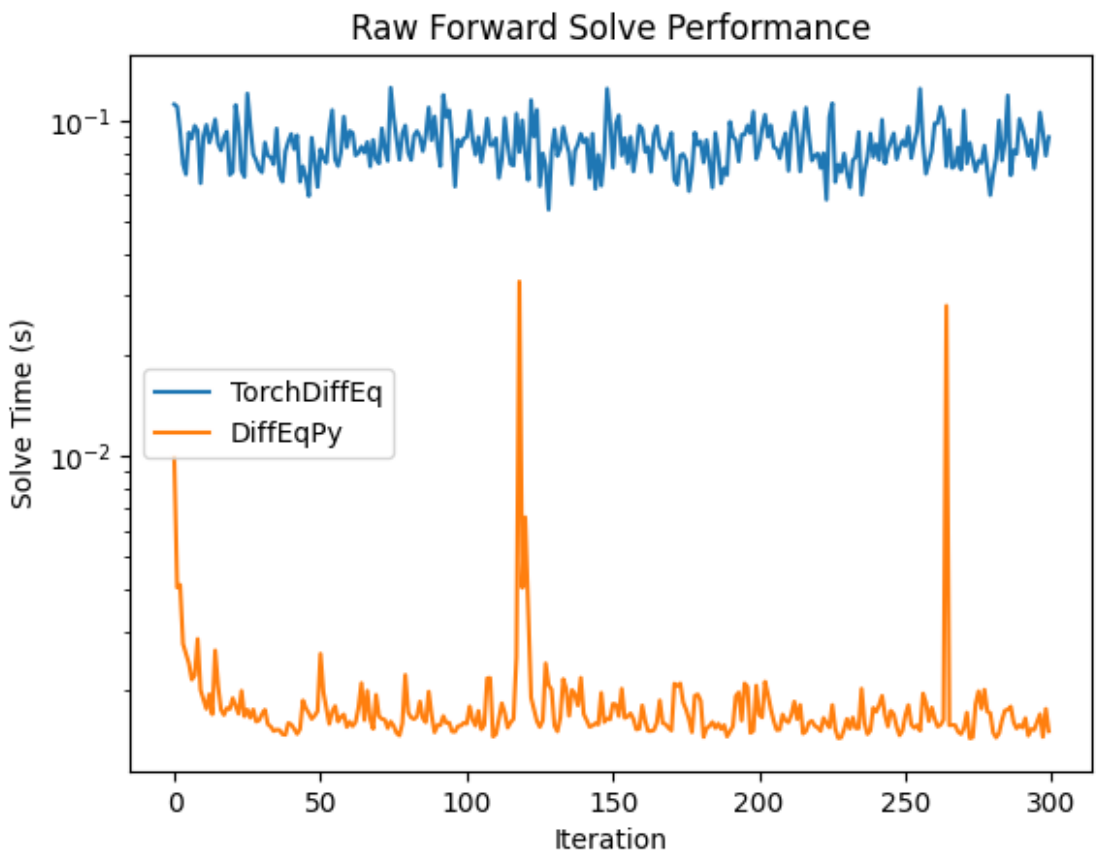


Figure 48. Difference in solve time with and without JuliaHub's backend solvers (DiffEqPy), for a Lotka Volterra system.

Influenza forecasting challenge accuracy metrics for single and ensemble models

- See Forecasting Challenge Results section for more detailed results,.
- Figure 49 and Figure 50 show forecast results for Florida and include CIEMSS results and results for FluSight forecasters (the Baseline in this context). ***The CIEMSS forecasts track very closely with the ground truth data and result in the lowest WIS scores for the last two timepoints of the forecast period, outperforming all the Baseline forecasters for these two timepoints***.
- Figure 51and Figure 52 show forecast results for Pennsylvania, for CIEMSS and FluSight forecasters. ***The CIEMSS ensemble forecasts track very closely with the ground truth data and result in the lowest WIS scores for the middle two timepoints of the forecast period, outperforming all the Baseline forecasters for these two timepoints***. However, their single model forecasts do not perform as well, falling in the middle of the neighborhood of Baseline forecaster scores.

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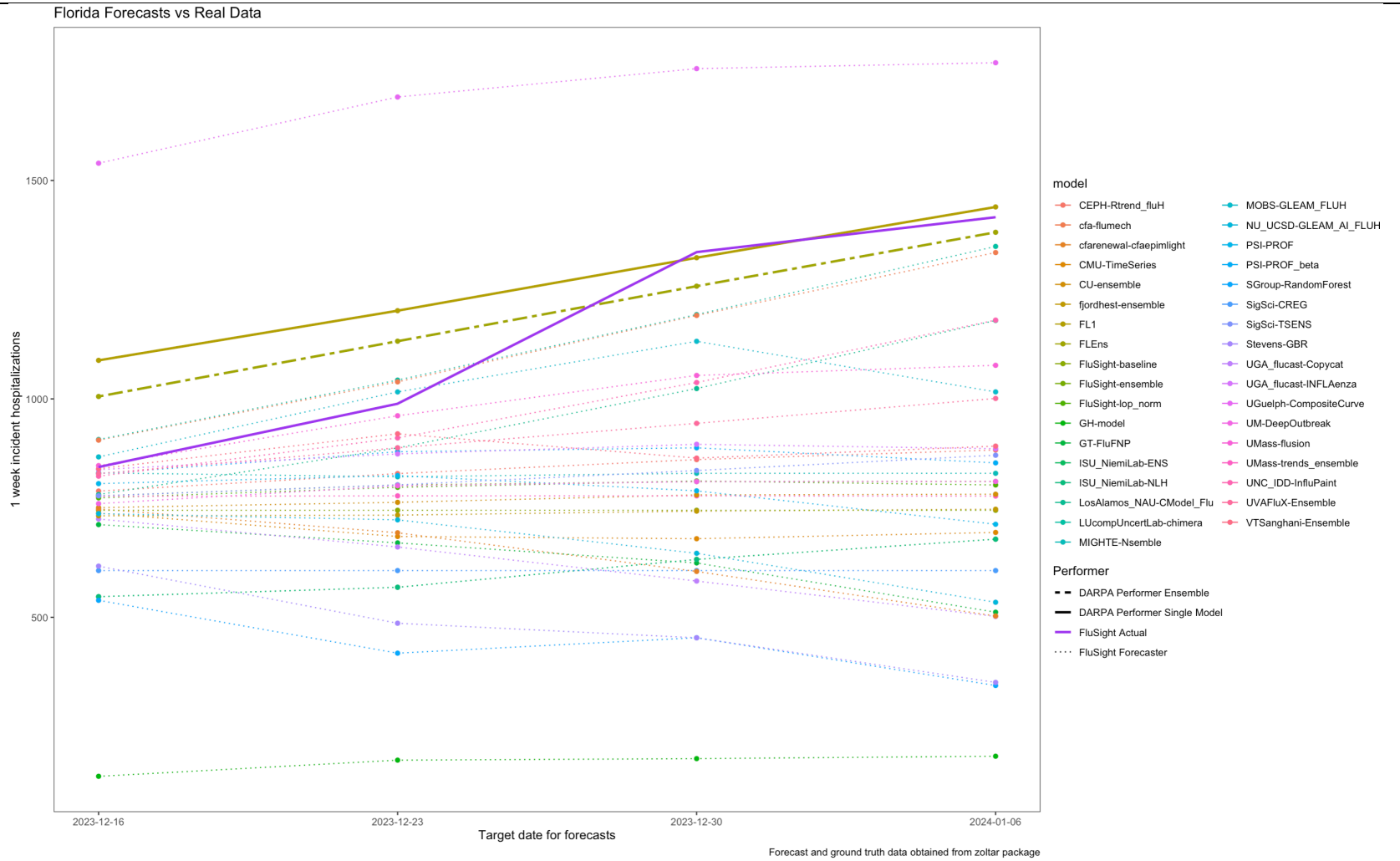


Figure 49. Florida Forecasts vs Ground Truth Data (bold purple line), for ASKEM performer team (bold tan lines) and FluSight forecasters (thin dotted lines in various colors). Copy of Figure 38, reproduced here for ease of reference.

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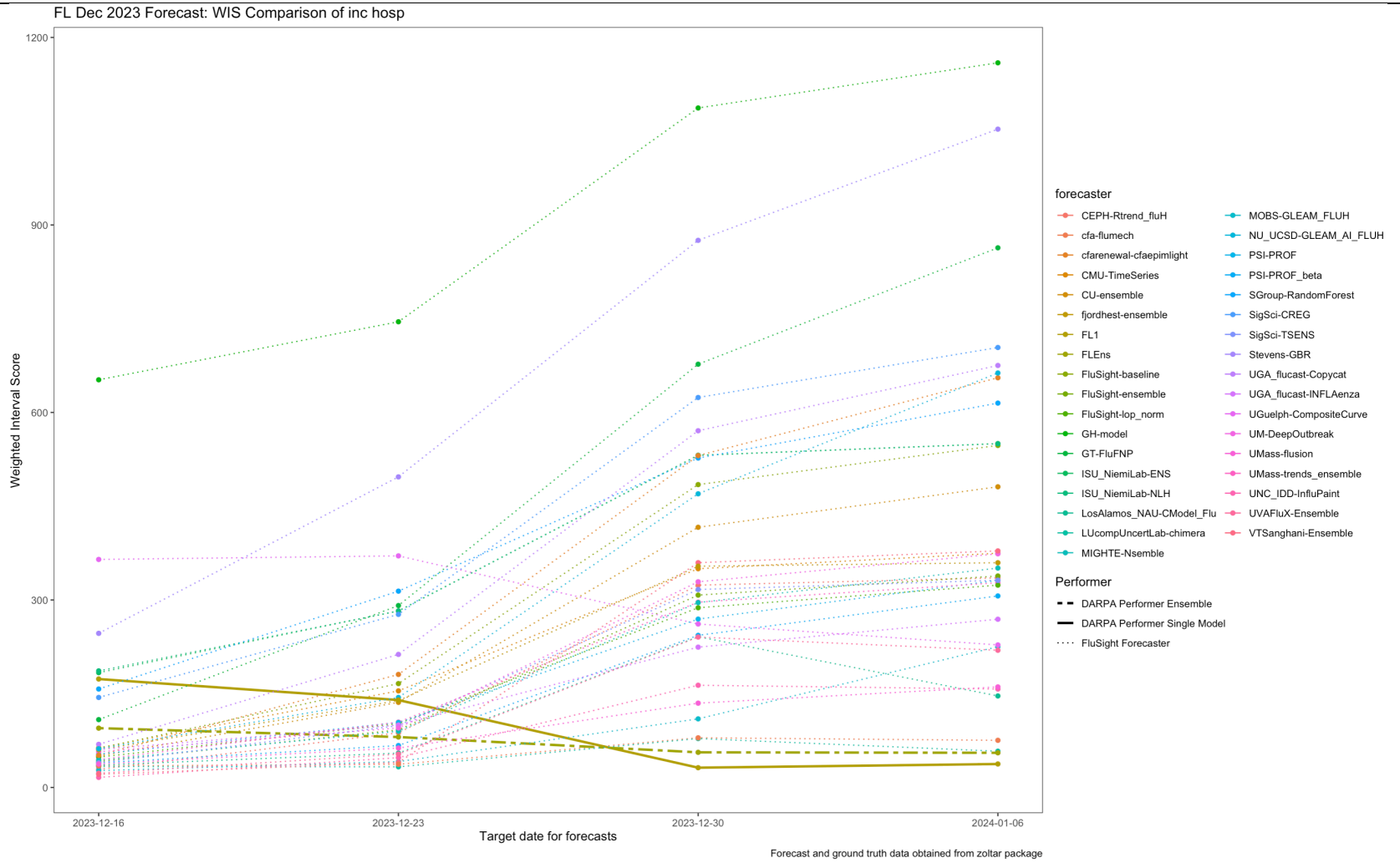


Figure 50. WIS scores for ASKEM performer (bold tan lines) and FluSight forecasts (thin dotted lines in various colors), for the state of Florida. Copy of Figure 37, reproduced here for ease of reference.

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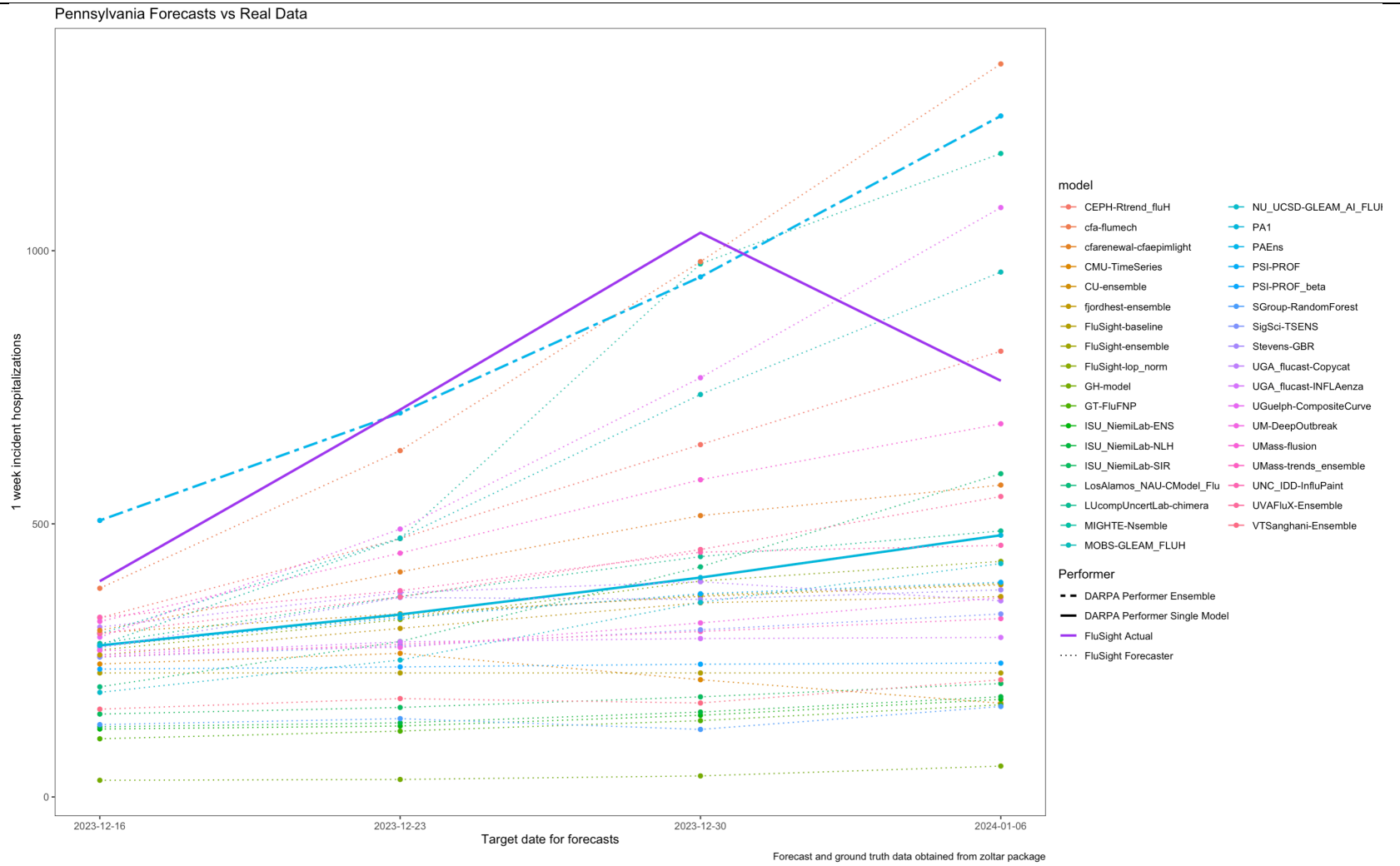
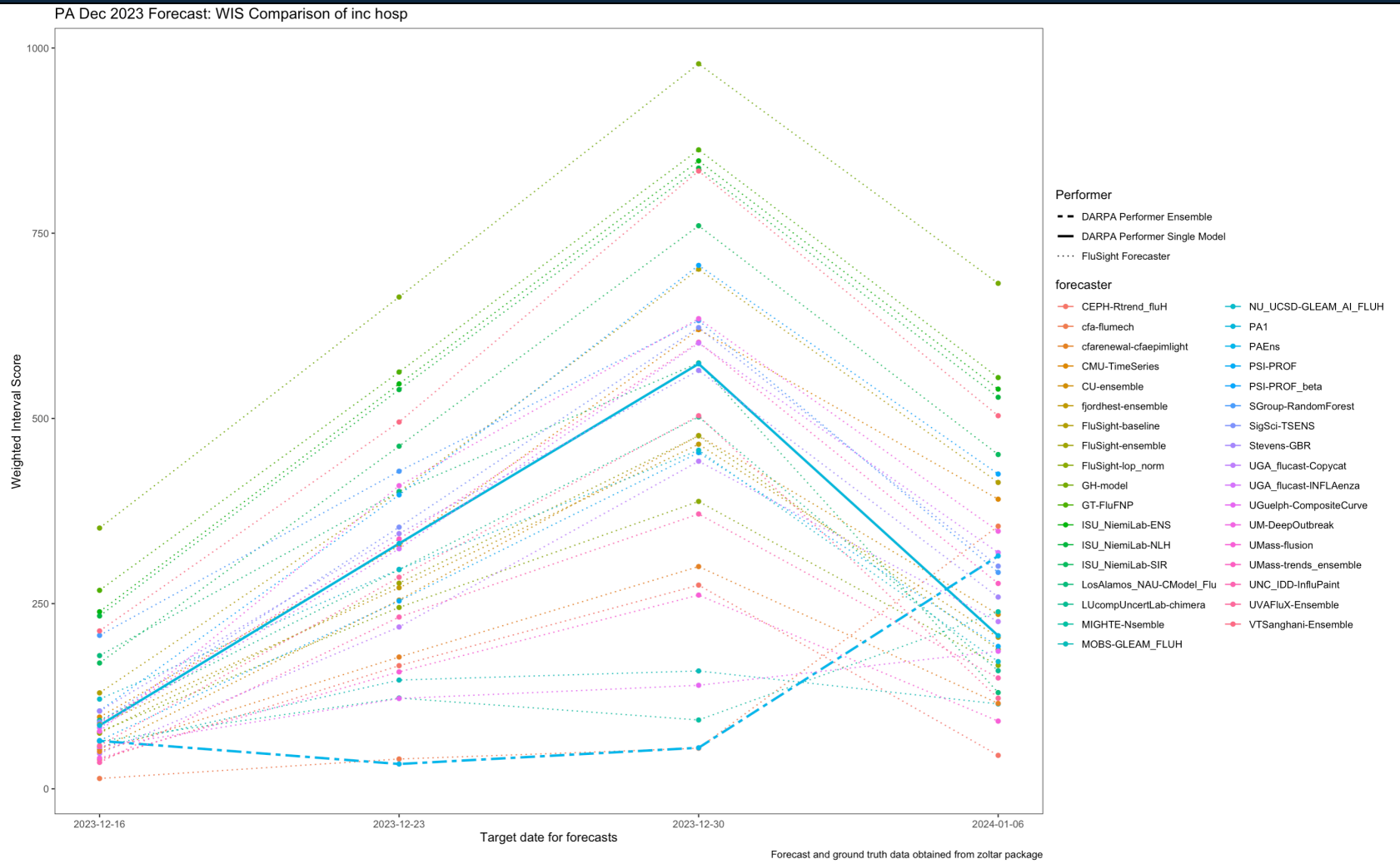


Figure 51. Pennsylvania Forecasts vs Ground Truth Data (bold purple line), for ASKEM performer team (bold blue lines) and FluSight forecasters (thin dotted lines in various colors). Copy of Figure 40, reproduced here for ease of reference.

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TA	Performer Team	Metrics
		 **Figure 52. WIS scores for ASKEM performer (bold blue lines) and FluSight forecasts (thin dotted lines in various colors), for the state of Pennsylvania. Copy of Figure 39, reproduced here for ease of reference.**
3	**JuliaHub**	**See Figure 53 for Climate IQ Section 2 (TA3) timing and quality scores for forecasting, sensitivity analysis, and calibration** with Decapodes physics models (developed by TA2 performer UFL), utilizing JuliaHub's simulation capabilities through the Beaker interface (developed by TA4 performer Jataware). ***See Climate IQ Section 2 (TA3) for more detailed discussion on outcomes***

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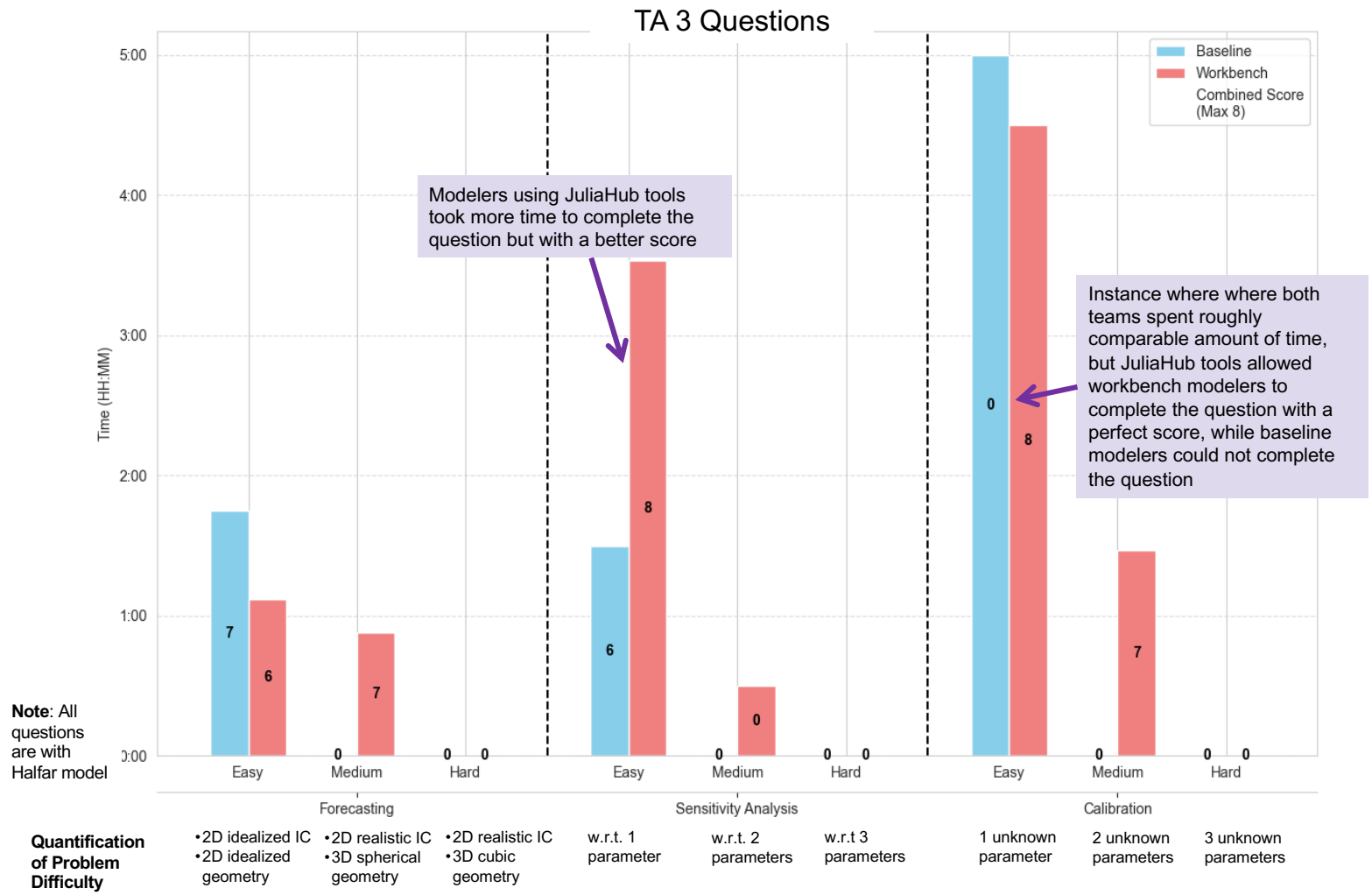
TA	Performer Team	Metrics
		

Figure 53. Climate Independent Questions (Climate IQ) Section 2 timing and quality scores for forecasting, sensitivity analysis, and calibration with Decapodes physics models (developed by TA2), but utilized JuliaHub's simulation capabilities through the Beaker interface (developed by TA4). Copy of Figure 21, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
		For more detail on speedup in CIEMSS simulation before and after integration with Julia backend solvers, see Figure 47 and Figure 48 under CIEMSS section above. Julia's SciML benchmarks page: https://docs.sciml.ai/SciMLBenchmarksOutput/stable/ **JuliaHub's Sparse Auto-Differentiation Capabilities Developed in Phase 2:** • Figure 54 and Figure 55 show how sparse Auto-Differentiation (AD) capabilities show much faster execution time compared to non-sparse approaches, in both log and linear scales. Sparse AD approaches scale very well with mesh density (which correlates to model size). • Figure 56 shows a work precision plot where each point represents a different absolute and relative set tolerance level, and resulting simulation error and execution time (x and y axes). For some range of points, the Sparse AD approach results in less error and faster execution time. However, there is a range of points at a vertical segment of the 'Roda5: Sparse' line where set tolerance is changing but error doesn't change; we hypothesize that the solutions being found have an error that is much lower that the tolerance level, and while the tolerance level varies, the actual error does not. • JuliaHub has also integrated with Enzyme for the AD backend, resulting in very comparable solver time for non-Enzyme solvers with standard ODE models (integration currently doesn't include Decapodes models), but with much greater flexibility to use GPUs and avoid preallocation and other constraints in writing the simulation and solver code.

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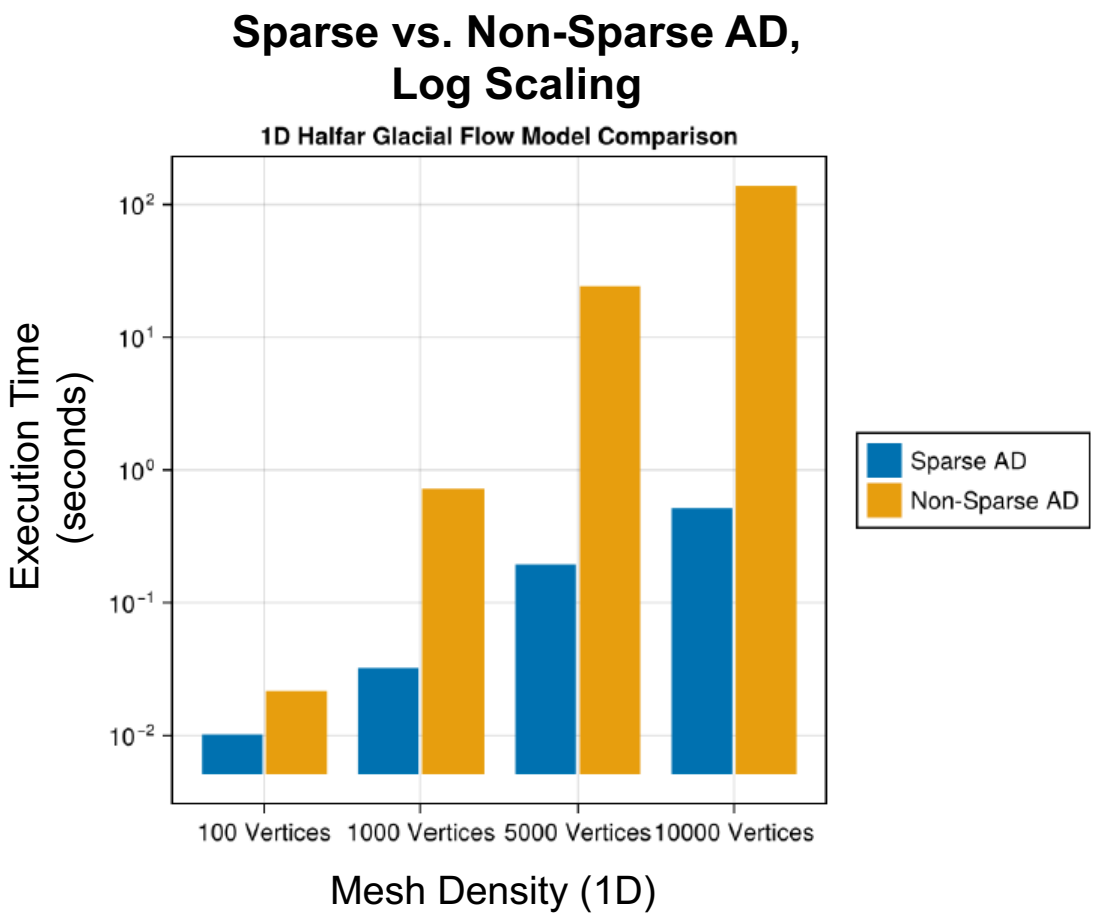


Figure 54. Sparse vs Non-Sparse Auto-Differentiation, showing how execution time scales with mesh density, with log-scaling for the y-axis for better visual comparison. Analysis and image provided by the JuliaHub team.

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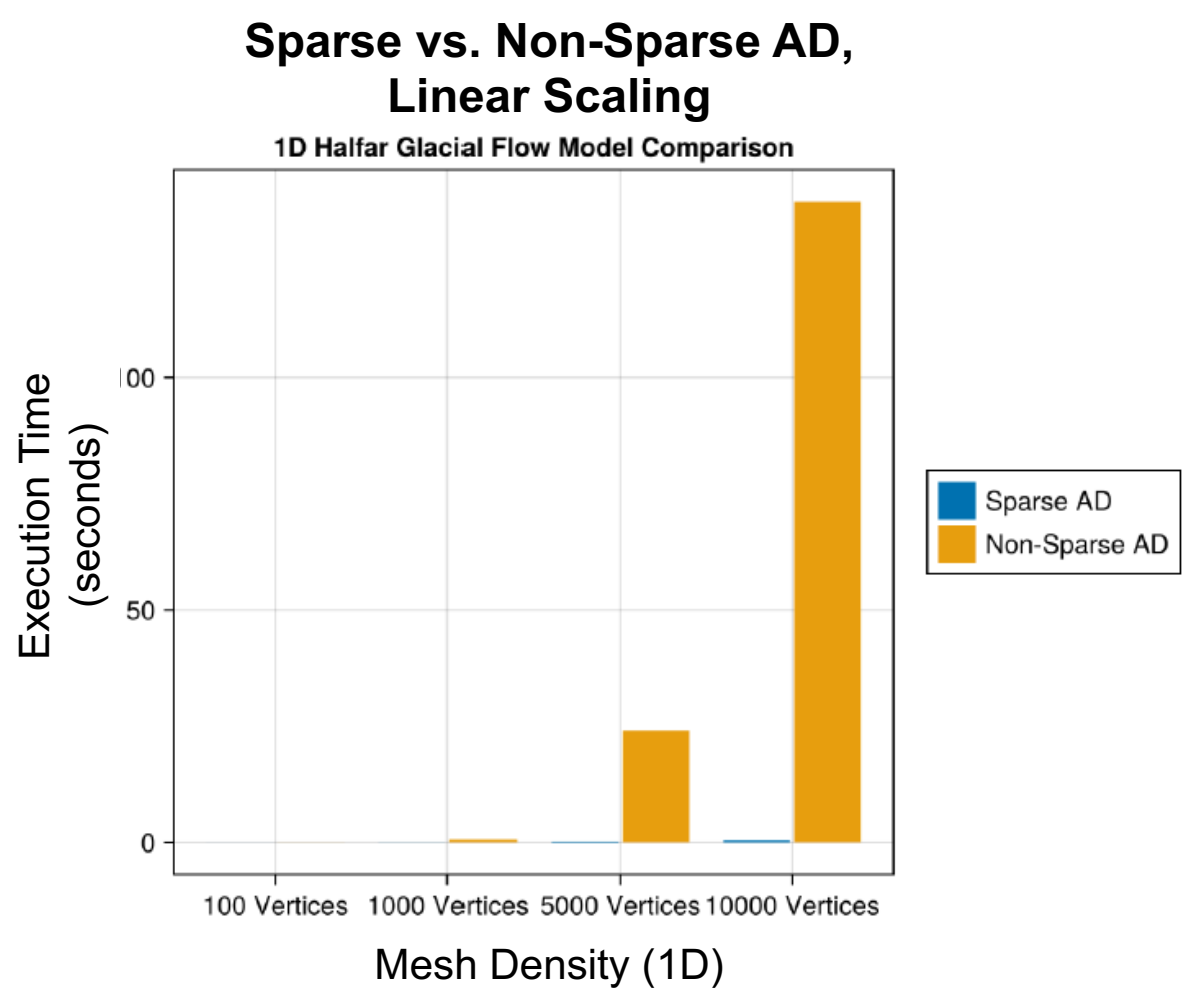
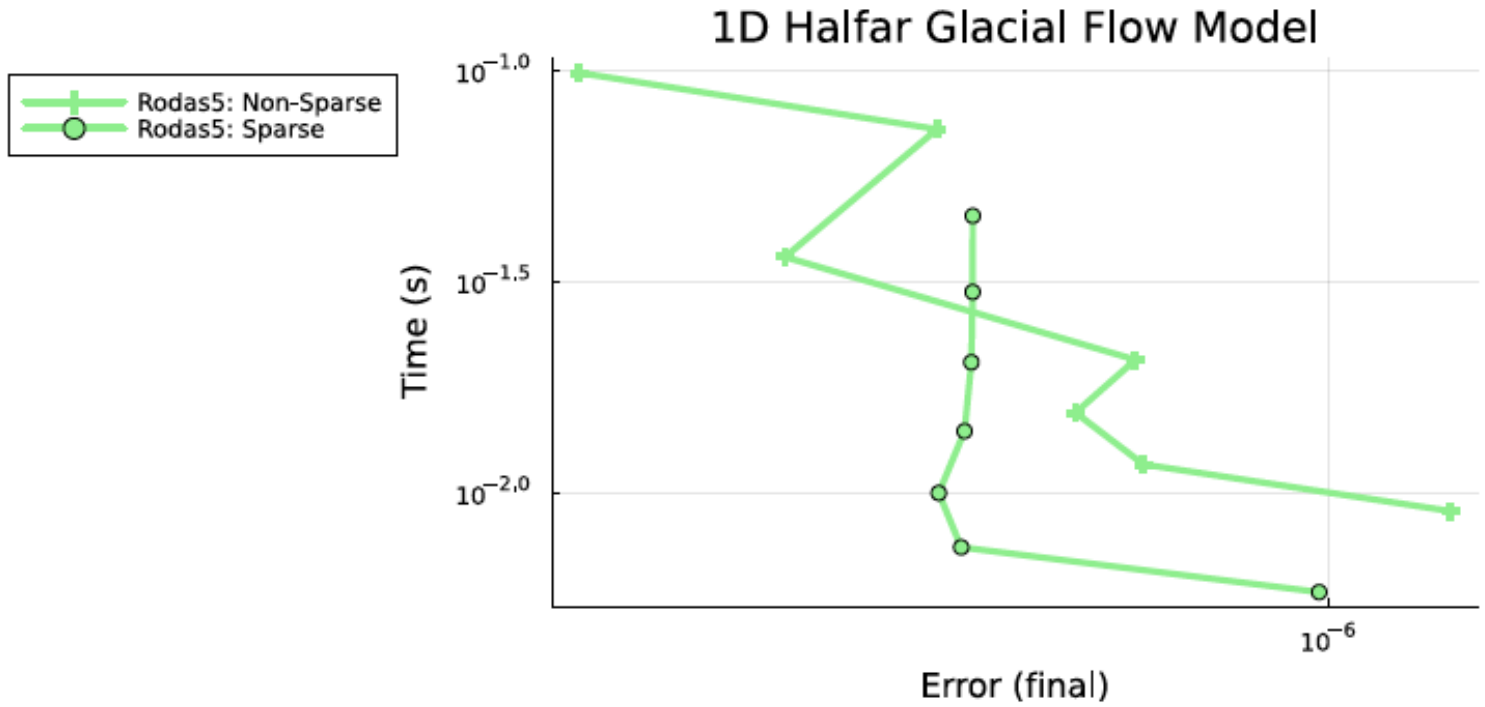


Figure 55. Sparse vs Non-Sparse Auto-Differentiation, showing how execution time scales with mesh density, with linear-scaling for the y-axis, to better understand how much faster Non-Sparse AD is, especially as the density/size of the problem scales up. Analysis and image provided by the JuliaHub team.

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TA	Performer Team	Metrics
		 **Figure 56. Work precision plot where each point represents a different absolute and relative set tolerance level, and resulting simulation error and execution time (x and y axes). Analysis and image provided by the JuliaHub team.**
4	**Uncharted**	**See Figure 57 for Epi IQ Section 1 timing and quality scores for model extraction**, which involves some TA1 (SKEMA) and TA2 (Mira) technology as well as TA4 capabilities developed by Uncharted. • It should be noted that the Baseline modeling team didn't choose to use any state-of-the-art LLM capabilities (possibly due to unfamiliarity with using such tools for these types of modeling problems); in using commercial LLM tools, the difference between Workbench/Baseline in timing/speed might have turned out to be much less. • In general, times for both Baseline and Workbench teams increase with difficulty/size of the extraction problem, with the exception of the hard Covid model. • See Epi IQ Section 1 (TA1) and Appendix A: Metrics Tables for more detailed discussion and summary on how TA4's model extraction capabilities met TA1 targets

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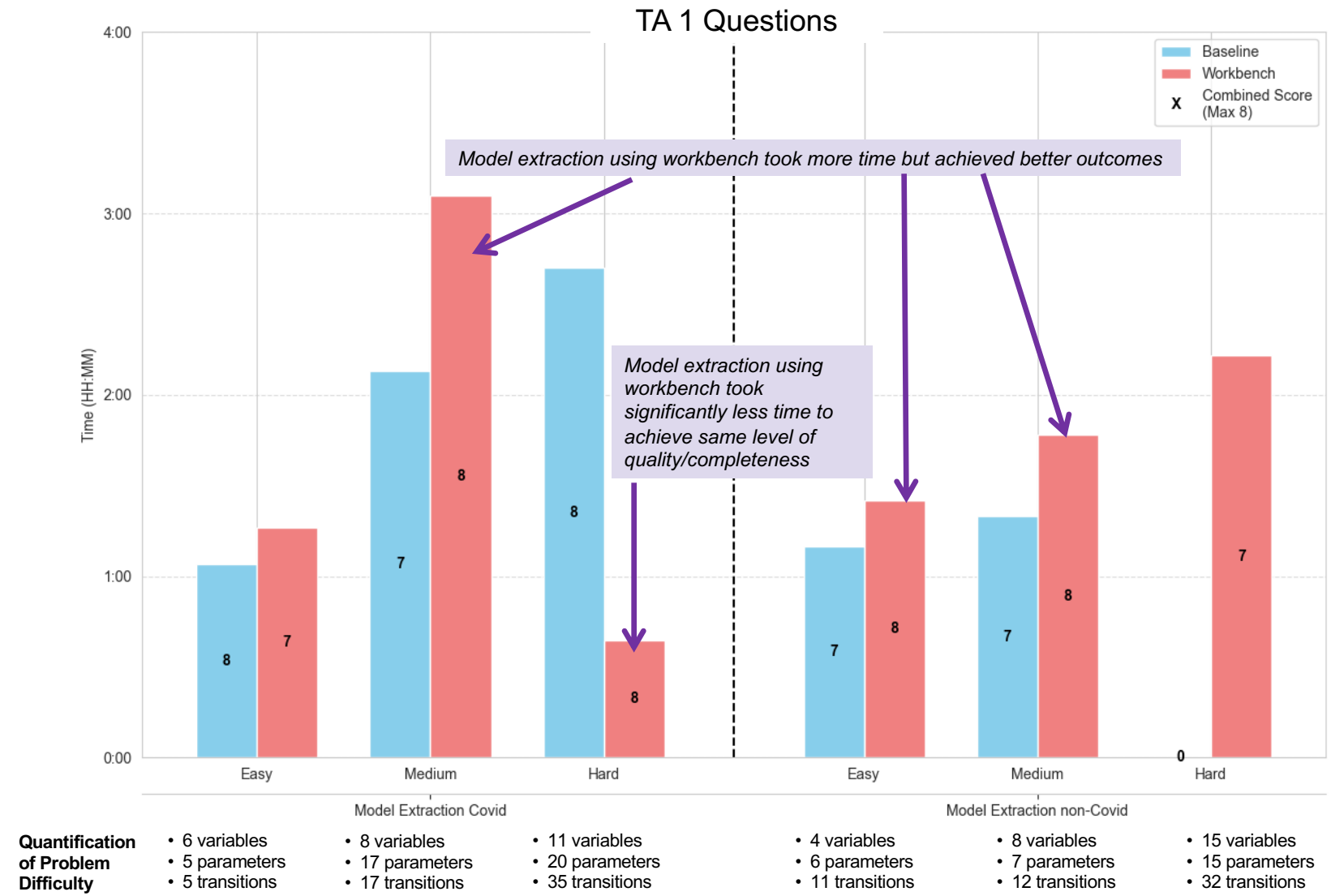
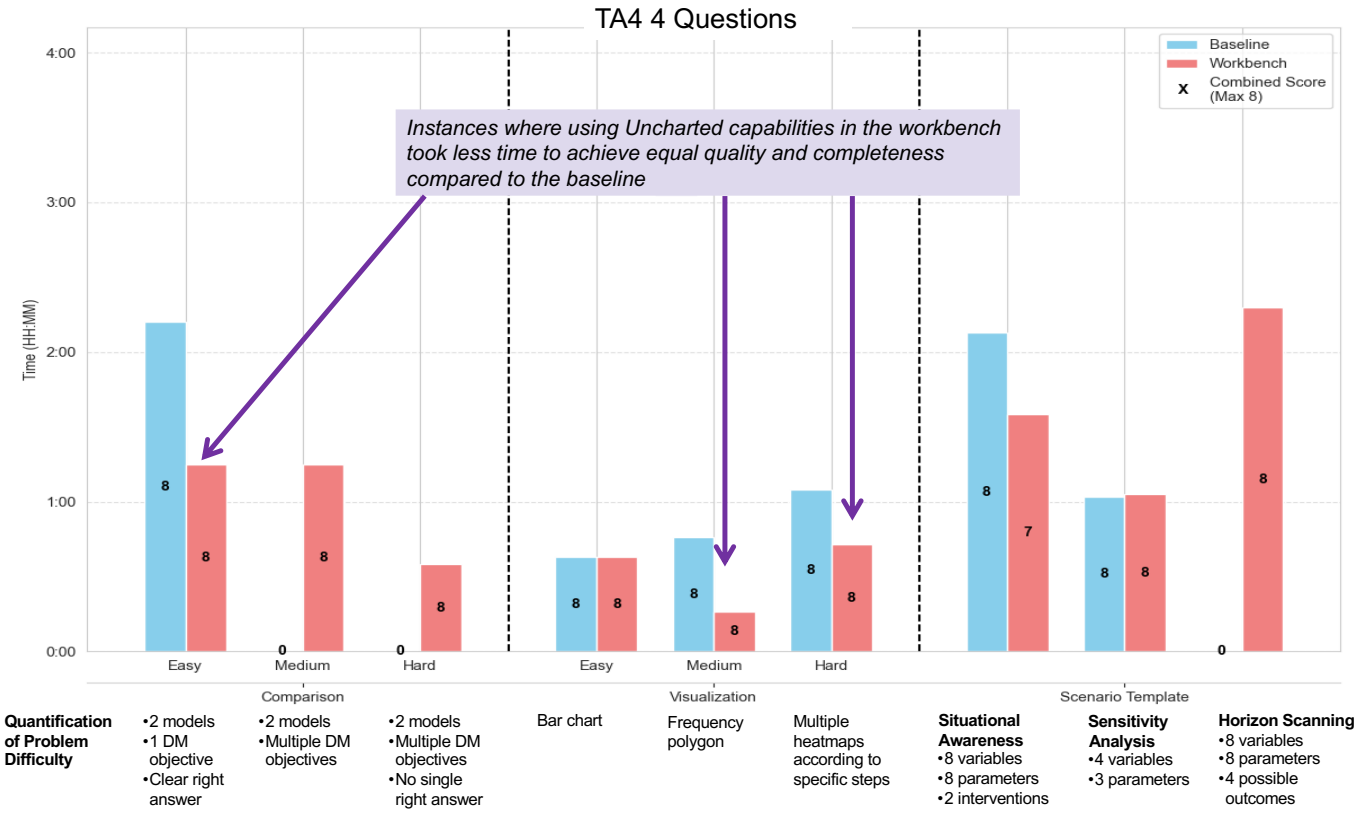


Figure 57. Epidemiology Independent Questions (Epi IQ) Section 1 timing and quality scores, which involves some TA1 and TA2 technology, as well as TA4 capabilities developed by Uncharted. Copy of Figure 5, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
		See Figure 58 for Epi IQ Section 4 timing and quality scores for model summarization and comparison, visualization, and scenarios templates, which are solely based on Uncharted's capabilities. • See Epi IQ Section 4 and Appendix A: Metrics Tables for more detailed discussion and summary on TA4 outcomes with Terarium • Note that for the Scenario Templates, the questions do not scale with difficulty, and simply represent different types of analyses; therefore we do not label those problems with easy/medium/hard difficulty. • There are no clear trends, except for the visualization questions, where the Baseline team spent increasing amounts of time, as the complexity of the visualization task increased  **Figure 58. Epidemiology Independent Questions (Epi IQ) Section 4 timing and quality scores for template scenarios, model summarization and comparison, visualization, and scenario templates, which are solely based on Uncharted's capabilities. Copy of Figure 11, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.**

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TA	Performer Team	Metrics
4	**Jataware**	**See Figure 59 for Climate IQ Section 3 timing and quality scores for general understanding/retrieval tasks with Integrated Assessment Models (IAMs), forecasting with IAMs, interventions with IAMs, optimization with IAMs, data search, and data transformation**, with all questions utilizing Jataware's technology stack. • Interestingly, for the questions where both teams could be compared, there seemed to an inverse relationship between problem completion time, and problem difficulty. The exception was the data transformation questions, where the Baseline team spent increasing time as the questions became more difficulty, but the opposite was true for the Workbench team. • There were 3 sets of IAM questions that the Baseline team did not attempt at all. • See Climate IQ Section 3 (TA4) for more detailed discussion on outcomes

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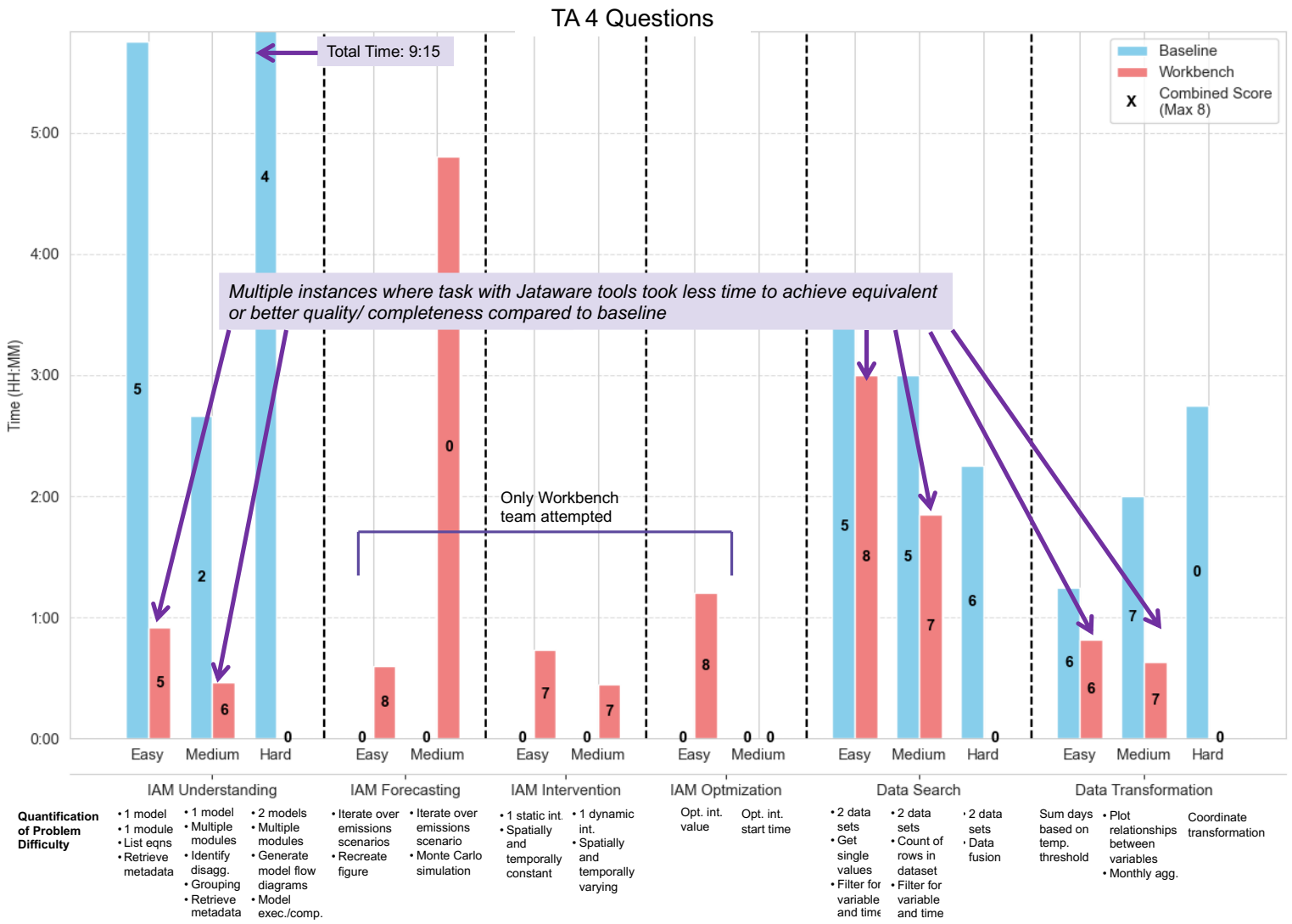


Figure 59. Climate Independent Questions (Climate IQ) Section 3 timing and quality scores for general understanding/retrieval tasks with Integrated Assessment Models (IAMs), forecasting with IAMs, interventions with IAMs, optimization with IAMs, data search, and data transformation, with all questions utilizing Jataware's technology stack. Copy of Figure 22, reproduced here for ease of reference. The heights of the bars represent 'critical path' time, and the numbers within the bars are the sum of quality and completeness scores. See Timing, Quality, and Completeness Logs for more detail on scoring methodology.

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TA	Performer Team	Metrics
		Qualitative examples with Beaker, from Climate DM scenario • **ESGF data**: Using Beaker, Workbench modeler were able to find the most relevant data more easily than Baseline team. Modeler was able to relay what they wanted into Beaker (e.g. 'I want data that meets xyz criteria'), and get list of files • **File management**: Workbench modeler used Beaker to inspect and clean up files no longer needed • **Complex geospatial tasks**: Modeler used Beaker to help with a complex geospatial task (unrelated to eval), on embedding lines from a shapefile onto a geospatial tiff with a given resolution and boundaries. Beaker was able to do this fairly easily where other common SOTA tools (ArcGIS) were not able to (or it was difficult to know how exactly to do it in those specialized programs). • **Beaker as an onboarding tool**: Beaker was helpful for onboarding with Decapodes and becoming acquainted with documentation. This became less useful once the Workbench modeler delved deeper into the tech stack and questions. • **Questionable/incorrect outputs from LLM highlights need for QC**: Sometimes workbench modeler would have to question Beaker output as a scientist, even as LLM agent very confidently gives incorrect results.

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